Federal Democratic Republic of Ethiopia

Ministry of Agriculture

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MINISTRY OF AGRICULTURE

Climate Resilient Wheat Value Chain Development

Project (CREW) Implementation Manual (PIM)

June 2024

Addis Ababa

Acronyms

ACC	Agricultural commercial Clustering
ADF	African Development Fund
AEFFP	
AGPII	Agricultural growth program II
ATI	Agricultural transformation institute
ATVET	Agriculture & Technical Vocational Training
AWP&B,	Annual work plan and budget
BoA,	Beauro of Agriculture
BoANR,	Beauro of Agriculture and Natural resources
BoAL	Beauro of Agriculture and Live stock
BoLPD	Bureau and Livestock and Pastoralist Development
BoPAD	Beauro of Pastoral development
BoQs	Bill of Quantities
BREFONS	Program to Build Resilience for Food and Nutrition Security
CA	conservation agriculture
CQS	Selection Based on Consultants' Qualifications
CREW	Climate Resilient Wheat Value Chain Development
CRGE	Climate Resilient green economy
CSA	Central Statistics Agency
DPs	DPs Development Partners
EAA	Ethiopian Agricultural Authority
EA	Executing Agency
ECC	Ethiopian cooperative commission

EGS	Early generation seed
EIAR	Ethiopian institute of Agricultural research
EPA	Environment protection Authority
ESAP	Environment and Social Assessment Procedure
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental and Social Management Framework.
ESMP	Environmental and social Management Plan
ESS	Ethiopian Statistical Service
FA	Financial Agreement
FAO	Food and Agricultural organization of the united nation
FBS	Selection under Fixed Budget
FSC	Federal Steering Committee
FFS	Farmer Field Schools
FPMU	Federal project management unit
FPPAA	Federal Public Procurement and Property Administration Agency
FTC	Federal technique committee
FTC	Farmer's training centre
GAP	Good agronomic practices
GCA	Global centre for adaptation
GDP	Growth domestic product
GoE	Government of Ethiopia
GHGs	Greenhouse gases
GPN	General Procurement Notice
GRC	Grievance Resolution Committee

GRM	Grievance Redress Mechanism
IC	Individual Consultant
ICB	International Competitive Bidding
IFB	Invitation for Bid
ISF	Irrigation Service Fee
ISFM	Integrated Soil Fertility Management
IPM	Integrated pest management
IMISET	Irrigation Management Information System for Ethiopia
IT	information technology
LCS	Least-Cost Selection
LOI	Letter of Invitation
MILLs	The Ministry of Irrigation and Low Lands
MIS	Market information system
MOI	Ministry of industry
MT	Metric tone
NCB	National competitive bidding
NWFP	National wheat flagship program
OCP	Office Chérifien des Phosphates, A state owned phosphate rock miner. Phosphoric acid manufacturer and fertilizer producer.
PEC	Procurement endorsing Committee
PIM	Project Implementation manual
PMP	Pest Management Plan
PMU	Program Management Unit
PPE	Personal Protective Equipment
PTC	Pastoralist training centre

PY	Project Year
QBS	Quality Based Selection
QCBS	Quality and Cost based Selection
REOI	Request for Expression of Interest
RFQ	Request for Quotation
RPF	Resettlement Policy Framework
RPMUs)	Regional project management unit
RSC	Reginal steering committee
RTC	Regional Technical Committee
SA	Social Assessment
SA	Safeguards, Social Assessment
SBDs	Standard Bid Documents
SHF	small holder farmers
SHIDLE	Smallholder Irrigation Development Lead Executive
SMS	Subject matter specialists
SSA	Sub-Saharan Africa
SSI	Small-scale Irrigation
SSS	Single sources Selection
TAAT	Technologies for African Agriculture Transformation
TEC	Technical Evaluation Committee
ToR	Terms of Reference
TOT	Training of trainers
ToRs/JoDs	Ternes of references / Job descriptions

TNA	Training Need Assessment
UNDB	United Nations Development Business
WHO	World health organization
WOAD	Woreda Office of Agriculture Development
WOANR	Woreda Office of Agriculture and Natural Resource.
WSC	Woreda Steering Committee
WTC	Woreda Technical Committee
WUAs	Water Users' Associations /Cooperatives,
WUO	water users' organisation

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1. Introduction

1.1. Background

Agriculture is the Dominant sector in Ethiopian economy contributing 35.8 percent of Ethiopia's GDP and employs 72.7 percent of its 120+ million people. The sector contributes 75% of export earnings. Crops take the highest share of total production and area coverage and contribute 68% of the agricultural GDP

Wheat provides a large proportion of the world's nutrition compared to nutrition from other cereal grains and also it is primarily used as a staple food providing more protein than any other cereal crop. Wheat ranks first in the world and sub-tropical regions of the cereal production and also, the world annually wheat feeding by human may be one-fifth of the population and is a staple food of about one third of the worlds (FAO, 2014). According 2024 EC MoA annual report, Ethiopia produces about 29.1 million tons of wheat from 8.15 million hectares of land, annually using rain feed and irrigation, making it the largest wheat producer in terms of acreage and output in sub-Saharan Africa (SSA). The crop is produced by more than 5 million smallholder farmers, making it the fourth most widely produced crop and taking up about 17% of all the arable land in Ethiopia. It is largely grown in the highlands of the country and constitutes roughly 10 % of the annual cereal production. The major wheat producing areas in Ethiopia are Arsi, Bale, Shewa, Western Harerghe, Sidama, Tigray, Northern Gonder and Gojam zones and currently it is produced under irrigation at lowland areas of the country like Somali and Afar regions.

Wheat production in Ethiopia is limited to smallholder farmers who use rain fed agriculture and are not able to produce enough to feed the growing population of the country, due to different constraints like Land fragmentation, inadequate infrastructure, and climate change impact further hinder agricultural productivity. Additionally, reliance on rain-fed agriculture makes the sector vulnerable to erratic weather patterns, leading to fluctuations in yield. The ministry of agriculture developed National wheat flagship program (NWFP) to support the ambitious goal of the Government of Ethiopia (GoE) to ensure food security for a large proportion of the population; increase small holder farmer household income; a sustainable input

supply to agro-processors; create large number of jobs for rural youth and save the foreign currency that can be utilized for other national priorities.

Therefore, the CREW Project aimed to address the gap of production and productivity through the climate resilient wheat production approach in the targeted areas of the country.

1.2. Project description

Wheat being one of the strategic crops in Ethiopia, there has been concerted effort to increase production and productivity over the last five decades. Productivity levels have increased substantially with an average national productivity of 3.11 t/ha. However, there is a huge potential for yield improvement given the 3.5 t/ha yield attained by model farmers and the 5 to 8 t/ha recorded at the research stations. Even though such potential is existing in wheat crop, there are some obstacles to Low agricultural productivity that can be attributed to smallholder farmers. These are limited access to agricultural inputs, financial services, improved production technologies, irrigation, and agricultural markets and, importantly, to poor land management practices that have led to severe land degradation and climate change.

Wheat demand has been growing at a faster rate than wheat supply due to emerging Agro processing industry and population increment in the country that leads to wheat supply deficit. The government has been spending a lot of money to import wheat for consumption per year. Although there are many constraints to the wheat subsector, the CREW project focuses on transforming wheat production through boosting productivity in rain-fed areas and expanding production into lowland areas through use of irrigation. CREW also focus on addressing wheat production through climate and marketing constraints. The choice of these interventions has been done based on government priorities to subsides the import wheat by producing in the country.

The CREW project will be strengthening actors in seed system and Mechanization to enhance production and supply of appropriate seed varieties that respond to the demand and to be enhance the mechanization service providers to boost the productivity of climate resilient wheat production at rainfed and irrigated area of the country. The reality is that some of the smallholder farmers do not afford to buy the production inputs needed for increasing crop productivity such as seeds, fertilizers, lime for acidic soils and agrochemicals, on cash basis, (herbicides, insecticides, fungicides) necessitates the establishment of a workable and sustainable credit

mechanisms. The needs of smallholder farmers in the country for timely access to the required productive resources and their proper use underscores the need for support for an efficient extension system and knowledge on transformational innovations and the most of the wheat producers cannot easily access the the mechanization due to limitation of mechanization service provider limited to already well known and conducive area and. On top this the CREW project was aimed boost up the participation of women and youth group in the intervention woredas through increasing by 50% the women participation in different activities related to Climate resilient and climate smart Agriculture and creating jobs for youth also the project was implemented through the following approaches .

- Use of improved wheat varieties tolerant to both abiotic and biotic stresses together with science-based agronomic practices. Developing such varieties may include the use of wild relatives and synthetics to develop improved wheat germplasm for resistance to abiotic and biotic stresses and enhancing yield and quality. Continuous varietal development in responding to emerging new rust races and supporting the deployment of newly released resistant varieties could help smallholders in dealing with rust challenges and maintaining improved yields in the rust prone environments of Ethiopia.
- Adopting climate-smart wheat technologies such as drip irrigation, conservation agriculture, fertigation, and site-specific nutrient management for resilient wheat production even with the changing climate.
- Integrated pest management (IPM), an approach that blends cultural, physical, chemical, and biological tools to control pests in a manner that minimizes economic, health and environmental risks.
- Establishing stronger seed systems and scaling up methods. New and improved seeds may be produced but an effective seed system ensures that they reach the farmers at the right time. In most developing countries, the seed systems are constrained by poor policy and regulatory frameworks; inadequate institutional and organizational arrangements; deficiencies in production, processing, and quality assurance infrastructure; poorly trained personnel, and limiting technical and managerial capacities.
- Building capacity for value addition to increase the earnings by the value chain players and motivate the farmers to invest more in wheat production.

- Introducing enabling policies and business environment. Subsidies on improved seed, irrigation equipment and water, and agro-chemicals encourage farmers to adopt improved technologies and inputs. Coupled with good infrastructure and marketing systems, these lead to a more vibrant wheat sub-sector characterized by high production and productivity, and value addition.
- Strengthening regional and international networks for collaborative research, production and dissemination of improved varieties and agronomic practices

1.3. Rationale

Ethiopia's wheat production and productivity has shown increasing rate in the past two decades at the same time, the gap between production and consumption has kept widening, due to high growing wheat demand for different purpose. The upward growing demand for wheat is attributed by population growth, growth in agro processing, urbanization, and improved household incomes. The situation is not expected to improve as wheat supply and demand.

Over the past 30 years, droughts have been increasing in severity and frequency in the Horn of Africa and their impacts have created conditions of chronic vulnerability to climate change with persistent food insecurity, water scarcity, widespread economic hardships, conflicts, and migration. The outbreak of the COVID-19 pandemic compounded by Russia's invasion of Ukraine induced disruptions in global wheat supply chains leading to a sharp price increase by 50% globally. In addition to the Bank's launch of the USD 1.5 billion AEFFP initiative in May 2022, the Bank co-organized in January 2023 the High-Level Summit in Dakar on Feed Africa: Food Sovereignty and Resilience (Dakar 2), to mobilize political, financial, and technical support for agricultural transformation on the continent at scale. The Dakar 2 Summit reinforces the Bank's Feed Africa Strategy (2016-2025) that seeks to transform African agriculture into a competitive, inclusive, and agribusiness-oriented sector that provides food and nutrition security for Africans, creates economic opportunities, and improves livelihoods of rural African communities. The GoE presented its Country Food and Agriculture Delivery Compact at the Dakar 2 Summit focusing on the opportunities for agricultural productivity, infrastructure, processing, value addition, markets, and financing to accelerate the transformation of agriculture and food systems in Ethiopia. The Compact prioritized wheat as one of the strategic commodities

for the rapid transformation of agriculture and food systems at scale. The outcomes of the Dakar 2 Summit were endorsed at the 36th African Union Summit of February 2023 calling for global support for its immediate implementation to address rising food prices, disruption in the global food supply, and worsening of food insecurity in Africa.

1.4. Purpose of the manual

The Project Implementation Manual (PIM) guides implementers of the climate Resilience wheat value chain project (CREW) and provides transparency about procedures and criteria to all project stakeholders. The PIM provides detailed procedures to guide stakeholders in the implementation of activities at all levels (Federal, Regional, Zonal, Woreda, Kebele and community). The PIM is also a tool for supervision and monitoring by the Steering Committees; Senior Management and Development Partners (DPs). It provides guidance for all areas of project execution including financial management, procurement, disbursement, monitoring and evaluation, social safeguards and reporting system.

The PIM defined the project Geographical area of implementation, eligible beneficiaries and targeting and eligibility criteria, as well as the detailed mechanism behind each activity and their sequencing and roles and responsibilities for implementation. It depicts the required processes both for Government of Ethiopia procedures, approvals and annual planning, as well as those of the African Development Fund (ADF) (such as Annual Work Plans and Budgets, ADF, for non-objection) and any specific requirements of other Development Programs.

This PIM is based on the government program design as well as the African Development fund documents, including the Project Appraisal Document, the Environmental and Social Safeguards, Social Assessment (SA), Environmental and Social Management Framework (ESMF), the Resettlement Policy Framework (RPF) and the Governance and Anti-Corruption (GAC) guidelines. The original guidelines remain the official reference for the implementation of these aspects during the program.

A number of detailed manuals accompany this PIM (the list is found in Annex). The PIM is a dynamic document that should be continuously updated in consultation with project stakeholders, in particular the Africa Development Fund to ensure that it remains valid and useful over time. However, any provisions in the PIM cannot be amended or waived without the

approval of the Federal Steering Committee (FSC) and the non-objection by Africa Development Fund (ADF)

1.5. Project Objective

1.5.1. General Objective

The objectives of Ethiopian Climate Resilient Wheat Value Chain development project (CREW) are to enhance the production and productivity thereby increasing the incomes of small-scale wheat producer farmers in Ethiopia.

1.5.2. Specific Objectives

- ✓ Enhance the production and productivity of wheat crop across agroecology
- ✓ Enhance soil productivity across the intervention areas through applying integrated soil fertility management.
- ✓ Ensure the women living standard and Job creation for youth across the intervention woredas of the country.

1.6. Target areas and beneficiaries

CREW targeted regions: The project will, be implemented in four regional states of Oromia, Amhara, Somali and Afar region. Those regions have been selected in consultation with the government of Ethiopia based on the following criteria: a) conducive Agro-ecology for wheat production, b) highly Experienced on Cluster farming; c) presence of economic/market opportunities relevant to the project.

The project will benefit a total of 500,000 smallholder households or 2,300,000 people. The project targets 50% women. These will include 15% of women headed households (representing 75000 households with 345000 people) in 76 Woredas/Districts. From this 38 District from Oromia, 24 district from Amhara, 11 districts Somali and 3 from districts from Afar Regional States. Amhara and Oromia regions are the largest highland rainfed and irrigated wheat producing areas of the country while Afar and Somali regions have great potential for lowland irrigated wheat. According to the (CSA, 2021) Oromia and Amhara produced 50.03 million quintals with in a season by rainfed and in Somali and Afar region wheat produced by irrigation as well as by rain fed in the small amount compared to the Oromia and Amhara but those Afar and Somali region have a potential of irrigable land for lowland production during the off season as the whole the four regions produce approximately 5.6 million tons of wheat 1.87 hectares of

land was covered under rainfed also about 2.8 millions of ha produced under irrigation at lowland and during offseason. The beneficiary Woredas were identified by the Ministry of Agriculture in consultation with the four regional governments. Other key beneficiaries included in the CREW project like the wheat millers and traders, women Agro booster groups and Support for youth and start up Support for youth and startups An initiative dedicated to support idea-holders and startups in the Ag-Tech sector. The main goal is to enable African youth to tackle the great challenges faced by African farmers through innovation, technology, and entrepreneurship. At project preparation and appraisal, stakeholder consultations were undertaken at federal, regional and community levels, with national and district local administration, millers' cooperatives, traders, financial institutions and farmers cooperatives and farmers groups to provide inputs in project design. Targeting mechanisms for the selection of beneficiary farmers and value chain actors detailed in the project implementation manual, based on the results of the environmental and social impact assessment studies.

2. Institutional arrangements and responsibilities

2.1. Implementation Arrangements & Project Governance

The Ministry of Agriculture (MoA) will be the Executing Agency (EA). The Project will use the existing government structures at the Federal, Regional and Woreda levels through a project Management Unit (PMU) at the federal level and regional PMU all of which will be constituted with recruited staff. A Federal Project Management Unit (FPMU) will be established within the Crop Development Department at the Ministry of Agriculture, for the day-to-day implementation of the Project. The FPMUwill coordinate implementation of the project with the four Regional Project Management Units (RPMUs) in Afar, Amhara, Oromia, and Somali Regional States.

A Federal Project Steering Committee (PSC) will be established to ensure inter-ministerial coordination and provide overall coordination, policy guidance and oversight for the implementation of the project. The Federal PSC, chaired by the State Minister of Agriculture and Horticulture (MOA) will include membership from the relevant sector ministries and institutions.

The implementation of the activities financed by OCP Africa will be ensured by the OCP Ethiopia Representative Office. A dedicated team will be set up within the office in Addis Ababa and in the regions concerned under the supervision of the country manager. The Office will work closely with the Project Management Units and the Ministry of Agriculture to ensure good

coordination and consistency in the implementation of the various components of the Project.. The Technologies for African Agriculture Transformation (TAAT) will provide technical assistance through its wheat and policy compacts in the following areas: (i) Strengthening the national wheat seed system; (ii) Enhancing the skills and capacity of seed technicians in the private sector; and (iii) Review and recommend policy and institutional reforms. The Global Centre on Adaptation (GCA) will also provide technical support in areas of climate change.

2.1.1. Federal CREW Steering Committee

The project will be overseen by Governance structures established at different levels of the program implementation and coordinated by MoA but with wide representation of stakeholders. CREW Steering Committee's will be established at Federal, Regional and Woreda levels, responsible for:

- Overall oversight, decision making a strategic guidance for Project and implementation
- Review and approve AWP&B, submitted by the PMU, including subsequent re-allocations or corrective actions as needed.
- Review and approve the annual implementation performance report prepared by the FPMU and overseeing the implementation of corrective actions;
- Ensure coordination, harmonization and alignment amongst the different implementing institutions;
- Ensure that gender inclusiveness, climate, youth employment and other cross-cutting issues are sufficiently considered and addressed.
- The zonal bureau Agriculture head will be member of the regional SC and Regional Chairs persons will be members of the federal SC to ease communication between and create transparent and participatory decision-making processes at all level.
- Selected DPs Working on the area of wheat value chain (like millers Association) could be member of the federal SC.
- The Federal and Regional SC will meet biannually and the meetings at regional level will be scheduled before the Federal SC meeting.

Federal CREW Steering Committee members

1. State Minister, Agriculture & Horticulture sector
2. State Minister, Ministry of Irrigation and Low Lands
3. State Minister, Ministry of Water and Energy
4. State Minister, Ministry trade and regional and regional integration
5. State Ministers, Ministry of Industry (MOI);
6. Director General, Ethiopian Institute of Agricultural Research (EIAR)
7. Director General, Agriculture Transformation Institute (ATI)
8. Director General, Ethiopian Agricultural Authority (EAA)
9. Director General Federal Environment Protection Authority (EPA);
10. Commissioner, Federal Cooperative Commission (FCC)
11. Representatives from the Regional Bureau of Agriculture in Afar, Amhara, Oromia and Somali Regions
12. Regional CREW project managers.
13. Representative from the Millers' Associations
14. Federal PMU: - Secretary

The main roles of FSC are:

- Policy guidance on project implementation;
- Overseeing project management;
- Federal Level Coordination with line ministries and national agencies to ensure consistency with national standard and policies, to avoid duplication and to underpin sustainability.
- Approving annual work plan and budget;
- Monitoring implementation progress.
- The FSC will meet by bi annually. The secretarial function for the FSC will be done by FPMU including preparation of detailed agenda notes, maintaining minutes and following up on important decisions of the FSC.

2.1.2. Federal Project Management Unit (FPMU)

The implementation arm of MoA will be the FPMU which will undertake the day-to-day project management and coordination activities. The FPMU will be established with a multi-disciplinary dedicated team headed by national Project Manager.

Key CREW staffs for FPMU

1. The Project Manager
2. National Agronomist/Value Chain Officer
3. Senior Irrigation Specialist
4. Environmental and Social Safeguards Officer
5. Gender, Youth and Community Engagement (GYCE) Officer
6. Monitoring and Evaluation Officer,
7. Procurement Officer
8. Finance Officer
9. Accountant
10. Social Safeguard Officer
11. Administrative Officer & Office secretory
12. Other support staff.

Roles and functions of FPMU:

The FPMU shall be responsible for the day-to-day management, implementation, monitoring and evaluation including fiduciary oversight of the project at federal level. The FPMU being the key implementing body, the leadership and management team collectively remain accountable to achieve the outputs and results envisaged under the project with the following major roles.

- Organizing Project Implementation set up at all levels;
- Managing the implementation of project activities at federal level;
- Ensuring compliance with financial management guidelines;
- Managing procurement;
- Ensuring compliance with environmental and social standards;
- Capacity building of implementing agencies at federal, regional, woreda and Kebele levels whenever necessary.
- Carrying out monitoring and evaluation activities including regular performance, financial and other progress reports;
- Liaise with other stakeholders and undertaking communication activities;
- Mobilizing external technical support as needed; and

- Collaborating with funding agencies and development partners for ensuring interaction.

2.1.3. Federal Technical Committee (FTC):

The FTC has membership comprising of the FPCU thematic experts and technical experts from the member institutions of FSC and observers. The FTC will be chaired by the National Project Manager. The FSC will be given technical backstopping support by the FTC.

Roles of FTC:

The main roles of the FTC are:

- Overall technical guidance to FSC;
- Prioritizing project implementation activities;
- Reviewing sub projects proposed for implementation.
- The FTC will meet as often as needed, but in any case, at least one meeting shall be held every quarter.

Line Ministries and Agencies:

The regional level and zonal level sector bureaus and offices will assist woreda and kebele level offices and institutions to implement project activities.

Their roles include:

- Implementation support to woredas for all the project component activities;
- Sensitization, awareness Mobilizing and establishing and strengthening community institutions;
- Supporting compliance with environmental assessment and implementing ESS;
- Facilitating design, construction and quality assurance of social and economic infrastructure that are used for climate resilient wheat value chain development.

2.2. Regional Level Institutional Arrangement

The facilitation, management coordination, planning and monitoring of project activities at the regional level will follow a similar arrangement of the federal level. The main regional level implementation mechanism consists of:

- Regional Steering Committee;
- Regional Project Management Unit.

- Regional Technical Committee; and

2.2.1. Regional Steering Committee (RSC):

The main policy making and overseeing entity at the regional level is the RSC. Depending upon the existing institutional structure in the region, the RSC will be chaired by the head of the Bureau of Agriculture (BoA). The membership of RSC is shown below.

Regional CREW Steering Committee members

1. Regional Head of bureau of Agriculture: -Chair Person
2. Regional CREW implementing institutions Bureau/Authority/ Agency /commission heads
3. Regional Agricultural Research Institution Director General
4. Head, Zonal Office of Agriculture
5. Regional PMU: - Secretary

Roles of RSC:

- Regional level strategic guidance;
- Overseeing project management;
- Coordination of regional implementing institutions, regional bureaus and regional agencies;
- Approving annual work plan and budget for the region and ensuring recurrent budget is allocated for operation and maintenance; and
- Monitoring regional project progress and removing implementation bottlenecks.
- The RSC will meet every quarter.
- The RPMU will act as the Secretariat, including preparation of agenda notes, maintaining minutes and following upon key decisions.

Regional Project Management Unit (RPMU):

Depending upon the structure prevailing in the region BoA, BoANR, BoAL or BoPAD will be the Regional Implementing Agency.

Key staff at the Regional PMU

1. Regional Project Manager,
2. Procurement Officer,

3. Monitoring and Evaluation Officer and
4. An Accountant.

Roles of Regional PMU

- The Regional Project Manager (RPM) would be responsible for planning, administering and coordinating the activities of the ADF financed Climate Resilient Wheat Value Chain Development Project (CREW) at the regional level
- Ensure that the Project is implemented in accordance with the signed Grant Protocol Agreement, approved Project Appraisal document and Annual Work Plan and Budget (AWPB).
- Responsible for the monitoring, overseeing and guiding the implementation of the different components of the Project by the project staff and other stakeholders in the region.
- Responsible for ensuring the project resources are used for eligible expenditures as per the Protocol Agreement, Project Appraisal Document and approved Annual Work Plan and Budget.
- Report on Quarterly and Annual Progress Reports.

2.2.2 Regional Technical Committee (RTC):

The backstopping support to project implementation at the region will be through the setting up of an RTC. The RTC will have the following members:

- Regional Project Manager (chair).
- Technical Directors or delegated experts from the relevant sector bureaus/agencies;

The key roles of RTC are:

- All technical guidance to RSC;
- Review and prioritization of project activities relating to all component activities;
- Review and recommendation of proposed sub projects;
- The RTC will meet as often as needed, but at least one meeting every quarter.

2.2.3. Woreda Level Arrangement

Similar to the regional level arrangement, the woreda/city/town administration level implementation arrangement mainly consists of:

- Woreda level sector office
- Woreda Steering Committee;
- Woreda Technical Committee;

Woreda Steering Committee (WSC)

- The woreda council will function as WSC and will be the apex implementation body at the woreda level.

The composition of the (WSC) consists of:

- Heads of sector Agriculture / Pastoral Development offices /Cooperative Promotion office/low land and irrigation office, Representatives Microfinance institutions; and

The roles and responsibilities of WSC:

- The woreda level guidance and policy support for project implementation include:
- Woreda level guidance and overseeing of project implementation;
- Approving annual work plan and budget;
- Monitoring project implementation at woreda level;
- The WSC will meet on a monthly basis.

Woreda Technical Committee (WTC):

- The technical backstopping for the smooth functioning of WSC will be provided by WTC.
- Composition and Membership:
- The WTC will be composed of technical staff from all the sector offices responsible for project implementation.

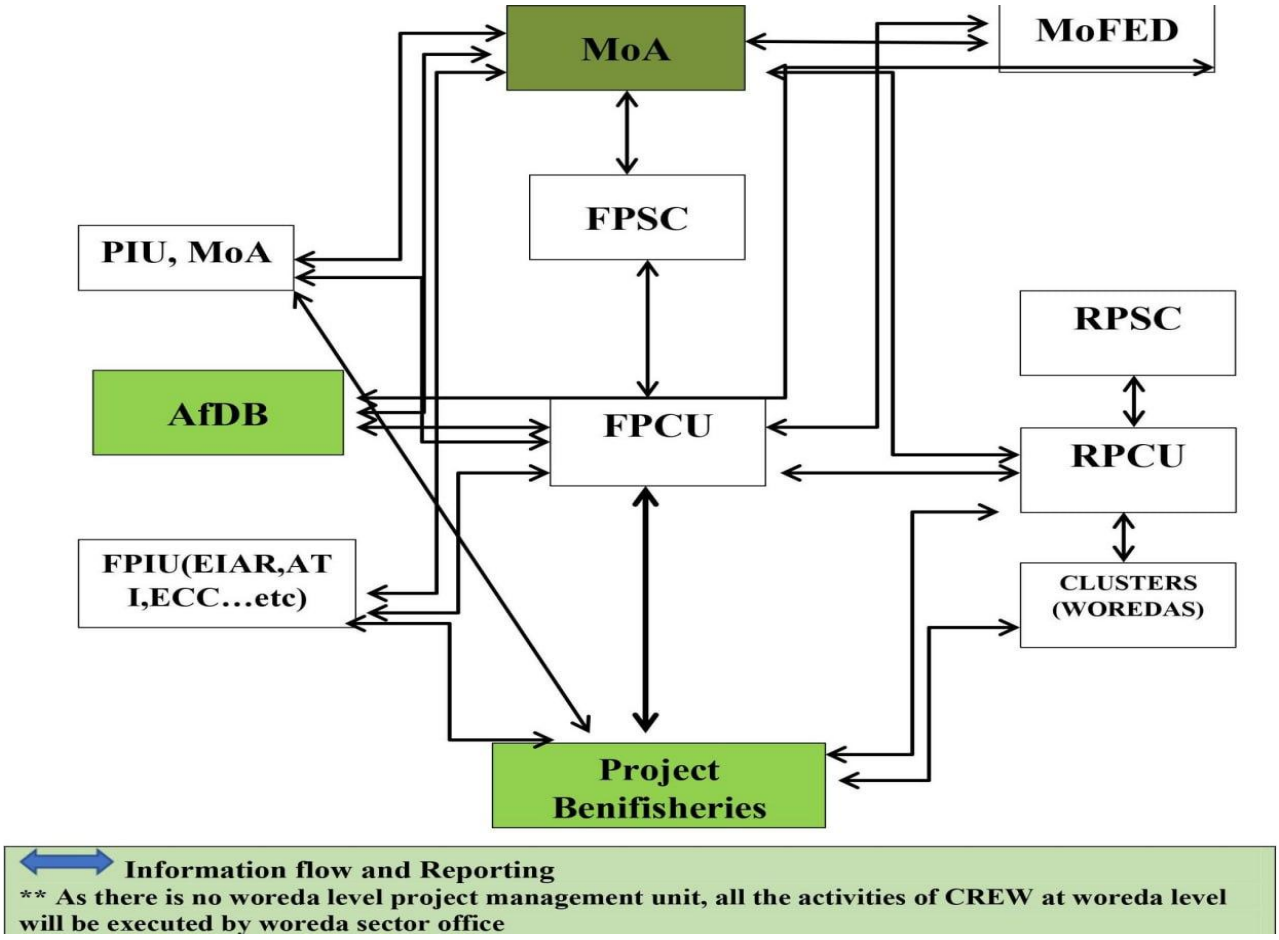
Roles and responsibilities of the WTC:

- Facilitating local level planning;
- Supervising sub project implementation;
- Compiling, Promoting and sharing best experience on climate resilient wheat value chain development.
- WTC will meet as often as needed, with at least one monthly meeting.

Woreda level Sector Offices:

- Support the implementation of all component and sub components described in the annual plan of respective CREW projects woredas
- Capacity support activities for sector offices
- Support to Extension advisory services like strengthening best experiences obtained in climate resilient wheat value chain development.

Figure 1 pictorial representation for CREW project implementation arrangement



3. Project components

3.1. Component 1: Climate Smart Wheat Productivity and Production

3.1.1. Subcomponent 1.1 Facilitate access to seeds for small-scale farmers

The main objective of this sub-component is to strengthen the national seed system for the production of early generation seeds (EGS), basic and certified seeds of high nutritional value (protein, gluten, and other micronutrients) in sufficient quality and quantity with affordable prices. The key activities implemented to ensure improved seed access to small scale farmers are listed below.

3.1.1.1. Enhancing operational capacity of research centers for EGS supply:

A/ Equipment for research stations (Federal-EIAR)

Currently, wheat research and source seed multiplication activities in the national research system are severely constrained by a shortage of farm machinery in terms of types, quantity and quality. The aim under this category is, therefore, to bridge the gap that exists in farm machinery and seed storage access across the national research system, to meet the efficiency in technology generation, adaptation, and source technology multiplication. The investment in farm machinery facilities focuses on special equipment that enhances land preparation, planting, harvesting, threshing, and seed cleaning operations of the source seed production of each center. On the other hand, research centers do not have conducive seed storage facilities to store their produced seed, so it gets attention to fill the gap by constructing seed stores.

B/ Procurement Implementation Arrangement:

The identified farm machinery will be procured based on the type; quality and detailed specification primary developed considering the soil condition of centers possess, following complete BID and procurement procedures. The MoA, through the Federal Project Coordination Unit, should be responsible for monitoring, supervising and coordinating project procurement activities. The Procurement Directorate in the MoA assumes primary responsibility for procurement activities at the federal level. The MoA through the Federal Project Management Unit shall procure targeted farm machinery goods, including tractor, disc harrow, moldboard

plough, row planter, seed drill, combine harvester, thresher and seed cleaner that require foreign currency. The major activities will be:

- Identify the suitable tractor regarding house power, disc harrow, moldboard plough, row planter, seed drill, combine harvester, thresher and seed cleaner types that suite to centers soil property.
- Prepare detail specification of the targeted machinery purchased
- Prepare international bid, procure for tractor, disc harrow, moldboard plough, row planter, seed drill, combine harvester, thresher and seed cleaner
- Build suitable seed storage facilities access
- Training seed professionals in the research system

3.1.1.2. Support EGS production operation within the research system:

Production and promotion of wheat varieties adapted to new irrigated areas and mid to high altitude of wheat producing belt areas of the country, requires regular supply of targeted varieties with comprising quality and quantity aspects at reasonable price, place and time. However, in many developing countries like Ethiopia farmers have not yet been able to fully benefit from the advantages of using improved quality seed due to multiple factors, including inefficient seed production, distribution, and lack of quality assurance systems.

Among others, the shortage of quality Early Generation Seed (EGS) is most cited as the cause for the inadequate performance of the national seed system. EGS, which encompasses variety maintenance, breeder seed, pre-basic and basic seed, is the critical connection between breeding activities and the eventual production and distribution of varieties to farmers. Therefore, to enhance stress tolerance wheat EGS production operation within the research stations and to confirm seed access for small holder farmers, seed enterprises, and other seed producers, multiplying pre-release candidate varieties seeds, demanded varieties maintenance, and source seed multiplication (breeder and pre-basic seed) will be operationalized during the project implementation period. Therefore, the major activities will be:

Variety maintenance:

Maintenance breeding is an obligatory step in the crop breeding programs and variety development process to ensure the sustainable development of improved varieties responsive to

the national seed demand. Growing high-yielding varieties with improved quality plays a significant role in Ethiopian agriculture by contributing to food and nutritional security, export market, and import substitution. However, one of the major constraints in enhancing crop productivity is the deterioration of varieties when they are multiplied in subsequent generations, contributing to low yield and poor quality. Thus, keeping the identity of a variety is a key role of breeding programs that developed the variety following the standard variety maintenance and breeder seed multiplication procedures. The major interventions to ensure variety maintenance should be done on:

- Maintain or ensure the genetic purity
- Maintain the identity (true to type) of the variety,
- Maintain other good qualities of the seed, and
- Ensure the seed supply chain with good quality seed

Pre-release multiplication of candidate varieties:

Pre-release multiplication of candidate varieties permits an adequate amount of seed for further multiplication to be employed pre-release seed multiplication during the main and off-seasons. This practice mainly deployed when, wheat disease remains one of the most important production challenges for wheat in the highlands and the country suffered from periodic rust epidemics. The development and deployment of resistant varieties has been the major strategy to control rusts of wheat worldwide. While resistance conditioned by major genes has been most widely used breeding strategy, pyramiding of several genes into a single cultivar is an attractive breeding strategy since it reduces stepwise accumulation of virulence by the pathogen against each gene. The intervention here is:

- the development and deployment of cultivars carrying durable or slow rusting resistance based on quantitatively inherited, multiple genes referred to as adult plant resistance.
- Implement pre-release multiplication of candidate varieties

Demand-driven breeder and pre-basic seed production of wheat varieties:

A sustainable seed system will ensure that high-quality seeds of a wide range of varieties and crops are produced and fully available in time and affordable to farmers and other stakeholders. However, in many developing countries like Ethiopia, farmers have not yet been able to fully benefit from the advantages of using improved quality seed due to multiple factors, including

inefficient seed production, distribution, and lack of quality assurance systems. Among others, the shortage of quality Early Generation Seed (EGS) is most cited as the cause for the inadequate performance of the national seed system. EGS, which encompasses breeder seed, pre-basic and basic seed, is the critical connection between breeding activities and the eventual production and distribution of varieties to farmers.

Moreover, the quality of the seed at this stage is highly critical, and hence EGS production is expensive and demands high resources (capital, skilled labor, land/media, and quality assurance support infrastructural facilities) allocation. The production modality can involve a cost recovery system and could be a source of revenue generation. Therefore, the interventions here will be:

- Demand assessment and production planning of EGS
- Produce demanded breeder and pre-basic seed production of wheat varieties
- Seed quality assurance
- Allocation and distribution of EGS timely with an affordable price

3.1.1.3. Operationalize variety licensing for better commercialization of new varieties:

Operationalizing variety licensing in Ethiopia is vital for the commercialization of new wheat varieties because it provides a clear legal framework for intellectual property protection, incentivizes investment in research and development by ensuring returns for innovators, facilitates the flow of varieties from federal to regional research, and ultimately helps seed companies more efficiently bring improved seeds to farmers. This process enables the timely availability of high-quality, productivity-enhancing seeds, a crucial step for Ethiopia to achieve food self-sufficiency and boost agricultural productivity.

Benefits of Operationalizing Variety Licensing :

- **Enhanced Research Incentives:** A robust licensing system protects intellectual property rights, which provides a crucial incentive for the Ethiopian Institute of Agricultural Research (EIAR), regional institutes, and even private sector seed developers to invest in breeding and releasing high-yielding, climate-resilient wheat varieties.
- **Clearer Pathways for Commercialization:** Licensing creates a transparent process for translating research innovations into commercially viable products. It defines how and by whom a variety can be used for seed multiplication and sale, facilitating the transition from research labs to the marketplace.

- **Streamlined Seed Production and Distribution:** A well-defined licensing process helps connect federal and regional research with seed companies and farmers more effectively. This ensures that successful research outcomes translate into the multiplication, processing, and distribution of certified, high-quality seed.
- **Increased Investment in the Seed Sector:** Licensing creates a more predictable and secure investment environment, attracting both domestic and international seed companies to invest more, expand their operations, and compete in the market. This leads to greater availability of improved seeds at affordable prices**.**
- **Improved Food Security and Economic Growth:** By enabling faster adoption of improved seed varieties, the licensing system contributes to higher yields, better disease and pest resistance, and enhanced climate adaptation. This directly supports Ethiopia's efforts to achieve wheat self-sufficiency and improve food security, while also generating income and opportunities in the wheat value chain.

The Role of Different Actors in the Variety Licensing;

- **Federal Research (EIAR):** Develops new varieties and provides initial basic seeds.
- **Regional Research Institutes:** Conduct localized testing and may participate in further seed multiplication.
- **Seed Companies:** License the varieties from research institutes, multiply the seed, and distribute it to farmers, contributing to a competitive formal seed sector**.**
- **Regulatory Bodies:** Oversee the licensing and certification processes to ensure quality and adherence to regulations, as highlighted by the Ministry of Agriculture (MoA)'s role.

Therefore, strengthening the seed quality monitoring and evaluation capacity of the research system is an important aspect of operationalizing newly released potential variety licensing. So, the major activities will be:

- Conduct an awareness creation workshop on variety licensing:
- Field monitoring on seed production from licensed varieties
- Support the sole organization Ethiopian Agricultural Authority to bring on board to enhance the Variety licensing operation in the country.

Implementation of enhancing the operational capacity of research centers for seed supply will be undertaken by the different research centers under the national and regional research institutes.

The implementing research centers and program leaders will report their performance to their respective institutes 'Ethiopian **Climate Resilient Wheat Value Chain development project' coordination** units and respective sector directors of EIAR on quarterly, bi-annually and annual basis.

Table 1 1Enhancing operational capacity of research centers for EGS supply expected outputs and responsible institutions

1.1.1.1 Equipment for research stations (Federal-EIAR) 1.1.1.2 Support EGS production operation within the research system: 1.1.1.3 Operationalize variety licensing for better commercialization of new varieties**:**
Lead Organizations/Institutions: MoA (Agricultural and horticulture sector and Agricultural Investment & Input Sector) Main Implementing Institutions: - Relevant EIAR Directorates, relevant research Centers at all levels and relevant lead executives of the MoA Supporting/collaborating institutions: - EAA, Regional Bureau of Agriculture and Relevant Regional Bureau, agency, ATI, Seed companies.
<u>Outputs (deliverables)</u> **Output 1.1.1.1: Equipment for research stations (Federal-EIAR**) ➢ Agricultural farm machineries comprising about: five tractors, five disc harrow, five moldboard, five combine harvesters, five row planters, five Seed drill, five seed thresher and five seed cleaner procured and distributed, ➢ 15 seed storage will be built, and ➢ 200 seed professionals will be trained for the short term **Output 1.1.1.2: Support EGS production operation within the research system:** ➢ 400 metric tons of Pre-release multiplication of candidate varieties will be produced ➢ 480 varieties maintenance breeding work will be carried out ➢ 4320 metric tons of Breeder and pre-basic seed will be produced **Output 1.1.1.3: Operationalize variety licensing for better commercialization of new varieties**:

- workshop on awareness creation with the participation 100 participants and variety licensing will be conducted
- 20 round Field monitoring and evaluation on seed production for licensed varieties carried out

Monitoring, Reporting & Evaluation Arrangements:

1 Monthly, quarterly, bi annually and, annual report

2 Field Supervision and Monitoring**.**

Joint Review and Implementation Support

Table 1Enhancing operational capacity of research centers for EGS supply

No.	Activity	Unit	Prog. Life	Unit Cost (UA)'000	Physical action plan implementation schedule						Budget allocated (UA '000') schedule					Total Budget (000) in "UA"
					2024	2025	2026	2027	2028	Total	2024	2025	2026	2027	2028	
1.1.1	**Enhancing operational capacity of research centers for EGS supply**															
1.1.1.1	**Equipment for research stations (Federal-EIAR)**					-	-	-	-							
	Tractors	Number	5	50,000	5	-	-	-	-	5	187	-	-	-	-	187
	Disc harrows	Number	5	20,000	5	-	-	-	-	5	75	-	-	-	-	75
	Moldboard plough	Number	5	20,000	5	-	-	-	-	5	75	-	-	-	-	75
	Row planters	Number	5	25,000	5	-	-	-	-	5	93	-	-	-	-	93
	Seed drill	Number	5	25,000	5	-	-	-	-	5	93	-	-	-	-	93
	Combine Harvester	Number	5	100,000	2	-	-	-	-	5	149	-	-	-	-	149
	Seed thresher	Number	5	15,000	5	-	-	-	-	5	56	-	-	-	-	56
	Seed Cleaners	Number	5	25,000	5	-	-	-	-	5	93	-	-	-	-	93
	Seed Stores at the research stations (build new seed store)	Number	5	15,000	15	-	-	-	-	15	168	-	-	-	-	168
	Short term training seed professional's in the research system	person	5	200	200		40		40	120	30	-	-	-	-	30
1.1.1.2	**Support EGS production operation with in the research system:**		5													
	Pre-release multiplication of candidate varieties	Metric Ton	5	92	-	50	50	150	150	400	-	3	4	11	11	29
	Variety Maintenance	Varieties	5	459	-	100	100	100	180	480	-	35	35	36	65	171
	Multiply wheat breeder/pre-basic seed	Metric Ton	5	55	-	500	1000	1,500	1,320	4,320	-	21	42	64	58	185
1.1.1.3	Operationalize variety licensing for better commercialization of new varieties		5													
	Conduct awareness creation workshop on variety licensing	Participants	5	184	-	100	-	-	-	100	-	14	-	-	-	14
	Field monitoring on seed production from licensed varieties	Lump sum	5	276	-	20	-	-	-	20	-	4	-	-	-	4

3.1.1.4. Increase certified seed production and supply of wheat varieties

3.1.1.4.1. Enhance contractual certified seed production capacity by strengthening out grower schemes

Out-grower systems are schemes that provide seed production and marketing services to farmers on their own land. It **is** agreement-based production arrangement with farmer or group of farmers or farmers' associations who are entered into agreement with an investor or investment company known as a nucleus farm where technical and follow up support are provided from field preparation to product harvest and as per the agreement the quality of crop production is maintained, timely payment is effected to farmers and the seed yield of which shall only be handed over to the investor. Furthermore, stakeholder forums will be organized to bring together buyers and sellers, and standardized contracts will be developed and popularized based on the contract farming proclamation. The major activities will be:

- Strengthen technical capacity of development agents and subject matter specialists in seed production
- Conduct a nationwide training of trainers on outgrower seed production skills
- Establish an incentive & rewarding system for productive & trusted out growers
- Reward best out grower groups as pilot for operating the reward system

3.1.1.4.2 Enhance seed firms average seed yield production

Currently the seed producing firm productivity potentials have been reduced their average annual production significantly from year to year, due to the emerging of new biotic and abiotic seed production stress, agronomic practice of seed production and improved agricultural inputs shortage. So to improve this seed production short comes, applying the following innervation is crucial. Therefore, the major activities will be:

- Train seed company/union farm agronomists, wereda seed experts on Production packages
- Reward the best yield and quality-performing seed firms

3.1.1.4.3. Strengthen contract-based seed production and marketing

This sub-sub component aims to focus on improving seed production and marketing through strengthening contract-based seed production and marketing, as well as enhancing the supply of climate-resilient wheat varieties. Along with this, the community or society will be benefited from technology package, which has a huge potential in strengthening those farmers will engage on contract-based seed production.).

It is practiced on agreement signed between farmers and seed producing enterprises or, farmer cooperative and/or unions for the production and supply of improved seeds with reasonable price, place and time to make secure seed availability. Through contract agreement can promote agricultural input (improved seed) production and guarantee a secure market for wheat seed, thereby allowing farmers to earn increased revenue and buyers to obtain a return on their investments (the so-called win-win situation)

In general, focused support to promote contract farming on seed production and marketing schemes will help to reduce transaction costs for both parties. Both parties stand to benefit from better linkages between farmers (buyers and sellers) through cooperatives or associations. As well as, it will be supported and promoted as a modern and systematic market linkage approaches to benefit small holder farmers and buyers in sustained supply, economies of scale with aggregation, transportation, and marketing, lower transaction costs in identifying and securing sources, while improving sustainability along agricultural supply chains/value chain actors.

CREW will invest in strengthening contract-based seed production & marketing in the wheat seed production & marketing value chains. The Project will critically review and identify key bottlenecks that constrain the efficiency and effectiveness of it. So, to strengthen this contract-based seed production and marketing scheme, the following four major intervention areas will be implemented:

- Develop a directive or Guideline for the establishment of outgrower Farmers' cooperative
- Organize out grower farmers in cooperative or association
- Enhance engagement and efficiency of out-grower- seed firm's relation on seed marketing through flexible pricing, and contract farming law enforcement.
- Incubate 2nd level contract based certified seed marketing between seed company/union and agro-dealers/cooperatives or private dealers

3.1.1.4.4. Strengthen Agro-dealership in seed business

Agro-dealership is a market-based solution to increase access to seeds for small-scale farmers. Smallholder farmers have limited access to and low usage of wheat seeds due to their dispersed lifestyle, huge logistics to respond to their demand in terms of diversity of agro-ecology, timing of supply, size of packages and technical advices. So to developed and established strong agro

dealer ship wing in current Ethiopian seed production and marketing system, incorporating the following major interventions is useful. Therefore, the major activities will be:

- Establish Agro-dealer association
- Conduct nationwide agro dealers training on seed business
- Develop a bankable business plan and project for Agro-dealers Association
- Promote, register, and engage new women & youths as agro-dealers
- Support Research and Development (R&D) for the development, production and promotion of wheat adapted varieties to new irrigated areas of the mid-highland regions

3.1.1.4.5. Scale out small seed packs based on marketing that suite favorable land holdings

Smallholder farmers in Ethiopia face a number of challenges in accessing seed in the traditional size bags. ISSD Ethiopia and partners work on initiatives to promote more accessible small seed packs. The small pack approach allows farmers to get more access to new varieties, and to test them at low risk. A small pack also encourages farmers to test one or several varieties of same crop or different crops on offer. Small seed packs are useful tool to access certified seed of improved wheat varieties to a large range of farmers considering gender issue and also provide an opportunity to build private sector capacity in major growing area. Moreover, further value chain study and analyses will be conducted for more prioritized wheat varieties to come up with clearly identified gaps and intervention recommendations. Therefore, to minimize such small scale farmer seed access bottlenecks, the project will intervene by doing the following major activities:

- Identify Possible seed pack sizes based on National Study
- Pilot Small pack preparation in collaboration with selected and willing seed firms
- Conduct validation workshops on small packs based recommendation
- Establish a Seed Marketing Platform for wheat
- TA through TAAT and its compacts to provide technical assistance in all the areas of seed development and fertilizer application
- Establish a Seed Marketing Platform for wheat
- Support to the regional seed R&D system - Amhara Region
- Support to the regional seed R&D system -Oromiyaq Region
- Support to the regional seed R&D system – Somali Region
- Support to the regional seed R&D system – Afar Region

Table 2 Increase certified seed production and supply of wheat verities expected out puts and responsibilities of institution

1.1.2.1 Enhance contractual certified seed production capacity by strengthen out grower schemes

1.1.2.2 Enhance seed firms average seed yield production

1.1.2.3 Strengthen contract-based seed production and marketing

1.1.2.4 Strengthen agro-dealership in seed business

1.1.2.5 Scale out small seed packs based on marketing that suite favorable land holdings

Lead Organizations/ Institution: MoA (Agricultural and horticulture sector and Agricultural Investment & Input Sector)

Main Implementing Institutions: - Relevant lead executives of MoA (Crop Dvt and AIS) and EIAR Directorate (Research Centers at all levels)

Supporting/collaborating institutions:- Federal cooperative commission (FCC), EAA, Regional Bureau of Agriculture and Relevant Regional Bureau/Commission/Authority/Agency, ATI,

Output 1.1.2.1: Enhance contractual certified seed production capacity by strengthening outgrower schemes

- Strengthen technical capacity of 8000 development agents and subject matter specialists in seed production
- Conduct 800 nationwide training of trainers on outgrower seed production skills
- Establish about 120 incentive & rewarding system for productive & trusted out growers
- Reward 800 best out grower groups as pilot for operating the reward system

Output 1.1.2.2: Enhance seed firms average seed yield production

- Train 2000 seed company/union farm agronomists, Wereda seed experts on Production packages
- Provide 80 Rewards to best yield and quality performing seed firms

Output 1.1.2.3: Strengthen contract-based seed production and marketing

- Develop one directive or Guideline for the establishment of outgrower Farmers' cooperative
- Organize 40 outgrower farmers in a cooperative or association
- Enhance engagement and efficiency of 20 out-grower seed firms' relations on seed marketing through flexible pricing and contract farming law enforcement.
- Incubate 200 2nd-level contract-based certified seed marketing between seed company/union and agro-dealers/cooperatives or private dealers

Output 1.1.2.4: Strengthen agro-dealership in seed business
➢ Establish 250 Agro-dealer association
➢ Conduct 500 nationwide agro dealers training on seed business
➢ Develop one bankable business plan and project for Agro-dealers association
➢ Promote, register, and engage 1000 new women & youths as agro-dealers
Output 1.1.2.5: **Scale out small seed packs based on marketing that suite favorable land**
➢ Identify 10 Possible seed pack sizes based on National Study
➢ Pilot 40,000 Small pack preparation in collaboration with selected and willing seed firms
➢ Conduct validation workshops on small packs based on recommendations for 120 participants
➢ Establish one Seed Marketing Platform for wheat
➢ Provide Support to the regional seed R&D system - Amhara Region
➢ Provide Support to the regional seed R&D system -Oromiyaq Region
➢ Provide Support to the regional seed R&D system – Somali Region
➢ Provide Support to the regional seed R&D system – Afar Region
Monitoring, Reporting, and Evaluation Arrangements. - Monthly, quarterly, biannually, and annual reports - Field Supervision and Monitoring**.** - Joint Review and Implementation Support

Table 3 Increased seed production and supply of wheat verities. Main activities and implementation schedule

No.	Activity	Unit	Prog. Life	Unit Cost (UA)'000	Physical action plan implementation schedule						Budget allocated (UA '000') schedule					Total Budget (000) in "UA"
					2024	2025	2026	2027	2028	Total	2024	2025	2026	2027	2028	
1.1.2	Increase certified seed production and supply of wheat varieties		5													
1.1.2.1	**Enhance contractual certified seed production capacity by strengthening out-grower schemes.**		5													
	Strengthen technical capacity of development agents and subject matter specialists in seed production	Trainees	5	-	2,000	2,000	2,000	2,000	-	2,000	-	389	395	401	407	1,593
	Conduct a nationwide training of trainers on out grower seed production skills	Trainees	5	-	200	200	200	200	-	200	-	39	40	40	41	159
	Establish an incentive & rewarding system for productive & trusted out growers	Participants	5	-	120	-	-	-	-	120	-	20	-	-	-	20
	Reward best outgrower groups as pilot for operating the reward system	Out-growers	5	-	200	200	200	200	-	200	-	139	141	143	145	569
1.1.2.2	Enhance seed firms' average seed yield production		5													
	Trains seed company/union farm agronomists, wereda seed experts on Production packages	Trainees	5		-	500	500	500	500	2,000	-	97	99	100	102	398
	Reward 80 best yield and quality performing seed firms	Seed Firms	5		-	20	20	20	20	80	-	28	28	29	29	114
1.1.2.3	**Strengthen contract-based seed production and marketing**		5													
	Develop a directive or Guideline for the establishment of Outgrower Farmers' cooperative	Lump sum	5		1	-	-	-	-	1	7	-	-	-	-	7
	Organize outgrower farmers in a cooperative or association	Workshop	5		-	20	20	-	-	40	-	33	34	-	-	67
	Enhance engagement and efficiency of out-grower- seed firm's relation on seed marketing through flexible pricing and contract farming law enforcement.	Lump sum	5		-	20	-	-	-	20	-	18	-	-	-	18
	Incubate 2nd-level contract-based certified seed marketing between the seed company/union and agro-dealers/cooperatives or private dealers	Participants	5		-	50	100	50	-	200	-	28	57	29	-	113

1.1.2.4	**Strengthen agro-dealership in seed business**	Participants	5													
	Establish Agro-dealer association	Lump sum	5		-	250	-	-	-	250	-	49	-	-	-	49
	Conduct nationwide agro dealers training on seed business	Lump sum	5		-	200	200	100	-	500	-	39	40	20	-	99
	Develop a bankable business plan and project for Agro Dealers Association	Lump sum	5		1	-	-	-	-	1	7	-	-	-	-	7
	Promote, register, and engage new women & youths as agro-dealers	Lump sum	5		-	200	400	400	-	1,000	-	39	79	80	-	198
1.1.2.5	**Scale out small seed packs based on marketing that suite favorable land holdings**		5													
	Identify Possible seed pack sizes based on National Study	Lump sum	5		10	-	-	-	-	10	2	-	-	-	-	2
	Pilot Small pack preparation in collaboration with selected and willing seed firms	Small Packs	5		-	10,000	20,000	10,000	-	40,000	-	3	7	4	-	14
	Conduct validation workshops on small packs based on the recommendation	Participants	5		-	120	-	-	-	120	-	23	-	-	-	23
	Establish one Seed Marketing Platform for wheat	Lump sum	5		-	1	-	-	-	1	-	23	-	-	-	23
	Form TA through TAAT and its compacts to provide technical assistance in all areas of seed development and fertilizer application	TA	5		0.2	0.25	0.25	0.3	-	1	45	57	58	70	-	229
	Provide Support to the regional seed R&D system - Amhara Region	Lump sum	5		0.2	0.2	0.2	0.2	0.2	1	75	76	77	78	79	385
	Provide Support to the regional seed R&D system -Oromiyaq Region	Lump sum	5		0.2	0.2	0.2	0.2	0.2	1	75	76	77	78	79	385
	Provide Support to the regional seed R&D system – Somali Region	Lump sum	5		0.2	0.2	0.2	0.2	0.2	1	45	45	46	47	48	231
	Provide Support to the regional seed R&D system – Afar Region	Lump sum	5		0.2	0.2	0.2	0.2	0.2	1	45	45	46	47	48	231

3.1.2. Sub-component 1.2: Soil Health and Fertility Management

The main objective of this sub-component is to address the serious challenge of soil acidity, salinity, and health through Integrated Soil Fertility Management (ISFM) throughout the wheat-growing areas.

The Key activities include: (i) Capacity development and training of development agents and district agricultural experts in critical skills of soil sampling, lime applications, and demonstration of ISFM technologies and innovations; (ii) Demonstration on Rehabilitation of Soil acidity and awareness (Lime supply, training, and follow-up), (ii) Installation of Lime crusher, (iv) Development of guideline for lime crusher implementation and Operational Modalities, (v) Assessment and Mapping of Soil Acidity and Prepare Management Plan for Acidic soil reclamation and Mitigation for Oromia and Amhara regions), (vi) Reclamation of strong Soil Acidity in Amhara and Oromia, Lime Supply and Lime application, Support and follow up), (vii) Assessment, Mapping and reporting of Soil Salinity in Amhara 14, Oromia 34, Afar 3 and Somali 5, and Provision of Training to regional experts who participate in the assessment, and Prepare a Management Plan for salt-affected soil Mitigation measures for the four regions), (viii) Rehabilitation of Salt Affected Soils: Improvement of Soil Salinity in Amhara, Oromia, Afar and Somali, Provision of Training to zonal, and woreda experts and DAs, Provision of Inputs /Salt tolerant wheat variety and Biological treatment (alpha alpha and rhodes), Monitoring and follow up), (iv) Verti Soil Management: Management of Excess Water for Verti soil in Amhara and Oromia, Supply of tractor mounted BBM to Amhara and Oromia, Supply IBAR BBM to Amhara and Oromia, and Monitoring and follow up, (x) Promotion and scale up of ISFM Technologies (Provision of Training to zonal, woreda experts and DAs, Provision of necessary inputs for Amhara and Oromia regions. Demo for Somali and Afar on Compost, Supply vermi compost, Supply in organic fertilizer (DAP and Urea), Supply improved seed, Consultative workshop, Monitoring and follow up), (xi) Capacity building for Regional DA and Follow up (Conduct Nation Wide ToT on Soil Health and Fertility, Conduct Workshop on Soil Health and fertility Management, Field Supervision, Monitoring, Evaluation and Learning on SHFM), (xii) Development of a digital soil mapping (DSM) and in-depth analysis covering 1 Million ha of acidic and salt-affected wheat growing soils in the major wheat production belts to complement; (xiii) Conduct R&D to develop innovative tailor made solutions integrating both soil fertility and health; (xiv) Setting up demo plots taking into account these diversities; (xv) Strengthening the Geospatial information system in collaboration with the Ethiopian Soil Information Service (EthioSIS)

to reach 1 M farmers, including AquaIndeX implementation (to reach 20K farmers), NPK-IndeX deployment (to reach 20K farmers), and PhytoIndeX implementation (to reach 20K farmers).

This subcomponent will be implemented by the Ministry of Agriculture in collaboration with OCP-Africa.

i) Capacity development of development agents (DAs) and district agricultural experts in critical skills of soil sampling, lime applications, and demonstration of ISFM technologies and innovations

To optimize agricultural productivity, knowledge of soil acidity and how to mitigate it must be applied appropriately. As a result, increasing the awareness, technical knowledge, and abilities of professionals who assist small-scale farmers through capacity building is critical. Based on this, training will be given to experts in the targeted project regions (Oromia, Amhara, Somali, and the Afar region) on soil sampling, lime applications, and ISFM technology and innovation demonstrations.

ii) Demonstration on Rehabilitation of Soil acidity and awareness (Lime supply, training, and follow-up)

Demonstrations on farmers' plots of land will be held to raise awareness, technical knowledge, and skills, as well as urge them to use lime (ag-lime) to reduce soil acidity. It aims to show farmers the benefits of using lime and integrated soil fertility management practices. This will highlight the positive impact on crop yields. Each demonstration plot is 0.25 ha in size, and will be implemented in Oromia and Amhara regions. To carry out the demonstration, the CREW project will provide free ag-lime at the suggested application rate. Training on its application will be provided, as well as timely follow-up and support.

iii) Installation of Lime crusher

Liming, namely the use of agricultural lime (ag-lime), is a popular approach for raising soil pH and reducing acidity. Lime crusher installation is crucial for providing an appropriate supply of ag-lime to the targeted areas. As a result, the CREW project has included this machine installation in its project documentation. Based on this, the installation of five lime crushers has been planned.

To proceed with procurement, the five lime crushers, environmental and social safeguard assessment, specification, design, and bill of quantities must be completed on time by the Soil Resource Development Lead Executive, who will consult with the regional BoA, RPMU, and other relevant stakeholders to ensure timely installation. The Regional BoA and PMU are in

charge of completing the site selection, environmental and social safeguard assessment reports, and other necessary information.

iv) Development of guideline for lime crusher implementation and Operational Modalities

The lime crushers to be procured will be distributed to the targeted project regions. These machines should be well managed and administered properly to provide the intended service to communities in the project areas. A guideline will be prepared for its implementation, operation, and administration by participating relevant stakeholders. The document will be validated and approved for implementation by the concerned bodies.

v) Assessment and Mapping of Soil Acidity

Assessing and mapping soil acidity is crucial for sustainable agriculture and land management because it directly impacts plant growth and productivity. Mapping soil acidity allows for targeted lime application only where needed and in the right amounts, preventing over-application and environmental damage. This assessment and mapping were previously planned to be carried out in 41 woredas (27 in Oromia and 14 in Amhara regions). The SRD Lead executive will develop a detailed management plan for acidic soil reclamation in the Oromia and Amhara regions and communicate it to the FPMU, regional BoA, and PMU.

vi) Soil Acidity Reclamation

Soil acidity reclamation involves managing and correcting acidic soil conditions to improve agricultural productivity. This is usually done through liming, which is the application of ag-lime that neutralizes excess acidity and increases the soil pH. Other methods are integrated soil fertility management (ISFM). This reclamation of acidic soil will be implemented in Amhara and Oromia regions, which will be identified as strongly acidic by the assessment mapping of the areas. For this identified hectarage, an ag-lime will be supplied based on the application recommended rare free of charge to the selected small-scale farmers in the targeted project areas. For its proper implementation and performance support, follow-up will be done by the responsible experts.

vii) Assessment and Mapping of Soil Salinity

Soil salinity is one of the major abiotic stresses affecting agricultural production worldwide, particularly in arid and semiarid regions. Soil salinity is an excess accumulation of soluble

salts in the soil that hinders the functions of plant growth. High salinity levels negatively affect crop growth and yield, leading to reduced agricultural productivity.

Assessing and mapping soil salinity is vital for reducing soil salinity problem and ensuring agricultural sustainability by enabling effective management of salt-affected soils, optimizing irrigation, and predicting future risks. Since the CREW project is implemented in the lowland and semiarid areas, it is critical to assess and map areas that are in the project areas to design and propose appropriate mitigation measures.

Based on this, it is planned to assess and map 56 woredas in the four CREW project regions: Amhara 14, Oromia 34, Afar 3, and Somali 5 by training 56 experts from selected locations before the assessment process, who will participate in the assessment. A management plan will be prepared by the SRD Lead Executive and will be communicated to all relevant stakeholders.

viii) Rehabilitation of Salt-Affected Soils

Rehabilitating salt-affected soils involves leaching salts from the root zone through improved drainage and excess irrigation, applying chemical amendments like gypsum for sodic soils, enhancing soil organic matter, and selecting salt-tolerant crops. Other strategies include managing irrigation practices to control waterlogging, land levelling, subsoiling, and the innovative bio-saline approach using salt-loving plants for fodder.

The majority of the CREW project woredas are located in semi-arid lowland areas that have salinity problems. To address the issue in these places, rehabilitation of salt-affected soils will be carried out in the four targeted project regions (Amhara, Oromia, Somalia, and Afar). The activities planned to mitigate the problem are;

- Provision of Training to zonal and woreda experts and DAs on salt-affected soil management to raise their knowledge and technical skills for effective implementation.
- Provision of Inputs /Salt-tolerant wheat varieties and Biological treatment (alpha alpha and Rhodes).
- Technical support and follow-up will be provided to ensure correct implementation and result evaluation.

ix) Verti Soil Management

Vertisols have a high clay concentration, causing the soil to shrink and swell in response to moisture fluctuations. Cracking and shrinking behavior reduce workability and internal drainage, causing waterlogging and hindering nutrient uptake. This further reduces

agricultural output and soil potential. They are especially important in Ethiopia's highlands, where they cover over 12.6 million hectares and account for roughly 24% of the farmed highland area. Vertisols can be very productive if properly managed.

Vertisol management in Ethiopia focuses on overcoming waterlogging through improved drainage, such as the use of Broad Bed Makers (BBM) to construct a ridge and furrow pattern. Other important techniques include early planting, soil fertility management (particularly nitrogen fixation via legumes), and the use of improved crop types and fertilizers. These measures increase crop productivity and contribute to food security by making Vertisols, which span extensive regions of the Ethiopian highlands, more productive.

The key activities planned to mitigate vertisoil problem are:

- Management of Excess Water for Vertisols: Waterlogging is a serious problem in vertisols. It is planned to drain excess water through different techniques from land in Amhara and Oromia regions.
- Supply of tractor-mounted BBM and IBAR BBM: To remove excess water, technologies like BBM and IBAR BBM are critical, and it is planned to provide eight (8) Broad Bed Makers (BBM), four (4) to Amhara and four (4) to Oromia. The procurement will be executed by FPMU. Similarly, 2000 IBAR BBM (1000 for Amhara and 1000 for Oromia) will be procured by each region and distributed to selected beneficiaries.
- Monitoring and follow-up: Technical support and follow-up will be provided to ensure correct implementation and result evaluation.

x) Promotion and scale-up of ISFM Technologies

Integrated Soil Fertility Management (ISFM) is a set of practices related to cropping, fertilizers, organic resources, and other amendments on smallholder farms to increase production and input use efficiency. ISFM delivers productivity gains, increased resilience, and mitigation benefits.

The key components of ISFM are improved seeds, organic inputs (manure, compost, vermi-compost, cover crops), inorganic fertilizers (targeted application), water harvesting techniques, soil and water conservation practices, crop rotation, intercropping, residue management, and mulching.

The strategies for Promotion and Scale-Up of ISFM include demonstration plots, farmer training and knowledge sharing, farmer-to-farmer learning, addressing market and input access, adapting to local contexts, and farmer-driven technology development.

The primary actions listed above should be carried out by each small-scale farmer through extension advice and follow-up. In addition, the CREW initiative provides support to implement the following activities:

- Training will be provided for zonal, woreda specialists, and DAs in the four targeted project regions (Oromia, Amhara, Somali, and Afar).
- Provision of necessary inputs/supply Vermi-Compost:
- Conduct Vermi-compost demonstration: In Somali and Afar regions, this vermicompost is not well-known. Demos will be demonstrated on vermicompost (50 in Somali and in Afar regions. Inorganic fertilizer (DAP and Urea) and improved seed will be provided at the approved crop package recommended rate.
- Consultative workshop with the participation of regional experts will be held to discuss on the implementation of the demo, and Monitoring and follow-up will be done on time.

xi) Capacity building for Regional DA and Follow-up

To enhance the capacity of experts and DAs in soil health and fertility management, trainer training (ToT) will be provided, as well as a workshop to bring all stakeholders on the same page and raise awareness and consensus on soil health and fertility management. The implementation of soil treatment and management activities should be supplemented by technical support and follow-up during the rainfed and irrigation wheat production season.

Table 4: Mitigating Soil Acidity and Salinity: Expected outputs and responsibilities of the institution

1.1.2.1 **Mitigating Soil Acidity and Salinity** 1.1.2.2 Assessment and mapping of soil acidity 1.1.2.3 Soil Acidity Reclamation 1.1.2.4 Assessment and mapping of soil salinity 1.1.2.5 Rehabilitation of salt-affected soils
Lead Organizations/Institutions: MoA (Natural Resource Sector and Agricultural Investment & Input Sector) and OCP Africa

Main Implementing Institutions: - Relevant lead executives of MoA and EIAR Directorate (Research Centers at all levels), OCP Africa, and project regions **Supporting/collaborating institutions**: - Regional Bureau of Agriculture and Relevant Regional Bureau/Commission/Authority/Agency, ATI,
Output 1.1.2.1: To address the serious challenge of soil acidity, salinity, and health through Integrated Soil Fertility Management (ISFM) throughout the wheat growing areas.
➢ Assessing and Mapping Soil Acidity – Amhara 14 and Oromia 27 Woredas
➢ Reclamation of 6,000ha of Strong Soil Acidity – Amhara and 8,000 ha of Strong Soil Acidity - Oromia
➢ Assessment and mapping of soil salinity in 56 project Woredas
➢ Rehabilitation of salt-affected soils 131,500 in four regions
➢ Vertisols management of 120,000 ha in project regions
Output 1.1.2.2: Enhance soil fertility and health
➢ Conduct a nationwide training of trainers for 160 experts on Soil Health and Fertility
➢ Conduct a workshop for 240 participants on Soil Health and Fertility Management
➢ Train 800 Capacity Dev of DA on Soil Sampling, liming, and demonstrations
➢ 200 ha of soil acidity demonstration plots and awareness
➢ Establish 5 Lime crushers

Table 4: Main activities and implementation schedule for soil health and fertility management .

No.	Activity	Unit	Physical action plan implementation schedule						Budget allocated (UA '000') schedule					Total Budget (000) in "UA"
			2024	2025	2026	2027	2028	**Total**	2024	2025	2026	2027	2028	
	Assessment and mapping of soil acidity	No.Woredas	41	-	-	-		41	240					240
	Soil Acidity Reclamation	ha		2000	3000	5000	4000	14000		824	1236	2059	1647	5766
	Assessment and mapping of soil salinity	No.Woredas	-	56	-	-	-	56		327				327
	Rehabilitation of salt-affected soils	ha	29500	25500	25500	25500	25500	131500	153	132	132	132	132	682
	Vertisols management	ha	24000	24000	24000	24000	24000	120000	89	89	89	89	89	445
	Promotion and scale-up of ISFM technologies	No. Woredas	24	29	23	-	-	76	74	89	71	-	-	234
	Capacity building for the regional development agents and follow-up													
	➢ **Conduct a nationwide training of trainers on Soil Health and Fertility**	No. of participants	40	40	40	40	-	160	9	9	9	9	-	36
	➢ **Conduct a workshop on Soil Health and Fertility Management**	No. of participants	60	60	60	60	-	240	8	8	8	8	-	32
	Soil Mitigation Activities													
	➢ **Capacity Dev of DA on Soil Sampling, liming, and demonstrations**	Number	200	400	200	-	-	800	39	79	40	-	-	158
	➢ **Soil Acidity Demonstration plots and awareness**	Hectares	-	50	100	50	-	40		80	60	40	-	160
	➢ **Lime crushers**	unit	-	2	2	1	-	5	-	159	160	81	-	400

3.1.3. Sub-component 1.3: Rehabilitation, Development and Management of Irrigation Schemes

In response to strategy of expanding and promoting irrigated wheat production of Ethiopian government, African Development Bank (AfDB) engaged in the implementation of the National Wheat Flagship Program (NWFP)/Climate Smart Wheat Value Chain Development Project. Rehabilitate and expand irrigation infrastructure in which improving efficiency of existing schemes through rehabilitation and maintenance is sub-component of the project. In order to prioritize and operationalize proposed irrigation schemes the CREW mainly focus on potential wheat producing Regions in seventy-six Woredas.

1. Objectives

The main objective of this sub-component is to improve the efficiency of existing irrigation schemes through rehabilitation and maintenance of infrastructure – desilting, drainage, extend canals, etc. and to build the national and local capacity of irrigation service providers and development agents (irrigation scheme operations, maintenance, irrigation tariff collections, water users associations, etc.) on water management and irrigation agronomy.

2. Scope and Coverage

The scope and coverage of the Ethiopia Climate Resilient Wheat Value Chain Program (CREW) are described based on the National Wheat flagship program, which focuses on the rehabilitation and upgrading of existing irrigation schemes to improve their efficiency and enhance wheat production. The geographical coverage of the program implementation area was identified based on the existing wheat production capacity and the future wheat production potential. Based on the above mention statements, the Amhara region (24district), 38 Oromia Irrigation infrastructure development district), Somali region (11district), and Afar region (3 districts) are selected to incorporate in this program. Detail of regions and woreda list refer from **Annex 1**

3. Irrigation & drainage infrastructure and systems of the Program

Table 1 2 Investment under sub component on irrigation & drainage infrastructure system

	Priority Activities	Results
a) Irrigation Infrastructure Development.	Development. New irrigation infrastructure development	New infrastructure developed.

	Improving/upgrading existing irrigation schemes.	Existing infrastructure
	Institutional strengthening and capacity building.	Institutions strengthened.
b) Irrigation water management	Irrigation water management services.	Irrigation water management services established.
	Capacity Building /Training for irrigation water management.	Improved management of irrigation water.
	Irrigation technology demonstration.	Technologies demonstrated
	Proper agricultural land drainage.	Improved land drainage
	Agronomic practices/ soil management.	Improved agronomic practices.
	Integrated management practices for reclaiming salt affected soils.	Salt affected soils reclaimed.
	Land levelling service.	Proper planting seed bed.

a) **Irrigation Infrastructure development** the NWFP will support the construction of selected new potential irrigation schemes and upgrade existing ones. The target is for farmers to have access to sustainable irrigation schemes for wheat development. The Rehabilitation Scheme's overall target is to develop 6620 hectares, plus 528 pumps that each develop 10–15 hectares and are distributed to people with access to water for irrigation. This project will not consider new irrigation scheme construction. The main target of this project is rehabilitation or upgrading of existing Irrigation Schemes that could not properly functioning due to various problems. This sub-component will support farmers' access to irrigation knowledge and equipment with reasonable cost-sharing and practical irrigation system.

b) **Irrigation water management** Major irrigation induced threats in irrigated agriculture is *waterlogging and soil salinization*. Due to current traditional and mismanaged irrigation practices at small holder irrigated agriculture, low irrigation efficiencies (*conveyance, distribution, and application*) are the major challenges. This subcomponent will support the improvement of on-farm water management practices

adopted by beneficiaries. The aim is to improve the water conveyance, distribution, and application efficiencies of the existing irrigation schemes. The NWFP will seek to; (i) avail water saving and efficient irrigation methods (pressurized system & hydro-flume), water measurement devices (Parshall flume, syphon etc.), lining of conveyance and distribution system, pipe irrigation water conveyance and distribution system and land levelling service; (ii) build capacity through field trainings on irrigation water and salt affected soils management, scheduling and water measurement systems; (iii) demonstrate optimal irrigation amount and frequency, irrigation water measurement devices (Parshall flume, syphon etc.), water saving irrigation water distribution systems (hydro-flume), and sprinkler (drag line, centre pivot) irrigation system; (iv) design and implement proper drainage system (bio-drainage, surface and subsurface drainage system), rehabilitate existing surface and subsurface drainage system, and maintain, the natural drainage; (v) introduce the practices of pre-irrigation system, minimum tillage (to avoid soil compaction, maintains good soil structure, improve surface and internal drainage, and to facilitate deep leaching); and (vi) reclaim salt affected soils through applications of biological, physical/mechanical, hydraulic and chemical amendment measures.

4. Project Activities

There are six major interventions to be performed under this sub-component

I. **Rehabilitation** of a total of 260 ha (260ha in Afar2,100 ha in Amhara; 3,400 ha in Oromia; and 860 ha in Somali regions) of irrigation schemes whose salient features are described in the Technical **Annex**---(TA-Section)
II. **Installation of a total of 528 water pumps** with a capacity 75 litres/second together with maintenance kits to groups of farmers in a cost-sharing arrangement;
III. **Rehabilitation of a total** of 50 km of bad sections of access roads to link the schemes with the nearest road network; Access Roads.
IV. Promotion of multipurpose use of water by providing livestock watering troughs and fisheries development;
V. provision of soil moisture sensors/enhance use existing Irrigation Management Information System for Ethiopia (IMISET)
VI. Strengthening the capacity of Water Users' Associations (WUAs/Cooperatives, experts, farmers, service providers, and extension agents) in areas of irrigated agriculture, through short term training and exchange visits, to ensure sustainability.

The capacity development intervention aims to enhance capacity of IWUAs for improved Irrigation scheme operation and management. It will support IWUAs capacity on Irrigation Service Fee (ISF) collection to enhance mobilization of funding to cover O&M costs. Thus, the project will finance: (a) strengthening public sector capacity for oversight of IWUAS through training, and technical assistance; (b) training of IWUAs and its members; and (c) provision of operation and maintenance equipment and tools. In addition, the project will also finance stakeholder consultation and review meetings, and guideline preparation for irrigation service fee collection.

5. Standard Definitions

This is crucial for illustrative institutional responsibilities and accountabilities at each level based on their mandates. Hence, for clear understanding and accountability during implementation, the following terminologies shall apply:

(i) Small-scale Irrigation (SSI) Systems

This "small-scale" refers not so much to the physical size of the command area of the irrigation system, or farm plot (although these are generally small), but rather to the fact that the plots and the irrigation schemes (where a scheme is present) are managed and owned by the farmers themselves. Therefore, the term small-scale irrigation (SSI) systems refer to a wide range of approaches and techniques for catching, storing and using rainfall at or near the place where it falls (usually called rainwater harvesting); river flood flow storing using small dams(up to 15m height); diverting flowing water by gravity usually using diversion weirs (concrete or masonry), or simple diversion structures (stone/earth/brushwood structures), intake structures, run-of-river or spate irrigation systems, lifting/diverting surface water or ground water by human or motor power (lift irrigation), conveying water by canal or pipe, and applying water to land by controlled or uncontrolled flooding (gravity), overhead sprinklers (pressurized), or low pressure drip irrigation;

(ii) Revitalization

Revitalization is an all-embracing approach, including the development of both hard ware (physical investment) and software (social investment) aspects that provide complex systems to use irrigation water. It refers to the re-design and restoration/reconstruction of existing infrastructure components and also comprises engagement with the organizational and social dynamics of water allocation and distribution, the agricultural production system, human

capital development and empowerment, finances, business strategy development and marketing;

(iii) Hardware

It refers to physical infrastructures developed/installed in an irrigation system for the purpose of facilitating availability of irrigation water from a water source for crop production. In a gravity flow irrigation system these include abstraction structures, conveyance structures, distribution structures, drainage structures and on-farm service roads. It also refers to stationary or movable facilities employed for supply of irrigation water in pressurized irrigation systems;

(iv) Software

It refers to the non-tangible aspects of an irrigation system such as irrigator(s), organizations, institutions, management, by-laws, and Capacity Development (trainings);

(v) **Rehabilitation**

Rehabilitation is mainly the restoration of physical structures to original design and specifications and it then includes engineering- centred re-construction of dilapidated infrastructure and is focused primarily on securing the water supply by repairing the irrigation abstraction, conveyance and distribution and application system. It mainly involves the rebuilding/restoration of non- functional/damaged structures of existing SSI subprojects to regain the services as per prior design;

(vi) **Repair**

It means major maintenance, fixing damage or deterioration of headwork and irrigation structures that is beyond the beneficiary capacity

(vii) **Expansion/Extension**

It includes expanding the command area through redesigning for provision of additional structures such as improvement of heads works, extension of canal networks, night storage, etc. on existing SSI subprojects;

(viii) **Upgrading**

It includes replacing traditional irrigation structures with modern structures to improve the durability of the structure, improve conveyance efficiency and hence increase the command area or improve the reliability, equity and adequacy of water delivery;

(ix) **Improvement**

It includes redesigning and reconstructing of structures (with new works being constructed or design changes made to improve and modernize the irrigation systems (head works, off-farm and/or on-farm such as canal lining, construction of crossing structures, etc.) in order to improve efficiency, operation and management of existing SSI subprojects through improved water conveyance, distribution and application systems.

This intervention is to make fully functional to its design capacity and/or beyond through rehabilitating, improving and/or extension of existing irrigation infrastructure and keeping the infrastructures functional. Rehabilitating, improving and/or upgrading the physical infrastructure includes a combination of both "hardware" and "software" solutions. The hardware development involves improving/rehabilitating existing irrigation water infrastructures (off-farm and on-farm) to improve reliability of supply as well as developing groundwater to augment lean season water supply. The intervention of "software" solutions is critical and requires the development of more efficient mechanism for delivery and management of irrigation system up to the field level through a clear delineation of responsibilities between the regional government, in charge of infrastructure development, and the Irrigation Water Users Associations (IWUAs), in charge of delivering irrigation services to farmers/pastoralists. This intervention focuses primarily on revitalization of existing (both modern and traditional) SSI systems.

6. Guiding principles or the rational for SSI Development

It will follow a Participatory Irrigation Development and Management (PIDM) approach which is outlined in detail in the National Guidelines for Small Scale Irrigation Development.

The main guiding principles underlying the activity design are:

(i) Irrigation systems shall be developed from the three sources of water (rainwater, ground water, surface water) depending on the potential and comparative advantage of the intervention to the target area;

(ii) SSI interventions will be prioritized based on the principle of community investment decision making processes in which beneficiary communities are consulted for the investment, but the final decision will be made by woreda, regions & federal considering technical, social, environmental and financial feasibility criteria even if demand is high for the investment in that particular community;

(iii) SSI is multi-disciplinary by its nature. Thus, SSI improvement or development is an intervention in which the integration of social, institutional, policy, engineering, agricultural, economic and environmental issues are paramount;

(iv) Integrated approach: the overall objective is the development of sustainable and profitable irrigated agriculture by male and female farmers, agro-pastoralists or refugees within the command area of SSI schemes. In addition to the construction of irrigation and drainage structures, it is also required to: a) establish and develop capacity of water users' organisation (WUO) for the operation and maintenance (O&M) of SSI schemers; b) provide appropriate irrigation and support services to support male and female farmers to develop their irrigated agriculture successfully within the command area of SSI schemes; and c) improve soil and water management.

(v) Participatory approach: Ensure that all male and female farmers are fully involved, informed and consulted during all stages of the SSI development process Ensure that male and female farmers are fully involved in decision making during all stage of the SSI development process Ensure that male and female farmers through their Water user organization take over the full O&M responsibility for their SSI schemes.

(vi) Gender responsive approach: Ensure effective participation of all female farmers in the management of their Water user organization and decisions related to the O&M of their SSI schemes Ensure full access of all female farmers to irrigation water as well as the provision of training, extension and support services Avoid/minimise any (potential) negative impact of the development of irrigated agriculture on female household members (i.e., increased workload) Facilitate the safe use of irrigation canal for non-irrigation purposes, such as fetching water for domestic use, washing clothes and watering livestock, by providing social structures in scheme design.

(vii) Watershed-based approach: Reduce the impact of siltation problem, unreliable water supply and (more) frequent floods on the performance and sustainability of SSI schemes within the watershed by improving soil and water management upstream of the SSI schemes.

7. Upgrading, Improving/ Rehabilitation Existing SSIS

This intervention is to make it fully functional to its design capacity and/or beyond through rehabilitating, improving, and/or extending existing irrigation infrastructure and keeping the infrastructure functional. Rehabilitating, improving, and/or upgrading the physical infrastructure includes a combination of both "hardware" and "software" solutions. The hardware development involves improving/rehabilitating existing irrigation water infrastructures (off-farm and on-farm) to improve the reliability of supply as well as developing groundwater to augment the lean season water supply. The intervention of "software" solutions is critical and requires the development of more efficient mechanisms for delivery and management of irrigation systems up to the field level through a clear delineation of responsibilities between the regional government, in charge of infrastructure development, and the Irrigation Water Users Associations (IWUAs), in charge of delivering irrigation services to farmers/ and pastoralists. This intervention focuses primarily on the revitalization of existing (both modern and traditional) SSI systems.

(a) Key activities:

The following are major activities in revitalization of existing SSI schemes

Upgrading, improving/rehabilitation of the existing SSISs shall be using the guiding principles of national guidelines which developed for small scale irrigation development in Ethiopia (MOA – Small scale irrogation guideline (SSIGL) of Ethiopia – 2018). This helps for the four mutually interlinked irrigation schemes systems; the physical system, the cropping system, the economic system and social- organizational system. There for the technical, social, environmental and economic aspect issues will be addressed properly.

(i) Inventory of existing SSI schemes

Inventory of existing SSI schemes for rehabilitation and/or improvement including operation and management will help to identify existing SSI schemes with problems that are functioning under capacity and don't serve the planned beneficiaries as per the design. It shall also include identification existing SSI schemes that have potential for improvement, expansion/extension and upgrading. It will have the objective to develop Woreda Inventory

Map of Existing SSI systems, which clearly shows the location, type and coverage of the existing small-scale irrigation systems in the refugee hosting Woredas. Also, develop Woreda Existing Small-scale Irrigation Renovation Plan, which shows the prioritized list of all existing small-scale irrigation systems in the Woreda to be rehabilitated and improved within the project implementation period;

(ii) Conduct diagnosis of the existing schemes and design the rehabilitation/improvement of schemes

Based on the inventory result and extent of the problems, prioritized implementation plan will be prepared considering the impact of the rehabilitation and feasibility of implementation of the rehabilitation. The biophysical and socio-economic conditions of the existing SSI systems will be thoroughly evaluated, redesigned and planned as per the actual need. Of the community Regional and woreda specialists will do this work with strong support from national FPCU in case of complex rehabilitation work;

(iii) Rehabilitating, improving, and expanding existing modern SSI systems

It includes the following interventions:

- Reconstruction of damaged/non-functional head works, conveyance systems, and/or on-farm infrastructure;
- Repair/maintenance of partially functional leaking headwork, conveyance systems, and/or on-farm infrastructure;
- Improvement of canals through lining or changing in to closed canal systems;
- Construction of additional canals/structures for expansion of command areas;
- Improvement of water distribution structures at the field level such as additional off-takes, crossing structures, etc.; and
- Changing the current water application methods to more efficient one such as surface irrigation systems into drip irrigation systems (water saving technologies).

(iv) Upgrading and/or improvement of existing traditional SS irrigation systems

This would be done through replacing temporary diversion structures with permanent ones; improving canal alignment and lining with masonry to improve conveyance efficiency; construction of on-farm irrigation structures such as off-takes, and crossing structures; etc. Woreda and regional specialists will take the responsibility for the execution of this activity with close support from the Federal Project Coordination Unit (FPCU).

(v). Key Implementation Procedures and Approaches

Steps 1 - Identify stakeholders and conduct extensive stakeholder consultations

Stakeholder consultations are conducted to enlist vital stakeholder support and input in the implementation of the revitalization process. The diagnostic analysis workshops shall identify the major constraints to optimum system performance and, give recommendations on how to alleviate the constraints. The other important reason for stakeholder consultations is to ensure that there is effective buy-in of the project by all levels. Project ownership by beneficiary institutions and communities is crucial to its success. Identification of and consultations with stakeholders shall be initiated at the start of the revitalization intervention;

Step 2 –Carry out site visit

The diagnostic studies shall be conducted by teams of professionals representing disciplines such as irrigation engineering, sociology, economics, environment and agronomy. The site visit helps to define what and how the diagnostic work shall be carried out. Based on the field visit findings, Terms of Reference (ToR) shall be developed. The ToR outlines the details of the diagnostic work. Such work may require the services of competent consultants. Hence, the procurement of such services involves the preparation of a Request for Proposal (RfP) and a bidding process which has to be conducted according to Selection and Employment of Consultants Guidelines of World Bank. In the RfP, it needs to be stressed that sustainable revitalization can be achieved only when both the software and hardware components of the scheme are worked on;

Step 3–Carry out diagnostic analysis and baseline studies

The baseline diagnostic and analytical studies provide an objective basis for defining the most favourable social-economic conditions and the most appropriate methodology for rehabilitating the schemes in a sustainable and profitable manner. At this stage, the following main study areas will be considered:

- Damage on the physical structure Conduct Water Resources Assessment: It involves water availability assessment, water demand analysis and irrigation water utilization analysis;
- Conduct Irrigation System Infrastructures Analysis: It involves assessment of the conditions of the physical infrastructure, the amount of physical infrastructure and quantify amount and cost of repairing or replacing physical infrastructure to assess the

practicability of irrigation system layout, the compatibility of distribution system with the community needs, and assess options and models;

- Conduct Socio-economic Analysis: Assess historical perspectives of the scheme, assess governance structures (scheme management/IWUA, institutional and legislative frameworks), assess gender issues, assess social-economic constraints, market linkages and develop appropriate governance/management model for the scheme;
- Agricultural Resource Assessment: It involves conducting crop production system analysis, and assessing value addition opportunities and options including assessment of existing agricultural and livestock resources, land-use, farming systems and cropping patterns, input supply and product marketing, other economic activities. It shall also include assessment of existing Agricultural Development Plan and suggest alternative production methods and models, crop suitability, cropping alternatives, cropping rotations, and mechanisation.

Note that it is imperative to stress that availability of sufficient water to meet the needs of the revitalized irrigation scheme as a pre-condition for proceeding with the revitalization intervention.

Step 4 - Define the bidding documents for the restoration of the SSI system:

The tendering/bidding document provides the basis for defining how and what shall be revitalized in the scheme. Revitalization process involves working on the hardware and software section of the scheme. The bidding document shall be prepared based on AfDB Standard Bidding Documents for small works;

Step 5 - Carry out software and hardware renovation work

Hardware rehabilitation as part of revitalization process may involve rebuilding, upgrading, improving or modernizing existing infrastructure. Such work requires services of competent contractors. Hence, the procurement of such services involves tendering process which has to be conducted according to small works guidelines and procedures of AfDB. The software restoration involves implementation of diagnostic findings such as establishing and/or strengthening scheme governance structures which includes establishing and operationalizing a viable IWUA to manage the scheme. It also involves overhauling and strengthening the IWUA based on the diagnostic recommendation. Software revitalization also involves reviewing and putting in place appropriate integrated crop and water management including

mainstreaming gender issues in scheme operations. It also involves reviewing market linkages and linking the SSI scheme to viable and sustainable markets for its produce as well as inputs. The final activities in software rehabilitation involve developing scheme operation, management and maintenance models which will characterize the functions of the rehabilitated irrigation scheme.

- Carry out hardware rehabilitation works: Produce bidding and contract documents for contractors, and advertise physical works, evaluate contract bids, select the best contractor based on selection criteria, and engage contactor. Establish contract administration mechanisms to support the contractor, who carries out detailed physical rehabilitation works which may involve redesigning, and then repairing, upgrading or modernizing existing infrastructure;

- Carry out software rehabilitation works: Develop/strengthen scheme governance structures including IWUA; develop/strengthen operation and maintenance systems; implement agricultural development plan; mainstream gender in irrigation operations to remove gender inequalities; and integrate MUS of water into scheme water infrastructure. (A detailed of objective, Management Structures and responsibilities of IWUA strengthening and organizing is attached in the annex VI);

- Capacity Development: Train extension workers at regional, woreda and kebele level in entrepreneurial skills; train farmers /pastoralists/agro-pastoralists in irrigation farming as a business; and train them in operation and maintenance of irrigation and water supply infrastructure.

Step 6: Develop and put in place a sustainability exit strategy

This step involves developing and implementing appropriate strategies that will sustain of the expected improved productivity of the rehabilitated/ improved/modernized irrigation scheme. Proposed strategies include:

- Putting in place a Monitoring and Evaluation (M&E) system: To be able to constantly track the performance of the rehabilitated scheme against predetermined baseline conditions; and

- Capacity Development: develop the capacity of the farmers/pastoralist/agro-pastoralist so that they will be able to utilize and benefit socially and economically from the rehabilitated/improved/modernized irrigation scheme.

A very important capacity element is to inculcate entrepreneurial skills into the irrigation farming culture so that they can able to take irrigation farming as a business. They are also trained in all basic operation and maintenance principles and practices. Hence, training of farmers/ pastoralists/agro-pastoralists in irrigation farming as a business as well as in operation and maintenance is an essential section of capacity development of the farmers/pastoralists/agro-pastoralists that will contribute to sustainability of the rehabilitated irrigation scheme. Development and implementation of sustainability strategies are generally on-going processes.

(vii) Implementation Time table

This intervention shall be mainly implemented at Woreda level except complex rehabilitation works (require critical diagnosis and redesign work, mobilization of machineries, involves significant volume of Reinforced Cement Concrete (RCC) concrete, mass concrete and masonry works) and works contract values above the threshold limit of shopping (national competitive bid is not feasible at Woreda level) for works, which shall be handled at regional level; otherwise the role of the region is limited to technical support and supervision work. MoA and FPCU shall support in potential identification and need assessment (inventory of existing SSI), feasibility study and detailed design preparation, design documents review, supporting on tender document preparation/review to final procurement process and construction supervision.

Upgrading, improving/rehabilitation existing SSIS will be implemented in four batches. At the beginning of the project year, inventory of existing and new potential assessment will be documented /collected for traditional and modern SSI Schemes will be done. Community request, identification and detail design will be carried out from Project Year-1 (PY-1) to Project Year -3 (PY-3), procurement and construction management work will be from Project Year -1 (PY-1) to Project Year-4 (PY-4) whereas the sub component achievement monitoring and evaluation will be from PY-1 to Project Year – 5 as summarized in the following table 5.6.1.

Table 5.3 Existing SSI scheme implementation time table

No	Activity	Batch1	Batch2	Batch 3	Batch 4
1	Inventory of existing SSI systems and new potential assessment will be documented /collected	PY-1,	PY-2,	PY-3	
2	Selection by and request from beneficiaries	PY-1	PY-1	PY-2	PY-3
3	Identification, screening and reconnaissance level study	PY-1	PY-1	PY-2	PY-3
4	Feasibility studies and detailed design	PY-1	PY-2	PY-2	PY-3
5	Design review and approval	PY-1	PY-2	PY-2	PY-3
6	Construction Procurement process including bid document preparation, advertising, bid evaluation and contract award	PY-1,	PY-2	PY-3	PY-4
7	Construction work and contract administration	PY-1	PY-2	PY-3	PY-4
8	Monitoring and evaluation	PY-1	PY-2	PY-3	PY-4, PY-5

The following points are to be considered in the Time table:

- If there is additional shelf document available at woreda or regional level, procurement and construction processes will also be done in the project year one (PY-1);
- And also, if the volume/type of work complexity is simple, then all activities starting from potential assessment (existing scheme inventory) up to construction and contract administration will be started in the project year one (PY1); and
- Operation and maintenance manual of each project must be prepared after the irrigation construction of each scheme was completed/commissioned using the guiding principles of national guidelines which developed for small scale irrigation development in Ethiopia (MOA – SSIGLs of Ethiopia – 2018)

- Implementation Time table - The activity will be implemented in four batches. At the beginning of the project year, potential assessment on traditional and modern SSI and MHI Schemes will be done. Community request, identification and detail design will be carried out from Project Year -1 (PY-1) to Project Year -PY-3), procurement and construction management work will be from Project Year -1 (PY-1) to Project Year-4 (PY-4), whereas the sub component achievement monitoring and evaluation will be from PY-1 to Project Year – 5 as summarized in the following.

8. Capacity Building

8.2. Physical Capacity Development

The intervention will support for the strengthening of regional and Woreda offices (BoWRs/BoANR). These include purchase and provide office facilities and field equipment like Hand GPs, safety materials like safety shoes, and technical books and prepare, print and distribute technical training materials on irrigation design and management. The implementation will follow the following procedures:

- Conducting capacity needs assessment;
- Preparation of physical and financial action plan as per need assessment findings and recommendations;
- Procurement planning, implementation and contract administration; and
- Equipping implementing institutions.

8.2. Human Resources Capacity Development

Training will be useful at all levels including the smallholder farmers/pastoralists/agro-pastoralists in capacitating the irrigation sub-sector's capabilities in planning, design and construction management, in introducing new ideas from other countries, in codifying existing procedures and in upgrading the standard of operation and maintenance for SSI systems. In addition, the project will support review meetings, short-term training programs to raise the knowledge and skill of technicians at all levels specialized in SSI development. It will also include support to short-term study tours on the planning, design, construction management, and management of irrigation systems for federal and regional experts.

The project will also support technical assistance to improve the implementation capacity of implementing agencies at all levels specialized in SSI development. The technical assistance intervention will include engagement of national technical assistants, who will assist

implementing agencies at all levels in the planning, engineering design, procurement, construction management and technical monitoring of SSI irrigation systems.

The key activities to be performed will be:

- Conducting Training Need Assessment (TNA);
- Prepare physical and financial action plan as per need assessment findings and recommendations;
- Procurement planning, implementation and contract administration;
- Practical training of public staffs at all levels in Study and Design, Engineering design soft wares, Construction Management/Contract Administration;
- Irrigation Water management Irrigation scheme administration
- Capacity building training to WUAs/Cooperatives, experts, farmers, service providers, and extension agents) in areas of irrigated agriculture, through short term training and exchange visits, to ensure sustainability.

10. **Implementation Arrangement and Responsibilities**

Lead Organization/ Institution: This project mainly managed under the Ministry of Agriculture MoA/ Agriculture and horticulture Development Sector. However, Sub-component 1.3: Rehabilitation, Development and Management of Irrigation Schemes Led by MOA/ Natural Resource Development Sector-SHIDLE. There is a Federal Project Coordination Unit (FPCU) to under way over all day today activities

Main Implementing Institutions: - The responsible body for technical aspects sub-component1.3 is Smallholder Irrigation Development Lead Executive (SHIDLE).

Supporting/collaborating institutions: Ministry of Irrigation and Low Land (MILL), EIAR, RARIs, ATI,

Regional Implementation: Regional Bureaux of Agriculture in Afar, Amhara, Oromia, and Somali regions.

The Rehabilitation, Development and Management of Irrigation Schemes activities will be executed directly using the existing structure at Federal, Regional and Woreda levels of the Ministry of Agriculture (MoA) through both the recruitment of supervision consulting

firms/contractors and on force account modality (through the involvement of various Federal, Regional and Woreda stakeholders).

Federal level

At federal level, the major sub-component activities components will be coordinated by the FPCU under MoA, who will inform and report to the National Project Steering Committee on policy matters and strategic issues. The FPCU is responsible for managerial, coordination, administrative and financial matters.

The Federal Project Coordination Unit (FPCU) has the responsibility of supporting the project activities under Rehabilitation, Development and Management of Irrigation Schemes among which are the construction, rehabilitation and supervision of SHI Schemes Rehabilitation work in Amhara, Oromia, Afar and Somali Regional States.

The major responsibilities of the FPCU under sub-component 1.3. include:

- Construction and rehabilitation work Support the Regional Project Implementing Unit (PIUs) while conducting baseline assessments and preparing respective reports for Rehabilitated Irrigation Schemes and survey for construction;
- Collect all relevant standards, guidelines and designs developed by concerned bodies (MoA, MILLS, Regional Bureaus, etc) for different Rehabilitated Irrigation Schemes and provide to the Regional PIUs.
- Collect outputs/deliverables (detail designs, technical working drawings, specification and Bill of Quantity (BoQ), engineering cost estimates and tender documents) Rehabilitation, Development of Irrigation Schemes to provide to the Regional PIUs.
- Review and comment Standard Bid Documents (SBDs) for recruitment of responsive contractors to Rehabilitation of Irrigation Schemes.
- Assist in the construction and supervision contract administration and management.
- Closely monitor and supervise the actual progress of the construction and rehabilitation works and ensure quality of the work as per the designs,

specifications, norms and contract agreements in collaboration with SHIDLE, the regions and the RPCUs.

- Facilitate the advance and interim payment requests from the Regional PIUs.
- Handing over the developed and/or rehabilitated Rehabilitated Irrigation Schemes in collaboration with the SHIDLE, Regional Bureau and PIUs.

a) Supervision (both development and rehabilitation works)

- Preparation of draft Terms of Reference (ToR) for the supervision consultancy service and submission to all relevant stakeholders (governmental and non-governmental institutions at all levels, RPIUs, AfDB, etc.) for review and comment.
- Compliance of all relevant comments and suggestions into the draft ToR and finalization of the ToR for submission and submission of the final ToR to the Bank (AfDB) for "No Objection".
- Recruitment of consulting firms in collaboration with SHIDLE based on the Bank's procurement procedures. Refer the procurement manual for further details.
- Submission of all compiled relevant design documents (detail designs, technical working drawings, specifications and Bill of Quantities (BoQs) and contract document) to the consultants pertaining to the construction and rehabilitation works.
- Contract administration and management.
- Closely monitor and supervise implementation of the supervision work by the recruited consulting firms in collaboration with the RPCUs.
- Facilitate the advance and interim payment requests from the consulting firms.

Regional level

At regional level, the sub-component activities under the Rehabilitation, Development and Management of Irrigation Schemes will be coordinated by the Regional PIUs under Bureau of Agriculture (BoA) in Amhara and Oromia Regions and Bureau of Agriculture; Afar NR and Pastoralist Bureau and Livestock and Pastoralist Development (BoA) in Somali region who will inform and report to the Regional Project Steering Committees on policy matters

and strategic issues. The RPIUs are responsible for managerial, coordination, administrative and financial matters at regional levels.

The Regional Project Coordination Units (RPCUs) have the responsibility of supporting the project activities under the BOA and BILL components among which are the construction, rehabilitation and supervision of Rehabilitation of SHI schemes, and other relevant works in Amhara, Oromia, Afar and Somali Regional States.

The major responsibilities of the RPIUs under these specific assignments are:

a) Construction and rehabilitation work (water schemes, feeder roads and livestock infrastructures)

- Conduct feeder roads assessments and road survey; and prepare detail designs, technical working drawings, specification and Bill of Quantity (BoQ), engineering cost estimates and tender documents for feeder roads and small bridges construction with the main responsibility of the Regional Rural Road Authority (RRRA).

- Collect and review all relevant standards, guidelines and designs developed by concerned bodies (MoA, MILL, Regional Bureaus, etc.) for different livestock water trough and Fishery-Aquaculture infrastructures development and rehabilitation.

- Conduct baseline assessments to critically identify and prioritize the construction and rehabilitation sites for new developments and existing livestock watering trough and Fishery-Aquaculture infrastructures in the project intervention areas.

- Modify the already collected designs in light of the NWFP contexts and any other considerations in collaboration with the Regional Bureaus.

- Prepare baseline assessment reports for feeder roads and livestock infrastructures development and rehabilitation; and submission to the Bank for “No Objection”.

- Compilation of outputs/deliverables (detail designs, technical working drawings, specification and Bill of Quantity (BoQ), engineering cost estimates and tender documents) from surface water studies, ground water investigations, baseline assessments for each proposed Rehabilitated SH Irrigation infrastructures to be developed and rehabilitated.

- Recruitment of responsive contractors for out-sourcing modality based on the Bank's procurement procedures. Refer the procurement manual for further details.
- Handing over the construction sites and commencement of the construction and rehabilitation works to the contractor.
- Contract administration and management.
- Procurement of construction materials and any supplies as per the need for on force account mode of construction and rehabilitation based on the Bank's procurement procedures. Refer the procurement manual for further details.
- Conduct consultative discussions with the concerned stakeholders at different levels (Regional, Woreda, Kebele and Community).
- Mobilization of resources to the construction and rehabilitation sites and commencement of the work for on force account method.
- Closely monitor and supervise the actual progress of the construction and rehabilitation works and ensure quality of the work as per the designs, specifications, norms and contract agreements.
- Confirm the advance and interim payment requests by the respective contractors in collaboration with the RPIU and sent to the FPCU for processing.
- Handing over the developed and/or rehabilitated Irrigation schemes to the communities in collaboration with the FPCU and corresponding Regional Bureaus.

b) Supervision (both development and rehabilitation works)

- Review and comment the draft Terms of Reference (ToR) for the supervision consultancy service submitted by the FPCU.
- Submission of all compiled relevant design documents (detail designs, technical working drawings, specifications and Bill of Quantities (BoQs) and contract document) to the consultants pertaining to the construction and rehabilitation works.

- Supervise all construction and rehabilitation works conducted through on force account modality in collaboration with concerned stakeholders (BILL, BoNRP in Afar and BOA in Ethiopian Somali;
- Assist in contract administration and management.
- Closely monitor and supervise implementation of the supervision work by the recruited consulting firms in collaboration with the FPCU.

Woreda level

The relevant Woreda Office of Agriculture Development (WOAD) in Amhara, Oromia and Afar and Woreda Office of Agriculture and Natural Resource. (WOAin Somali regions and their staff members will be responsible for the coordination and mobilization role. It works closely with the Kebele and communities for direct program component and sub-component activities implementation.

The major responsibilities of the Woreda Agriculture Office (s) under these specific tasks are:

a) Construction and rehabilitation work (Rehabilitated Irrigation schemes, livestock trough and Fishery-Aquaculture infrastructures)

- Assist the baseline assessments to critically identify and prioritize the rehabilitation sites for new developments in the project intervention areas.
- Conduct consultative discussions with the concerned stakeholders at Woreda, Kebele and Community levels.
- Support resources mobilization to the construction and rehabilitation sites and commencement of the work for on force account method.
- Facilitate handing over of the construction sites and commencement of the construction and rehabilitation works for out-sourcing method.
- Closely monitor and supervise the actual progress of the construction and rehabilitation works and ensure quality of the work as per the designs, specifications, norms and contract agreements in consultation with the RPCUs.
- Handing over the developed and/or rehabilitated irrigation schemes and livestock trough and fishery-Aquaculture infrastructures to the communities in collaboration with the FPCU, RPCUs and corresponding Regional Bureaus and Woreda Offices.

- **Supervision (both development and rehabilitation works)**

- Support supervision of all construction and rehabilitation works conducted through on force account and out-sourcing modality in collaboration with concerned stakeholders and consulting firms.

- Closely monitor and supervise implementation of the supervision work by the recruited consulting firms in collaboration with the RPCU SHIDLE and RPIUs.

Table 1.4. Sub-component 1.3 Rehabilitate and Expand irrigation infrastructure 5 years budget

Template Table for 5 years physical quantity and Financial Budget plan for each sub-component and activity																
Note: use UA in '"000" for the financial budget Plan																
No. /Ref. Code	**Activates**	**Unit**	**Programme Life**	**Unit Cost (UA)'000**	**Physical Plan (quantity) in Years of Implementation**						**Financial Budget Plan in Years of Implementation**					**Total Budget (000) in UA**
					2024	2025	2026	2027	2028	Total quantity	2024	2025	2026	2027	2028	
	Component 1: Climate Smart Wheat Productivity and Production															
	Sub-component 1.3: Rehabilitate and Expand Irrigation Infrastructure															
	1. Investment costs															
1.3.1	Rehabilitation of irrigation schemes in Amhara	Hectares		3889	-	200	900	1,000	-	2,100	-	835	3,814	4,302	-	8,951
1.3.2	Rehabilitation of irrigation schemes in Oromia	Hectares		2593	-	400	1,000	2,000	-	3,400	-	1,114	2,826	5,736	-	9,675
1.3.3	Rehabilitation of irrigation schemes in Somali	Hectares		4815	-	100	400	360	-	860	-	517	2,099	1,917	-	4,533
1.3.4	Rehabilitation of irrigation schemes in Afar	Hectares		4815	-	60	100	100	-	260	-	310	525	533	-	1,368
1.3.5	Provision of water pumps	Number		6296	-	200	200	128	-	528	-	1,352	1,372	892	-	3,616
1.3.6	Maintenance Kits for pump maintenance and repair-youth	Number		944	-	100	-	-	-	100	-	101	-	-	-	101
1.3.7	Training of youth on maintenance kits and establishing cooperatives	Trainees		370	-	100	200	100	-	400	-	40	81	41	-	161
1.3.8	Rehabilitation of access roads to the irrigation schemes	Km		14814.8	-	-	50	-	-	50	-	-	807	-	-	807
1.3.9	Consulting firm for supervision of works	Lumpsum		74074.1	-	0.3	0.4	0.3	-	1	-	24	32	25	-	81
1.3.1	Development of livestock troughs	Lumpsum		370370	-	0.3	0.4	0.3	-	1	-	119	161	123	-	404

0																
1.3.11	Development of fisheries - aquaculture	Lumpsum		370370	-	0.3	0.4	0.3	-	1	-	119	161	123	-	404
1.3.12	Training of water users associations - Amhara	Lumpsum		222222	-	0.3	0.4	0.3	-	1	-	72	97	74	-	242
1.3.13	Training of water users associations - Oromia	Lumpsum		222222	-	0.3	0.4	0.3	-	1	-	72	97	74	-	242
1.3.14	Training of water users associations - Somali	Lumpsum		148148	-	0.3	0.4	0.3	-	1	-	48	65	49	-	161
1.3.15	Training of water users associations - Afar	Lumpsum		148148	-	0.3	0.4	0.3	-	1	-	48	65	49	-	161
1.3.16	Construction supervision costs	Lumpsum	3	180	-	1	1	1	-	3	-	180	180	180	-	540
1.3.17	Soil moisure measuring devices (Automatic soil moisture sensors)	Number	25	1.35	-	25			-	-	-	34	-	-	-	34
1.3.18	Parshal Flum for on-farm water measurement	Number	228	0.68	-	228			-	-	-	155	-	-	-	155
1.3.19	Hand GPS for SSI schemes data collection in all intervention Woredas	Number	152	0.19	-	152			-	-	-	29	-	-	-	29
	Total											**5,168**	**12,382**	**14,116**	**-**	**31,667**

3.1.4. Sub-component 1.4: Sustainable Farm Mechanization

Overview

The main objective sub-component 1.4 is to strengthen and establish full package mechanization service centres in the wheat growing areas along the wheat value chain (production, harvesting, processing, postharvest and storage) and identified spare parts as bottle necks to support private mechanization service providers, to diversify their services and areas of operation.

Increased use of mechanization service and technology enables decreased dependency on livestock power source. This indirectly, decreases the emission of methane gas to the air ensuring climate resilience. Additionally, it was aimed to support and organize youth groups on agricultural mechanization service provision, support and enhance quality on farm machinery maintenance training in ATVET colleges (Alage and Ardaita).

Working with agricultural mechanization training colleges to support and organize youth and women for service provision. Furthermore, through demonstration, promotion and awareness creation on agricultural machineries and equipment's enable diversified service provision, adoption and limited use of traditional system. Encouraging the development of inclusive and sustainable business models for farmer cooperatives and private sector-led mechanization hiring services will help achieve sustainable farm mechanization through increased adoption and use of the services.

The skills of mechanization service center managers and technical teams will be among the critical issue considered during the execution of this project. Skilled managers act in the ways better expenses saved, wastages decreased and boost service business on use of mechanization technologies during service provision and enable enhanced productivity the machine as well the human resource. Thus capacity building, training and awareness creation is considered for mechanization service providers. Supporting R&D initiatives for the development of locally and agro ecologically suitable implements such as rotavator, lime spreader and ridger which enables soil health, improve soil condition and enhance productivity will be supported and promoted. New technology used on wheat production will be demonstrated and promoted. Tractor attached machines and implements such as Laser leveller which is very useful in

wheat production will be procured and demonstrated for the uptake of mechanisation services in service provision of wheat value chain.

The process of agricultural mechanization can be observed in different phases. For example, it can be seen as the process to replace power-intensive operations (ploughing) or control-intensive operations (weeding). The form of mechanization required could be location-specific and requires support services such as oil, fuel and parts. With mechanization, it is easy to increase area under production, improve timing of operations such as planting / weeding / spraying / top dressing, and the quality of seedbed preparation. Mechanization models include hire, contract or purchase.

Objective

The main objective sub-component 1.3 is to strengthen and establish full package mechanization service centres in the wheat growing areas along the wheat value chain (production, harvesting, processing, postharvest and storage) and identified spare parts as bottle necks to support private mechanization service providers, to diversify their services and areas of operation. Increased use of mechanization service and technology enables decreased dependency on livestock power source.

A) Establishing and Strengthen Agricultural Mechanization Service Centres

This sub title is planned for establishing standard agricultural mechanization service centre that give conducive environment for farmers services on agricultural mechanization service provision, quality maintenance and repair services machineries, Training for farmers which holds offices, training rooms, garage, workshops, spare parts storage, toilets, and machinery warehouse (shade).

One of the main problems in Ethiopia on wheat production is absence quality mechanization service provision. Increased brokers between farmers those buy services and service providers who sell services because of limited service provision system challenges the sector. Accordingly, this project is designed to establish standard agricultural mechanization service center in wheat potentials districts to give fair and quality services to smallholder farmer and others. Thus, from the total 8,321UA budget allocated for this sub component 5,365 UA was

from Netherland government and 2,956 UA is from African Development Fund. Establishing mechanization service center is the main component to ensure the visibility of this component. Thus, as stated in table below a required budget is allocated to establish service centers in four region of the country where rain fed and irrigation wheat production is among the top listed areas. Particularly, two lowland areas, Afar and Somali, where there is a high potential for production of irrigation wheat which require full package service provision. To ensure this 1,755.83UA budget is allocated to establish 2 agricultural mechanization service center in 2023/24 for Amhara region, 1,755.83UA budget is allocated to establish and strengthen agricultural mechanization service centers in 2023/24 for Oromiya region, 877.91UA budget is allocated to establish 1 agricultural mechanization service center in 2023/24 for Afar region, 877.91UA budget is allocated to establish 1 agricultural mechanization service center in 2023/24 for Somali Region.

The main activity under this subcomponent will focus on establishing agricultural mechanization service centers in selected regions and strengthening available agricultural mechanization service centers for to increase service provision on wheat value chain in potential areas to increase production and productivity. This activity broadly ensures use of mechanization technologies and systems to enhance productivity and reduce air pollution through reduced use animal power. Limited use of animal power creates improved species which indirectly reduces number of livestocks those their by-products result in higher emission of methane gas.

This activity is planned to implement in 2024.

- Assessment and Mapping to identify potential districts from the selected regions, assessment of appropriate technology, preferred services in the area and favourable location to establish agricultural mechanization service centre especially along diversified wheat value chain.
- Consultation, design preparation for established agricultural mechanization service centre, bid preparation and construction (administration building, office, training rooms, Fences, shade for machinery and related infrastructure in the centre)
- Construction of workshop and garage infrastructure for maintenance service given for the centre as well as for the community

- Simultaneous follow up and supervision activity of experts and officials crucial to finish in the stated time frame according to design. The supervision team will be from federal, regional and district.
- The available farm mechanization service centre will be strengthened for further enhancement of farm services to farmers.

Key Implementation Procedures and Approaches during establishment

Steps 1: Assessment and mapping to identify and seclect potential areas

Stakeholder consultations are conducted to enlist vital stakeholder support and input in the implementation of the farm mechanisation center construction. The assessment activity will be undergone in collaboration with regional bureau of agriculture and woreda administration to identify the area for construction. In this process the guiding checklist will be prepared and used for mapping the area. The agreement with the regional and woreda will be made to ensure the working process of the ceneter. Checklist will be prepared different stakeholders such as from regional state; woreda and kebele will be also.

Step 2: preparing design of farm service Center

Based on the field assessment findings, Terms of Reference (ToR) shall be developed. The ToR outlines the details of the design work. Such work may require the services of competent consultants. Hence, the procurement of such services involves the preparation of a Request for Proposal (RfP) and a bidding process which has to be conducted according to Selection and Employment of World Bank. The design preparation will help wheat type of the mechanization service center should be and what infrastructures and construction components it should have to have modern service service centre. The design will be prepared by consulting firm and submitted to the Ministry of Agriculture.

Step 3: Define the bidding documents for the construction of Mechanization service center.

The tendering/bidding document provides the basis for defining how and what shall be constructed in the center. The construction process includes site clearing and construction of infrastructure (office, Farmers training rooms, maintenance workshop Fences, shade and

related works) in the selected area. The bid may be prepared in two categories for two regions if in required. One bid for Somali and Afar region and one bid for Oromiya and Amara region. The bidding document shall be prepared based on AfDB Standard Bidding Documents for small works;

B) Establishment of new Mechanization service centre

In construction of new center the negotiation with regional and woreda administration will be undergone to secure the construction area. Site clearing process will be done to start the construction. The approval for construction and bidding process in accordance with the governing regulation will be done. The construction of the center is undergone on depending on the prepared Terms of reference after the assessment. The center will contain at office, Farmers training rooms, maintenance workshop, Fences, shade for machinery and water and electricity.

C) Strengthening of Mechanization Service centre

If service center is available in target area previously stated in the document the strengthening process will be done. In strengthening process the rebuilding, upgrading, improving or modernizing existing infrastructure will took place. Such work requires services of competent contractors. Hence, the procurement of such services involves tendering process which has to be conducted according to small works guidelines and procedures of AfDB.

Step 4: Considering and put in place a sustainability exit strategy

This step involves developing and implementing appropriate strategies that will sustain of the established or strengthen mechanization service center. The strategies

- Monitoring and Evaluation (M&E) system: To be able to constantly track the service given by the center, the quality of service given.
- Continuous capacity development: develop the capacity of the service center managers, technical workers and awareness creation to farmers so that they will be able to utilize and benefit socially and economically from the established center. Development and implementation of sustainability strategies are generally on-going processes.

Responsibility of different institutions

1. Federal agriculture office

- The federal government will facilitate for assessment and selection of potential in collaboration with regional bureau of agriculture
- Prepare design for constructed mechanization service centre,
- Prepare national bid for the construction of centre

2. Regional Bureau of Agriculture
   - Support for the land clearing and negotiations with collaboration with the selected woreda
   - Follow up the construction and do quality control in collaboration with federal office
3. Woreda Agriculture
   - Take full responsibility to keep and safeguard the center until it is finished

Ownership of the center: The ownership of the center will be the federal government of ministry of Agriculture

Implementation Time table

This construction will be done at Woreda level except special issue is given to kebele/village level. In case the national competitive bid not work in the woreda level is not applicable at woreda level, the regional offices will handle the process for works. MoA mechanization Lead executive office and project office shall support in every process of the center construction. Thus, the assessment and mapping of the area to select potential area is done in the first year of project life cycle. The Other process will be done starting from the second year (PY2). The Budget for construction was allocated on the PY2.

Table 6 Existing SSI scheme implementation time table

No	Activity	PY1,	PY2,	PY3	PY4
1	assessment and mapping to select potential area for construction	✓			

No	Activity	PY1,	PY2,	PY3	PY4
2	Site clearing and negotiation	✓			
3	Design, review and approval and Bidding process		✓		
6	Construction Procurement process including bid document preparation, advertising, bid evaluation and contract award		✓		
7	Construction work and contract administration		✓	✓	
8	Monitoring and evaluation		✓	✓	✓

D) Supporting access to Agricultural Machinery and equipment provision

Agricultural mechanization machineries greatly support to increase farm land productivity by facilitating timeliness and quality of cultivation. Agricultural mechanization helps to reduce draught animal power which is basically not environmentally friendly due to emission of large amount of greenhouse gases.

Studies show that crop production is more beneficial in areas where agricultural mechanization is used. According to the research conducted in mechanized agricultural areas, yields were significantly higher than non-mechanized areas. This machineries and equipment's also reduce drudgery and improves the timeliness and efficiency of farm operations. The use of pesticides was also more effective, and the land and fertilizers were used optimally with reduced environmental impacts and wastages. This is will be a wise way for our continent in need of achieving food security by preserving natural resources for future generations.

Agricultural Mechanization technologies are argued to be a key instrument to reduce harvest and post-harvest losses and contribute to food security. In Ethiopia, most of the activities in agricultural production from field preparation to harvesting and storage are traditionally carried out by draught animal power that are inefficient (time, resource, energy consuming for less productivity and quality). Utilization of it increases agricultural productivity and production, reduce labour requirement that can be channelled towards to other high value adding activities and sectors.

Over the last two decades, there has been an increase of farm implements into the farming community driven by private commercial service providers. The first of such service provision was on direct wheat combine harvesting operations. This service has become popular and there is high demand by farmers willing to pay for the service. Later other machineries have been introduced and demands have risen dramatically. The efficiency and quality of mechanization services are low despite higher available demands for service. There are however several constraints to wheat mechanization that will need to be addressed through effective implementation of this project.

After the centers are fully established some machineries and implements will be procured and installed for to increase service provision on the wheat belt areas on support of the agendas of food security. The Procurement and distribution of machineries and basic implements (Tractor, combine harvester, implements such as disc harrows, Mouldboard plows, Row planters, Seed drill, Tractor PTO shaft operated thresher, Seed Cleaners, Tractor mounted boom sprayer, Trailer, Ditcher, Fertilizer sprayer, Lime spreader, PTO and Baler and irrigation pumps), maintenance and Training tools for mechanization service centre will be according to national procurement plan and carried out by federal government ministry of agriculture.

E) Technical and business management training

To increase use of agricultural machineries and equipment's on business-oriented service technical and business management training will be provided. Training participants will be selected in collaboration and coordination with the service centres management officers and ministry of agriculture Mechanization lead executive consultation. Technical trainees will be operators, technicians and service provision operation heads. Business management trainees will include managers, service provision operation coordinators, finance head and farmers training coordinators.

F) Supporting and organizing youth groups through training for machinery maintenance and cooperative organization in four regions

More recently, tractors have been introduced to perform number of primary and secondary tillage activities such seedbed preparations, ridging and intercrop cultivations in limited areas which encourages strategic activity for future work. Planting different crops by organized youths, cooperative unions, private and public sectors institutions are more demanding and profitable for high productivity and ensured food security. The main problem in mechanization is not only insufficient use of technology but limited availability of maintenance services are also among the key problems. Supporting youths to organizing and work on maintenance service will greatly reduce the problem listed as a bottle neck in use of agricultural mechanization machineries and implements. This activity was aimed to support youths to organize and work on agricultural machinery maintenance and repair services and tackle the problems raised in use of technology.

The activity will be more effective if executed with regional bureau of agriculture and selected woredas on the target areas. Assessment of demand driven maintenance and repair services in four regions will be conducted with regional bureau of agriculture and selected werada.

G) Existing gap of agricultural machinery maintenance technicians in the country

In the past number of years a lot agricultural mechanisation machinery and equipment's were distributed by importers, manufactures and research centres. Small agricultural machinery importers give full package maintenance service for their customers. Most of the machineries and technologies distributed in the country were out of use for limited problem.

H) Provide training for youth groups on agricultural machinery maintenance and repairs.

Capacity building for group of youths will be given to increase the skill and know-how on agricultural machinery maintenance. Rural youth machinery operators will be identified from project targeted regions and the training materials will be prepared on agricultural machinery operation and maintenance and training will be organized. This will be executed with ATVET colleagues teaching machinery maintenance and private sectors for practical training. Capacity building for mechanization service center technicians is also included

under this activity. It will be challenge on grouping in Somali and Afar region because of shortage available trained human resource on mechanization. This can be tackled by capacity building of service center and private sectors demanding to work in the regions. Capacity building-of tractor operators for mechanization service center in four regions to increase wheat production machinery use in the targeted areas.

To ensure this a total of 192,000UA budget is allocated to train 80 maintenance technicians and operators in Oromiya region, 80 maintenance technicians and operators in Amhara region, 40 maintenance technicians and operators in in Somali region, 40 maintenance technicians and operators in Afar regional state in 2023/24 calendar. Half of the trainees will be trained in the second project implementation year and the other half in the third project implementation year.

Trainee selection criteria

- Trainees will be selected either from previously grouped youths, mechanization service providers, agricultural machinery maintenance service providers, related maintenance service provision centres those deemed to be eligible and have demand to work on sector.
- Graduate of Agro mechanics, Agricultural machinery Mechanics, Agricultural mechanization and Auto mechanics, related disciplines
- The participant should have at least 2 years' experience on machinery maintenance
- For those who have no education on the discipline and have reasonable experience, he/she should have to bring experience certificate and should demonstrate basic know-how of agricultural machinery and equipment's
- Should have basic know how of technical drawings

Implementing structure

- Arrangement of federal have basic know how of technical drawings

Federal Government

The federal government will arrange the whole facilities for training, select training center, trainers, and manage the process.

Regional government

The regional bureau of agriculture mechanization directorate will support selection of proper trainees working on mechanization maintenance service provision or have strong willingness to work on farm machinery maintenance service.

Participants: are youth groups need to work on mechanization service, with above listed stated criteria's

I) Supporting Youth groups through maintenance equipment provision

In the past number of years a lot of farm mechanisation machinery and equipment's were distributed by importers, manufactures and research centres. But there were limited number skilled man power in sector as whole, specially maintenance services. Dealers and local manufacturers are not very much good at after sales service. The government is now in performance great role by preparing directives to control service provision standard of farm mechanization service providers. Limited agricultural machinery importers give full package maintenance service for their customers. Most of the machineries and technologies distributed in the country were out of use for limited problem. To confront this problem a 1500 UA budget is allotted for the support of youth groups on farm machinery maintenance. It was aimed at supporting the youth groups through training and small scale maintenance equipment to work on the sector.

Implementation institution structure

- **Federal**

The whole process in procurement of equipment used for maintenance to be provided to the youth groups will be administered ministry of Agriculture agricultural mechanization lead executive and project manager. Do feasibility study on farm machinery and equipment and prepare Terms of reference.

- **Regional bureau of Agriculture**

Follow up and supervision with federal office, prepare local guiding regulations regulation

- **Maintenance equipment**

The maintenance equipment is used for basic farm machinery maintenance that can be used for routine maintenance. The equipment will be listed after the feasibility study has been done.

J) Support Manufacture and demonstration of soil health and condition improving Machinery

Broad bed makers, lime spreaders, and ridger are among the implements improving soil health through zero tillage, mixing and applying limes to acidic soil. It is important to increase the adoption and use of these technologies in the country. Thus, it will support to manufacture broad bed makers which will be done by research centers and demonstration is done on lime spreader, rotavator and rigders by Ministry of Agriculture. To execute this activity 235 thousand UA is allocated to be coasted for an activity.

Definition of terms

a) Broad bed maker: The BBM itself is a type of a plough that was developed from the traditional dual oxen- drawn plough, the maresha, in order to more efficiently make raised seedbeds and furrows at the time of seed covering—thus reducing water logging and encouraging early planting of improved cereals which could then be followed by a ...
b) Rotavor: A rotavator is a large piece of agricultural equipment that is used to break up, churn, and aerate the soil before sowing seeds and bulbs in farm land
c) Lime spreader: tractor attached lime spreaders, also known as rotary spreaders, are well-suited for distributing lime over large areas. They work by flinging the lime outward in a broad pattern, which is particularly effective for applying lime to lawns or fields.
d) Spreading lime will balance your soil's pH level and fertilizer will continue to give your plants the nutrients they need. You cannot use one in place of the other and you should expect to use both resources.
e) Ridger: ridger is a plough used to split the field into ridges, prepare broad beds and furrows, especially on irrigated wheat in Afar and Somali regions. The watering is applied through furrows prepared by rigders.

Ministry of Agriculture

- Demonstrate Rotavator, lime spreader rigders attached to tractors and drawn as well in collaboration with regional agriculture bureau

Federal EIAR,

- Assist in manufacture of Broad bed makers in research centers

Regional Research Institutions RARI

- Manufacture of broad bed makers and distribute to users in wheat target areas to improve production.

Guiding principles

R&D principles applying to research centres where technology is manufactured and developed farm technology to be used either small scale farming system of large scale. Developmental institutions like Ministry of Agriculture enhances the coverage of developed and newly imported farm machinery and implement. With one third of the budget allocated for this activity will be used for the manufacture of broad bed makers while the left budget is used for the demonstration of rotavator, lime spreader and ridger implement will be promoted and demonstrated on the farmers field, service providers field and models' farmers to enhance more adoption and use.

Promote and enhance the adoption of Mechanization services in the four regions

A. Demonstration of pre and post-harvest farm machinery and equipment

Mechanization service providers will demonstrate their machineries and improved practices on farmers' field with the arrangement of MOA mechanization lead executive. Post-harvest loss management such as metal silo and hermetic bag will be demonstrated. New technology used on wheat production will be demonstrated and promoted. Tractor attached machines and implements such as Laser leveller which is very useful in wheat production will be procured and demonstrated for the uptake of mechanisation services in service provision of wheat value chain.

- **Pre-harvest Farm Technology**: pre harvest farm technology and practices are agricultural farm operation activities carried out before harvest of product with use of machinery and equipment.
  - ✓ **Power tiller tractors**: These tractors are a multi-purpose machines used on small scale farm operations. They can be used with attaching different implements and attachments. It will be more preferable for small land holder farmers.
  - ✓ **Tractors**, Manual and tractor attached agricultural Sprayers machines, reversible ploughs and other implements, used on the protection of farm land and production.

- **Post-Harvest far technology**: Post harvest technologies are machineries and technologies used after the harvest of wheat in the value chain, PTO operated threshers, maize shellers, such ballers for the forage processing, metal sillo, hermatic bags and other loss minimized technologies and practices.

Capacity building for Agricultural mechanization experts

Capacity building for regions agricultural mechanization officers helps to better apply wheat project easily. Regional coordination with federal on every activity will be helpful, to meet the goal stated by the project. Thus, training regional officers and awareness creation will be done for targeted regions for wheat project. The capacity building will be done in collaboration with regional agriculture office. The federal ministry of Agriculture selects trainers, prepare training manuals, facilitate on the whole training process, and prepare inputs for the training such as fuel, oil for machineries and etc. The regional office selects trainees, arrange training centre, and may also prepare training equipment.

B. National bid, Procurement and demonstration of Laser leveller in the four regions

Wheat production is more productive on a property prepared farm land with the use of farm machinery and implements. Tractor attached machines and implements such as Laser leveller which is very useful in wheat production will be procured and demonstrated for the uptake of mechanisation services in service provision of wheat value chain. A laser land leveller is a machine equipped with a laser guided drag bucket- is much more effective and quicker in ensuring a flat table-top like surface. It is a new technology used to level irregular shaped land which is not proper for growth of wheat. It is used to break hardpan of the soil, loosening of the soil and helps the water to seep into the soil for improving drainage. An even land means irrigation water reaches every part of the field with minimal waste from run-off or water-logging.

- **Effect of laser levelling on wheat production time**

The laser leveller not only helped in reducing irrigation time in the laser levelled fields but also resulted in better utilization of available ground water resources. There was a significant saving in irrigation time on the field. Saving in irrigation application time is directly related to saving in electricity/diesel, Labour and most importantly saving in ground water. The grain yield of rice on levelled fields was indicated higher than traditional levelled fields. The

increase is due to more even crop stand, better efficacy of weedicides, fertilizer and other farm inputs. The laser land levelling is an essential farm operation towards precision agriculture which enhances water application efficiency, improves weed control efficiency, reduces labour requirement for irrigation.

Therefore, the laser leveller is more important for national irrigated wheat initiated by Prime minister, to boost the coverage of production on wheat potential areas in lowland areas tole natural resources and potential for wheat production.

- **Procurement Steps in laser leveller**

The procurement will follow the national procurement rules and regulation which stated also in this document.

In addition to national procurement rules and regulation the following activities will apply in the procurement process of the machine

- Preparation of technical specification to procure the laser leveller machine with senior Agricultural mechanization engineers
- Preparing national bid
- Procurement of the machine
- Technical inspection with group of experts on the procured machine in accordance with the previously prepared specification

Demonstration and promotion of laser leveller machine

The laser leveller is attached to tractor and used infrared radiation to receive and send information to the cut and fill machine. The laser levelling machine will be demonstrated in special wheat production potential woredas in the four regions. Especially on irrigated wheat field preparation it is much more demanding.

Table 7 Five years physical quantity and financial budget plan for each sub component and activity

No,of /Ref. Code	Component 1: Climate Smart Wheat Productivity and Production	Unit	Programme Life	Unit Cost (UA)' 000	Pysical Plan (quantity) in Years of Implementation						Financial Budget Plan in Years of Implementation					Total Budget (000) in UA
					2024	2025	2026	2027	2028	Total quantity	2024	2025	2026	2027	2028	
	Sub-component 1.4: Sustainable															-

No,of /Ref. Code	Component 1: Climate Smart Wheat Productivity and Production	Unit	Programme Life	Unit Cost (UA)' 000	Pysical Plan (quantity) in Years of Implementation						Financial Budget Plan in Years of Implementation					Total Budget (000) in UA
					2024	2025	2026	2027	2028	Total quantity	2024	2025	2026	2027	2028	
	Farm Mechanization															
1.4.1	Establish agricultural Mechanization service Center in four target regions															5,390.79
	Assessment and Mapping to identify and select potential area to construct service Center (from unallocated budget)	doc				1.00	-	-	-		5.79	-	-	-	-	5.79
	Design preparation, bidding, site clearing and construction of infrastructure (office, Farmers training rooms, maintenance workshop Fences, shade and related works) in two regions	number	-	-	0	6.00	6.00	-	-		-	2,593.56	2,634.56			5,228.12
	Followup and Supervision	lumpsum		3.92	0	20.00	20.00	-	-		-	78.44	78.44	-	-	156.89
1.4.2	Supporting and organizing youth groups through training for machinery maintenance and cooperative organization in four region															192.00
	Support and organize youth groups through training for machinery maintenance and cooperative organization - Amhara	number	5	741	-	40.00	40.00	-	-	80.00		32.00	32.00	-	-	64.00

No,of /Ref. Code	Component 1: Climate Smart Wheat Productivity and Production	Unit	Programme Life	Unit Cost (UA)' 000	Pysical Plan (quantity) in Years of Implementation						Financial Budget Plan in Years of Implementation					Total Budget (000) in UA
					2024	2025	2026	2027	2028	Total quantity	2024	2025	2026	2027	2028	
	Support and organize youth groups through training for machinery maintenance and cooperative organization - Oromia	number	5	741	-	40.00	40.00	-	-	80.00		32.00	32.00	-	-	64.00
	Support and organize youth groups through training for machinery maintenance and cooperative organization - Somali	number	5	741	-	20.00	20.00	-	-	40.00		16.00	16.00	-	-	32.00
	Support and organize youth groups through training for machinery maintenance and cooperative organization - Afar	number	5	741	-	20.00	40.00	-	-	40.00		16.00	16.00	-	-	32.00
	Support youth groups through provision of basic maintenance equipments in four regions (from unallaocated budget)	number of group	20	75	-	-	-	-		1,500.00	1,500.00	-	-	-	-	1,500.00
1.4.3	Support the manufacture of Broad Bed Makers and demonstrate other soil condition improving equipments`	lumpsum	5.00	-	3.00	-	-	-	-	3.00	235.46	-	-	-	-	235.46
1.4.4	Promote and enhance the adoption of Mechanization services in the four regions (from unallocated budget)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	628.24
	Demonstrate pre and post-harvest agricultural	-	-	-	18.00	-	-	-	-	18.00	533.76	-	-	-	-	533.76

No,of /Ref. Code	Component 1: Climate Smart Wheat Productivity and Production	Unit	Programme Life	Unit Cost (UA)' 000	Pysical Plan (quantity) in Years of Implementation						Financial Budget Plan in Years of Implementation					Total Budget (000) in UA
					2024	2025	2026	2027	2028	Total quantity	2024	2025	2026	2027	2028	
	machinery encouraging wheat production and productivity															
	Capacity building for regional mechanization officers on wheat production technologies through training	number	25	-	-	-	-	-	-	25.00	25.00	-	-	-	-	25.00
	prepare national bid, Procure and demonstrate laser leveller in the four region				5.00					5.00	69.24	-	-	-	-	69.24
	sub component Total															7,946.01

(**Note: use UA in "'000" for the financial budget Plan)**

3.1.5. Sub-component 1.5: Extension and Climate Advisory Services

The main objective of this sub-component is to facilitate rapid uptake and large-scale adoption of climate smart wheat production and postharvest management technologies (certified varieties, good agronomic practices, ISFM, mechanisation, postharvest loss reduction) to achieve multiple goals of improving production and profitability of small-scale farmers, managing climate and market risks, while enhancing soil health, ecological services of landscapes and livelihoods, through systemic approaches.

The key activities include:

(i) Development of wheat agronomy extension content and capacity building of experts and development agents on wheat agronomy, irrigation, marketing, nutrition, etc.;

(ii) Establishment of E-extension services for wheat advisory, data collection, climate, and market information;

(iii) Conduct large-scale demonstrations and provision of extension and advisory services to farmers on Good Agricultural Practices, ISFM, and postharvest management;

(iv) Support of Farmer Training Centres with digital extension technologies, and climate information sharing and early warning systems; and

(v) Establishment of Soil and Plant Health School Labs to reach 50,000 farmers, Farmer Hubs (one pilot by Region to reach 30,000 farmers), and Women in Agri-booster (to reach 10,000 women). The activities under this subcomponent will be implemented by the Ministry of Agriculture in collaboration with OCP Africa.

Strengthening the capacity of beneficiaries through promotion of climate resilient & sustainable agricultural practices and technologies that help farmers adapt to changing weather patterns such as access to climate resistant crop varieties, early maturing and disease resistant crop varieties and seeds will be expanded, at the same time access to agro-met advisory services in to the existing extension system through FTCs/FPC and on smallholder farmers' fields be supported.

Adaptation strategies based on the principles of climate-smart agriculture can counter the impacts of climate change, such as the promotion of conservation agriculture (CA), the sustainable management of natural resources, and the promotion of climate-smart crops. Access to climate-smart production inputs – The FFS/FBS will be linked to access to inputs from agri-prenuers supported by the Program (seed multipliers) or through credit arrangements with off-takers. DA, Woreda-level experts, and watershed/landscape committees will identify potential private sector players that can support input supply and be accredited by the Project. CREW will also facilitate access to relevant, affordable, climate-smart Wheat inputs and technologies from farmer groups that demonstrate (at least 50% membership of the groups will be women) by prioritizing interventions to them. Farmer organization supported by CREW will have to meet the minimum requirements for social inclusion, including female membership of at least 30%. Group memberships have shown to be an important medium for women to access various services, including inputs.

A) Technical support on Climate Resilient Wheat production and productivity improvement

The objectives of this sub component activities is increasing climate resilient wheat productio n & productivity through implementing improved agronomic practice, using new technologies and promoting climate resilient wheat cluster farming (ACC) and competitiveness through improvement of the systems supporting the development, dissemination and adoption of modern farming practices and cropping system to ensure the wheat production and productivity in the selected intervention areas of the project implementation; CREW support on the following major intervention activities are considered*:*

B) Support for climate-resilient cluster based wheat crop production & productivity

Increasing wheat crop production & productivity and promoting cluster farming in the selected Agroecology and competitiveness through improvement of the systems supporting to the development, dissemination and adoption of modern farming practices and cropping system. The activities will be practiced on selected area of project intervention of small holder farmers plot, and FTC (PTC) in the Two rainfed season (Belg and Meher) and through irrigation during off season.

Support will be cause to become for the production of climate wheat resilient to enhance food nutrition self-sufficiency and climate resilient production, fulfil the demand of targeted domestic consumption, raw materials for agroindustry, and export market of wheat commodity. The government of Ethiopia has planned to increase the production and productivity of wheat crop by prioritizing the most important national food commodities and strengthening wheat flagship programs. At different regions in different Agroclimatic conditions through rainfed and irrigation by applying integrated crop pest management (IPM) and appropriate pre-and post-harvest management, and minimizing post-harvest loss will be as much as possible to 5 % in 2030. To set the aforementioned government ambitions & targets, the following priority intervention areas should be addressed.

- Undertaking demonstration of improved crop technologies of wheat crop in farmers' fields, and FTC/TPC
- Support the adoption of crop technologies outreach.
- Promoting irrigation agronomy and irrigation scheme management
- Supporting initiatives to integrate cross-cutting issues into crop services
- Increase the production and productivity of export wheat and import substitution wheat
- Strengthen cluster/ACC and contractual farming (SHF especially for seed producers)

- Minimize grain post-harvest loss
- Supporting Value Chains that Put Innovations into Widespread Use.

C) Support for the demonstration of the newly released wheat variety

To improve & increase the production and productivity of Small holder Farmers; uses different wheat production technologies like improved seed, lime demonstration for Acidic soil & new released varieties, which are resistant to different wheat diseases and has high productivity as well as short growing season is necessary to enable the producers to use & apply these technologies, such technology has to be released either through research or adapted and promoted through demonstration.

Most of the highland areas of the Country were known for producing wheat crop without a crop rotation system, and those areas also have high rainfall during the main season. Due to those problems, the soil found in the area is highly affected by Acidity. So, to overcome the challenge of soil acidity, it is must to demonstrate the liming of soil by lime at FTC and on individual farmers' fields according to the demonstration manual. In the last five years, many new varieties and improved wheat technologies have been released and developed through research. But the outreach & dissemination of these new/improved technologies, especially to the small holder farmers (SHF), was highly resistant/very low. Therefore, to break this resistance, demonstration of improved crop technology at FTC/PTC, on small holder farmers' plot, and scale up under three production seasons (Meher, Belg, and irrigated farming conditions) is very important & has a vital role to disseminate new/improved technologies easily to the users.

The demonstration site will be selected based on the suitability of agro-ecology in the selected landscapes for the commodities, availability of land & also rain/ irrigation water resources. The farmers/agro-pastoral and FTC/FPC will be selected voluntarily by convincing them with Development Agent and local agriculture offices & administrative. The following improved/new crop technology demonstration activities will be carried/ considered: Demonstration of new released crop varieties (on selected commodities) in FTC/PTC and (SHF cluster group).

- Demonstration of climate-smart wheat production systems in selected target area

- Establish crop technology demonstration platform to involve SHF in identification, demonstration and evaluation of crop technologies.

All of these activities will be carried out using improved wheat varieties that are resistant to both abiotic and biotic stressors, as well as science-based agronomic approaches

D) Support adoption of irrigated wheat technologies outreach for CREW target area

Enhancing rural households' income and food security through improving access to improved irrigation Agronomy technologies is a key development strategy in agriculture, especially in wheat crop production development. Despite continued progress in the development and promotion of improved crop technologies, adoption rates are relatively low, especially with poor farmers/agro-pastorals and women. Many recommendations and experiences indicated that more involvement of the smallholders in evaluating promising agricultural technologies through adoption process brings a good result in the dissemination and outreach of new/improved agricultural technologies the activity will be conducted through the approach of Adopting climate-smart wheat technologies such as drip irrigation, conservation agriculture, fertigation, and site-specific nutrient management for resilient wheat production even with the changing climate.

E) Support to rain-fed wheat-producing farmers: –

Rain-fed farmers will largely be producing wheat crop and other crops, which have generally received a lot of support from the government. However, yields are still below world averages.

According to the CSA report (2021), Cereal's productivity is currently at Qt/ha were 3.0 tone ha· These figures show Ethiopia's productivity ranks poorly globally and in SSA; for example, average productivity in Ethiopia has reached 3.11 tons per hectare, which is lower than Egypt (6.7 tonnes) and South Africa (3.5 tonnes). The low productivity has been blamed on over-reliance on rain-fed systems and use of traditional production methods.

So, the CREW project aims to increase the productivity of climate-resilient wheat production in different Agro-ecologies and crop rotation, where the monocropping system is the dominant cropping pattern, like the highlands of Arsi and Bale areas of Oromia with legumes

or crop diversification and good agronomic practices (GAP) through Demos at the FTC and continuous technical support and follow up.

The producers will also access mechanization services for land preparation/minimum tillage, seed sowing, weeding, and harvesting provided through a paid service provided by the entrepreneurs supported in sub-component 1.4. Cooperatives, youth and unions will be a key to providing these services in ensuring return in investment through leasing services and/or pay back loans for mechanization services. Adopted mechanization will also consider women priority value chains, needs, and ease of access and use.

New innovations, technology and improved practices of crop production will be demonstrated at FTC by conducting assessment in integration with Ethiopian institute of agriculture research and prioritized by their positive impact on enhancing productivity both by irrigation and rain fed seasons. In addition to the demos, farmers will receive theoretical training at the demo site on the specific subject, two times per year to ensure all production activities are covered in line with the production season. Rainfed demonstration will started for 2024 crop year and irrigation demonstration will be started immediately after October 2024 at low land areas of Somali and Afar regions, and in Oromia and Amhara, the irrigation will take place after the main crop is harvested and the start of the summer season.

They will be trained on seed selection, emphasizing the importance of using high-quality seed varieties that are well-suited to local conditions.

Fertilizer management is another crucial aspect of agronomy support. Agricultural experts develop customized fertilizer recommendations based on soil nutrient analysis and crop requirements. Farmers are trained on proper nutrient application techniques, including timing, rates, and placement, to optimize nutrient uptake by the wheat crop. For irrigated wheat production, experts provide guidance on irrigation management, including scheduling, water application methods, and efficient water use, to ensure that crop water requirements are met throughout the growing season.

Crop protection support focuses on identifying and managing pests, diseases, and weeds that can significantly impact wheat production. Experts train farmers on integrated pest management (IPM) practices, emphasizing the use of cultural practices, biological control agents, and, as a last option, targeted pesticide application when necessary. Farmers are taught

how to identify common pests and diseases affecting wheat crops, monitor their presence, and develop appropriate control strategies. Effective weed management strategies, including cultural practices and judicious use of herbicides, are also shared with farmers.

F) Supporting/ Promoting Irrigation Agronomy

Recently, Irrigation is one of the fundamental factors to bring a significant change on wheat productivity & increasing cropping intensity in the irrigated area in Ethiopia.

Producing wheat through after one crop left the land is intensive and effective use of land and higher production can be achieved through irrigation. As we learned from the previous lesson assured irrigation facilities not only help in increasing productivity but also a fundamental condition for wheat production intensification and building resilience to climate change i.e minimize the effects of climate.

Provision of irrigation is needed as a pre-condition for application of other productivity increasing inputs. Recently, Ethiopia has attempted to develop irrigation facilities and produce crops in low land/pastoral & semi pastoral areas which were idle for long time and the country been learned to focus more and more at the lowland areas of the country. Though the country has high potential for irrigation development to different crop production but still the country is not yet fully developed the irrigation infrastructure and used the potential.

There fore particular emphasis on supporting/promoting irrigation through increasing the cultivated area of irrigated land, which will make the farmers/semi-agro pastorals less dependent from rain-fed and decreases the problem of frequent drought seen at low land areas. And which allow them to produce more than two times that ensures the production and increase the income of the farmers/ semi-agro pastorals/ pastorals which will improve the livelihoods of the community found in the CREW intervention woredas.

Therefore, by assessing and Identify irrigation potential areas for wheat production (including low land/ pastoral areas, Mid land and Highland areas).

- Plan wheat production according to agro-ecology and regionally based on irrigation potential
- Revision of Irrigation wheat Crop production agronomy package and manual
- Prepare hand book of wheat irrigation agronomy for Development Agent found in the intervention area.

- Practical based TOT on wheat Irrigation agronomy packages.
- Practical based Training on improved/revised Irrigation wheat agronomy packages for experts at all level and Producers.
- Follow up and support the project intervention on irrigation Agronomy

G) Supporting Wheat Crop Protection

The intervention explores ways to strengthen wheat crop health services. The wheat crop protection service focuses on providing wheat crop pest (Disease, insect, weed and Vertebrate) management support services that could significantly reduce losses due to pests and ensures productivity, quality produces and products for domestic and export markets.

- **Support wheat crop protection services**

This intervention focuses on providing wheat crop pest management support services that could significantly reduce losses due to pests and ensures productivity, quality produces and products for domestic and export markets. Thus, the activities that are identified can mainly be summarized as follows:

- **Conduct continuous Surveillance**

The absence of full-fledged wheat crop protection services brought about challenges, like known major economic pests have not been routinely surveyed, monitored, and their current status properly communicated; informed pest management control decisions are not being exercised. Under this intervention, the deliverable includes the survey and surveillance of economic pests of wheat crop and monitoring their status and infestation areas to inform effective control operations and identification of affected areas to take effective and efficient control measures based on IPM. In order to successfully implement the assessment and surveillance activity, the following activities will be conducted: -

- Survey and surveillance of major wheat economic pest infestation areas/major wheat growing areas.
- Recording and reporting possible occurrences of wheat pest outbreaks.
- Prepare and distribute Pest Management Plan (PMP) to the regional BoA
- Conduct Community-based monitoring and forecasting through training

These activities will be conducted by the federal plant protection lead executives in collaboration with the regional BoA. Based on the prepared PMP. Training will be provided to the regional BoA plant protection SMS and will be cascaded to the woreda SMS, DAs and farmers.

- **Support Private Wheat Crop Protection Service Providers**

Wheat production requires well-programmed and integrated production and pest management systems. There are several plant disease species progressively attacking wheat production. However, the major ones are rust and Fusarium head blight fungi aggressively infest the crop. Both fungal disease-causing pathogens are very virulent to break the resistance of wheat varieties. Insect pests and weeds are also very important yield-reducing pests.

To mitigate the problem os pests, the integrated crop production and pest management system will be implemented:

i. Selection of a relatively resistant crop variety.
ii. Sowing early-maturing varieties to escape the disease and other pests' invasion period.
iii. Apply the early warning and alert provided by CIMMYT for wheat rust early management.
iv. Regular survey to identify the disease initiation period to take control measures early.
v. As a last option, select the least toxic nationally registered fungicide or others and apply as per the research recommendation dose and time of application.
vi. Train experts and development agents, and push the training to be cascaded to farmers.
vii. When applying fungicides advise the users to use PPE (Personal Protective Equipment) to minimize exposure to chemicals by trained spray service provider.

Pesticides application for the protection of agricultural/wheat crops farmers without using personal protective equipment (PPE), standard sprayer equipment and unregulated use of pesticides by farmers is a common practice of wheat pest control in Ethiopia. This however creates human, animal, aquatic organisms, Honey bee and environmental risks. Pesticides are being mishandled by smallholders, which has become all too common practice across the country due to lack of availability PPE and standard spray equipment. Unless the pesticides and application equipment are properly used, their health effects are too numerous and complicated. Short-term effects and chronic effects of long-term exposure are common on smallholder farmers/owner-operators and neighbours, which have much less protection. So,

Promote and incentivizing private sector delivery of plant/wheat protection services with a standard training, licensing, standard sprayer, using only registered pesticides, when to spray, interval of spray, recommended dose, least toxic pesticides (used only WHO classification class U, III, II) and Personal protective equipment (PPE) will minimize application of pesticides. This will increase production and productivity of wheat crop, protect humans, animals, aquatic organisms, honey bee and the environment from unsafe use of pesticides (to reduce the impact of Mitigation).

Spray Service Providers promote the safe and responsible use of pesticides and there are many benefits for farmers hiring service providers. Using spray service providers also means there is reduced human exposure to pesticides, specially protecting women and children, misuse/abuse of pesticides and lack of tracking mechanism will be avoided and small holder farmers can get pest management services with a reasonable price. This activity was implemented by MOA, RBOA, and Districts.

H) Establishment of E-Extension Platform for Wheat Advisory, Data Collection, and Market Information:

The objective **of this intervention** is to provide timely advisory services, facilitate data collection, and disseminate market information to wheat producer farmers and stakeholders.

The activities that will be implemented are:

- Identify suitable technology platforms for the e-extension system.
- Develop the infrastructure for the platform
- Collaborate with relevant stakeholders such as agricultural experts, farmers, and IT professionals to design and implement the platform.
- Customize the platform specifically to wheat farming needs, including advisory services on cultivation techniques, pest management, weather updates, etc.
- Integrate features for data collection and analysis to monitor wheat production trends and identify areas for improvement.
- Establish mechanisms for real-time market information dissemination, including prices, demand forecasts, and market trends.
- Conduct training sessions for farmers and extension workers on how to use the platform effectively.

- Monitor and evaluate the platform's performance regularly, gathering feedback from users to make necessary improvements

A) Strengthening of wheat rust early warning software

To strengthen the software known as Wheat Rust Early Warning System already developed by the fund provided from the World Bank through AGPII program. This software is developed in collaboration with the Ethiopian Institute of Agricultural Research (EIAR), ATI and CIMMYT. The software is giving early warning to frontline regions and woredas of the wheat rust outbreak. Based on the early warning alert the Ministry of Agriculture plant protection lead executive give advisory and recommendations to the wheat growing area about type of wheat rust should be given due consideration (Leaf, Stem, and Yellow), the incidence, and the severity level of the disease in the intervention area. For this software, the information collection flows through Mobile up from the grassroots level to the national level, and the information is transferred and analyzed by information technology (IT) expert and finally advisory produced by national technical team from the above-mentioned institutions and advisory sent to wheat rust outbreak areas as an early warning information for quick survey and early control.

The Implementing Institutions: - MoA/Agriculture and Horticulture Extension, Crop development and Crop protection lead executives / jointly, Relevant Regional Bureau, woreda office of Agricultural, Kebele institutions and collaborating with other relevant partners / Digital Green,ATI, Lersha etc who can provide different platforms for advisory service delivery like IVR, Telegram chatbot etc./

B) Production of Media Training Materials on Extension Topics

Objective: To enhance the capacity of extension workers and farmers through informative and engaging training materials.

The specific activities that will be implemented are:

- Identify key extension topics relevant to wheat farming, such as best practices in crop management, disease control, post-harvest handling, etc.
- Develop multimedia training materials including video based extension, presentations, digital co.ntent creation and printed materials.

- Collaborate with subject matter experts, agricultural researchers, and media production professionals to ensure accuracy and quality of content.
- Utilize a variety of communication channels such as online platforms, digital studio, social media and printed materials to reach a wide audience.
- Organize training workshops and seminars to disseminate the materials and provide hands-on guidance on their use.
- Monitor the uptake and effectiveness of the training materials through surveys, focus groups, and feedback mechanisms.
- Update and refresh the materials regularly to reflect new developments, emerging challenges, and feedback from users.

The Implementing Institutions: - MoA/Agriculture and Horticulture Extension, Crop development and Crop protection lead executives / jointly, Relevant Regional Bureau, woreda office of Agricultural, Kebele institutions and collaborating with other relevant partners

C) ICT equipment (tablets, TVs, etc. for FTCs)

Information and communication technology in agriculture (ICT in agriculture), also known as e-agriculture, focuses on the enhancement of agricultural extension through improved information and communication processes. More specifically, e-agriculture involves the conceptualization, design, development, evaluation and application of innovative ways to use information and communication technologies (ICTs) in the rural domain, with a primary focus on agriculture. ICT includes devices, networks, mobiles, services and applications; these range from innovative Internet-era technologies and sensors to other pre-existing aids such as fixed telephones, televisions, radios and satellites. Provisions of standards, norms, methodologies, and tools as well as development of individual and institutional capacities, and policy support are all key components of e-agriculture.

Many ICT in agriculture or e-agriculture interventions have been developed and tested around the world to help agriculturists or development Agents/DAs/as well as farmers to improve their livelihoods through increased agricultural productivity and income, or by reducing risks. Some useful resources for learning about e-agriculture in practice are the most important for wheat crop information/In related to market information, climate information /and connection farmers or Experts to knowledge, networks and institutions ICT uses for inclusive value chains. ICT uses for inclusive value chains and Success stories on information and

communication technologies for agriculture and rural development have documented many cases of use of ICT in agriculture.

The digital technology or Information and communication equipment (ICT equipment) helps to quickly and easily transfer information from place to place and that improve the Wheat crop production and productivity. The ICT devices are giving the way for farmers to use digital technology to address some of their most persistent challenges with the help of digital technologies that enables to apply precision agriculture techniques.

Enhance agricultural knowledge and information systems: Properly documenting and managing knowledge and information enable to reach smallholder farmers and other practi'tioners in a timely manner and this consequently increase agricultural production and productivity and addressing food insecurity. Successful agricultural knowledge and information management requires strong institutions, infrastructure, facilities and skilled human resources to generate, capture, store and disseminate tailored made services to all farming communities.

The specific activities implemented in related to ICT equipped for Farmer Training Centers/FTC/ and PTC: -

- Selection appropriate digital tools.
- Conduct a needs assessment survey or consultation with agricultural experts, extension workers,
- Conducting procurement

Activity 1.5.6.1. Purchase and distribution:

D) ICT equipment/Digital equipment procurement and Demonstration agricultural inputs/technology for FTC/PTC.

The aim of the Information and communication technology in agriculture /ICT/ equipment helps to quickly and easily transfer information from place to place. However, the CREW project is scheduled to be completed in the second-year plan of the project in <supporting of equipment to the Development Agent. However, it was planned as part of the project's first-year plan, as it was believed that the early use of these tools would help in early gathering of basic information that would help to achieve the project's objective ICT equipment (tablets,

GPS, Pico projector, GPS, Agri-tech solution tool kits (nutrient & Phyto Index (for pest) detector), soil PH meter for for FTC/PTC able to use by Development Agents/DAs. In 2024 FY planning to deliver a set of ICT equipment for 22 FTC/PTC) in the selected woredas of CREW project supported areas.

E) **Conducting procurement Agricultural inputs/technology**/wheat seed variety, lime and recommended fertilizer, Metal silos, airtight/Pics bag/ containers etc for Demonstration on FTC/PTC

F) Conducting procurement Personal Protective Equipment (PPE)

Pesticides have a drawback specially on Human Health both in short-term (acute effect) and long-term (Chronic effect carcinogenic and Mutagenic) b/cs of spraying of any pesticides without PPE. So, to protect human health from pesticides injury PPE is very important specifically Sprayer Service Provider. So, PPE tools/cap, glove, eye goggles, rubber boots, overall, face mask and face shield/ procurement and Demonstration is conducted.

G) Airstrip construction (Maintenance)

Currently the government of Ethiopia decided to substitute wheat grain import by domestic wheat production. For this reason, it is planned to produce wheat also in potential irrigated areas in all regions where wheat can grow. The initiative showed promising result and the country is earning intended production from wheat crop.

However, there are several production constraints causing yield loss. The major challenge causing production reduction or yield loss among many pests is Quelea birds. Quelea birds can cause 50-100% if not controlled properly.

Quelea birds control operation is carried out in their night roosts starting from sun set when the birds collected and occupy their night roosting sites. The successful and effective control can be achieved by using spray aircrafts or aerial operation. The basic requirement for aerial control is availability of local airstrips or airports near the roosts and since spray is conducted around sunset, the airstrips should be around 30-50 km nearby the traditional roosting sites of quelea birds.

Airstrip construction for quelea birds' control in the intervention area of Afar region (Amibara), in Somali region and in Oromia region and the like is compulsory to harvest the planned production without significant yield loss.

H) Strengthening implementation capacity of the Ministry of Agriculture, EIAR and Regional research systems

- **Scholarships for PhDs in soil health**

As part of OCP-Africa contribution to the Government of Ethiopia Climate Smart Wheat Value Chain Development Project, two PhD's scholarships will be financed for potential PhD candidates holding masters (or equivalent) in Plant science, Agronomy and soil sciences, with solid credentials and excellent academic background. Both candidates will conduct their thesis research on soil health at the University Mohammed VI Polytechnic (UM6P) with the collaboration of local Universities doing research on soil health in Ethiopia. While focusing on areas with high wheat production potential in Ethiopia, the first candidate's research topic will be related to soil acidity and the second one's research topic will be related to soil salinity.

- **Executive Certificate: Smart crop irrigation**

A training of 10 days will be organized for 10 participants seven senior expert from crop development, two Expert from extension and one irrigation Agronomy from Small holder irrigation department will be participate on this training to enhance their implementation capacity. It will give them a general overview of the theoretical and practical aspects of precision irrigation, allows them to understand the advantages of this technology and to be able to use it accurately.

The course aims to enable participants to fully understand the basics of smart irrigation management, through the calculation of crop water needs, knowledge of data collection technologies such as IoT and remote sensing, processing data, review of decision support models and the development of digital interfaces that allow smart irrigation.

This training is useful for any participant, with basic knowledge in agronomy and crop production, who needs to properly manage water resource, optimize its effectiveness and efficiency.

- **Executive Certificate: Smart crop fertilization**

A training of 10 days will be organized for 10 participants seven senior expert from crop development, two Expert from extension and one irrigation Agronomy from Small holder irrigation department will be participate on this training to enhance their implementation capacity to know the different forms of fertilizer and modes of application; to be familiar with digital hardware devices; to understand the processes of data collection, analysis and processing; to understand the notion of input modulation, the methods of measuring this modulation and the different equipment needed to carry out this operation; to learn about the decision support models and the development of digital interfaces that allow intelligent management of fertilizers. Participants will ultimately understand the advantages of this technology and be able to use it properly.

This training will allow participant, with basic knowledge in agronomy and crop production, who needs to properly manage crop fertilization, and optimize its effectiveness and efficiency.

- ## Support to farmers For Access to Various service

  - Farm & fortune hub

The Farm & Fortune Hub is an all-inclusive last mile distribution solution that aims to address the challenges of farm inputs availability and accessibility by bringing together all basic farm inputs, training on Good Agricultural Practices (GAP) and extension services needed to ensure the perennial increase of farmers' yield and revenue under one physical roof close to small holders' farmers in the rural communities. Four (04) Farm & Fortune Hub program will be developed as a pilot in an underserved agricultural potential area, to achieve arm reach for farm inputs and services. It will include a warehouse, training hall and office space. Agricultural inputs (fertilizers, CPP and Seeds) are sold at the Hubs and in addition to that, farmers attend trainings on Good Agronomic Practices for different crops. the training is conducted by Agri promoters.

Agri- promoters: These are foot soldiers called who are saddled with the responsibility of training farmers on good agricultural practices, promoting in field, distribution of farm inputs, conduct extension services, conduct soil analysis, give fertilizer recommendation, and develop demonstration plots for farmers to practically learn on the field. The Agripromoters are

equipped with working tools such as Tricycles; used to reach out to rural farmers, deliver farm inputs and monitor farmers' activities, Tablets; used to capture farmer details and their farm information and small handheld implements such as the planter, backpack sprayer, etc. The Hub is also designed to be an aggregation center where farmers can sell their farm produce.

The key objectives of the Farm & Fortune Hub are as follow:

- To reach smallholder farmers with quality affordable farm inputs and services in underserved rural farming community.
- Job creation especially the Youth in Agriculture.
- To provide good agricultural practices (GAP) training (classroom and on field) to small holders' farmers to ensure increase in yield and farmers' profitability.
- Access to financial literacy, financial inclusion, and micro credit provision with mobile money services.
- To provide practical physical farming training in form of demo plots & green houses.
- To provide access to knowledge, product, and market linkage through digital solution to farmers to help increase yield.
- To serve as aggregation center for farm produce after harvesting.
- To provide soil analysis and offer fertilizer recommendation to farmers using the digital labs in the Farm & Fortune Hubs.
- To provide access to internet and data access through WIFI connection.
- To provide access to mechanization and Irrigation services to small holder farmers.
- To provide Corporate social responsibility targeting rural dwellers in host communities such access to clean water.

o Women in Agri booster

The Women in Agri booster programmer is a value chain activity where the smallholder woman farmer has access to quality and affordable agricultural inputs, training on Good Agricultural Practices and ready and available produce markets.

The program will start by constituting an ecosystem composed of the various actors of the value chain (suppliers of inputs, financing services, mechanization, off takers, etc) to be associated with the implementation of the program.

It's expected that the Agri booster pilot will be deployed and benefit to 10000 women in two regions of the Project with the objective of increasing their yields and revenues by 40 to 50 %

- School Lab

The School Lab is a mobile school and a Laboratory that moves from community to community providing training to 100-150 smallholder farmers and offering free soil testing to 30-40 key selected farmers that were pre-identified with the help of local institutions.

It is a powerful tool that combines training on Good Agricultural Practices and soil testing laboratory using latest innovation technology to provide live information on soil and fertilizer recommendations.

It is expected that 50.000 farmers will benefit from this program in different regions of the Project intervention with the following objectives:

- To create awareness of the need by Smallholder Farmers to know their soil health and fertility status
- To develop localized soil maps to guide in any fertilizer program
- To ensure farmers are applying the four regions stewardship approach (Using right input with the right source and the right dose at right time).
- To improve the productivity and revenues of Smallholder farmers.

I) Establish demonstrations of newly released varieties and GAP (liquid organic fertilizer)

The main objective of the component is to increase the quality, relevance and coverage of agricultural services leading to increased adoption to improved technologies and Agricultural. Practices, (and enhanced skills) by male and female farmers (and rural youth) which in turn contribute to improve the climate resilience wheat crop, to facilitate rapid uptake and large-scale adoption of climate Resilience /smart wheat production and postharvest management technologies (certified seed varieties, good agronomic practices, Integrated crop protection managements, Integrated Soil Fertility Management, mechanization, postharvest loss reduction) to achieve melt

Strengthening extension and advisory services Through the NWFP, wheat specific extension system will be developed with emphasis on market-oriented wheat production. This will include supporting the development of customized (area specific) extension communication packages, trainings on market-oriented wheat production, wheat full package, and extension communication package (ICT 4 Extension), specialized training for Development Agents

(DAs) and organization of field days. The package formulations will be carefully identified, compiled, disseminated, and scaled up. The extension infrastructure will be facilitated for demonstration and pre-scaling of the packages and motorcycles and other personal equipment to be procured and supplied to DAs. Guidelines for incentive mechanisms for DAs will be developed to maintain their continued services in wheat production areas, particularly in the lowland irrigation areas. The NWFP will seek to create and strengthen linkages between research, extension, and other stakeholders in the various wheat agro-ecologies and administrative levels. pile goals of improving production and profitability of small-scale farmers.

The sub -component major interventions namely, Establish demonstrations of newly released varieties and GAP (liquid organic fertilizer), Capacity Building, Technical support, and Support ICT equipment (tablets, GPS/ for Farmer Training centers /FTC/and Pastural Training centers/PTC) Demonstration of Integrated Pest Management (IPM) at Farmer Training CentersFTC/Pastural Training centers/PTC and Technical support on wheat production

J) Establish demonstrations of newly released wheat crop Technology/varieties, GAP and Post-harvest technology/

The aim of the program is to demonstration of new release and harsh climate resilience wheat crop varieties will be conducted at the Farmer Training Center /FTC/ and Pastural training centers/PTC/ by evolvement of local kebele communities like women, youth and other pastural and semi-pastural depend on rainfed and irrigation-based production. These activities mainly assisted by the kebele Extension workers/development Agents, the woreda agricultural office Subject Matter Specialist / SMS/ also support technical gap to the DAs.

The demonstration has a great role for women's skill empowerment to participate in agricultural extension services, that they adopt the improved climate resilience wheat technologies. The program will support the following interventions: -

- Demonstrations of newly released varieties.
- Post-harvest technology demonstration (metal silo and air tithed bags)
- Demonstrating Good Agricultural practice (application of liquid organic fertilizer)
- Farmers Field Day.

- Best practice identification and complaining
- Production of e-learning training materials on extension topics (agronomy, Plant protection, post-harvest, quality etc)
- Demonstration of Integrated Pest Management (IPM) at Farmer Training Centers FTC/Pastural Training centers/PTC.
- Technical support on wheat production

K) Demonstrations of newly released varieties:

In 2024 it is planned to establish demonstration of Ten new released wheat varieties on 10 hectares of land with full package by small holder farmers & FTC/PTC both in rain fed and irrigated conditions to popularize under four project regions (in Afar three woreda through irrigation, in Amhara, five woreda by irrigation and one woreda by rainfed , in Oromia Eight woreda in both condition and Somali in five woredas which two woredas by rain fed and three in irrigation condition) have planned for implementation.. Since, there are shelved/unpopular new varieties released for both conditions and agro-ecologies; starting in the 1st year will have a better role in the production increment than waiting until to the 2nd budget year of the plan.

Implementing Institutions: - MoA/Agriculture and Horticulture Extension, Crop development and Crop protection lead executives / jointly, Relevant Regional Bureau, woreda office of Agricultural, Kebele institutions and collaborating with other relevant partners

The specific activities to implement the demonstration are: -

- Identifying the farmer training Centers/FTC/ and Pastural Training Centers/PTC/ done by collaboration of Ministry of Agriculture- and RBoA with the evolvement of worda Office agriculture.
- Awareness creation to the Development Agent/DAs.
- Supply and Distribution of new release wheat verities and other related Climate mitigation mechanism technology based on agro-ecology.
- Demonstration takes place at farmer training Centers/FTC/ and Pastural Training Centers/PTC/
- Demonstration activity done by kebele Development Agent and closely supervision of Woreda Subject Matter Specialist.

- The demonstration is also included both method and result demonstrations
- Regular Monitoring and Evaluation is conducted two time per month for woreda, quarterly for the region and twice yearly for MOA.

L) Post-harvest technology demonstration

Postharvest is the final stage of production and any losses experienced at this point are more severe as they represent wastage of all resources involved during production. Post-harvest loss of most cereal crops reaches from 10 to 30% of the total production Packing house operations can be as simple as moving produce from the field into a crop storage or may include a variety of handling practices, started from filed to the market.

The advantage of demonstration post-harvest technology to Farming community will be equipped with important postharvest handling techniques for climate resilience wheat crops from the field to the market, to gain the knowledge on importance of temperature management of harvested crops in extending their shelf life, and on equipment and tools used for extension of life and improving quality of harvested product as well as to provide effective storage solutions that protect wheat grains from pests, insects, moisture, and other factors that can lead to spoilage and deterioration. Demonstrations can be conducted to showcase the benefits and proper utilization of these technologies to farmers and other stakeholders involved in wheat production. Metal silos are durable, airtight storage containers made of galvanized steel. They offer several advantages in minimizing post-harvest loss.

The activities conducted on wheat post-harvest technology demonstrations detail bellow,

- Demonstration of metal silos and airtight bags (PICS bags) at FTC/PTC can significantly minimize this much of wheat post-harvest losses which can feed many peoples.
- Demonstration of **60 metal silo and 400 PICS** bags PHT in selected CREW Project woreda of FTC/PTC is planned in 2024 especially for rain fed wheat production.

The Implementing Institutions: - MoA/Agriculture and Horticulture Extension, Crop development and Crop protection lead executives / jointly, Relevant Regional Bureau, woreda office of Agricultural, Kebele institutions and collaborating with other relevant partners

M) Demonstrating Good Agricultural practice (application of liquid organic fertilizer)

Good Agriculture Practices influence the quality and quantity of the farmer's produce. The implementation of Good Agriculture Practices will help the farmer to reduce physical, chemical and biological risks that may destroy the produce.

GAP begins with site selection, seed selection, organic and mineral fertilizer application, pest and disease control and irrigation. In this module the participants will cover Good Agriculture Practices related to soil, plant nutrition and crop protection, chemical and biological control.

application of liquid organic fertilizer with full of Agronomic practice in **Amhara** and **Oromia** on rainfed wheat demonstration.

- Selection farmer training Centers/FTC/ and Pastural Training Centers/PTC/ done by collaboration of Ministry of Agriculture- and ReBAgriculture with the evolvement of woreda Office of agriculture.
- Awareness creation to the Development Agent/DAs.
- Supply and Distribution of new release wheat verities and other related Climate mitigation mechanism technology based on agroecology.
- Demonstration takes place at farmer training Centers/FTC/ and Pastural Training Centers/PTC/ and farmer plot.
- Demonstration activity done by kebele Development Agent and closely supervision of Woreda Subject Matter Specialist.
- The demonstration is also included both method and result demonstrations
- Regularly Monitoring and evaluation conducted two time per month for woreda, quarterly for region and two time yearly for MOA.

The Implementing Institutions: -MoA/Agriculture and Horticulture Extension, Crop development and Crop protection lead executives / jointly, Relevant Regional Bureau, woreda office of Agricultural, Kebele institutions and collaborating with other relevant partners

N) Demonstration of Integrated Pest Management (IPM) at Farmer Training Centers FTC/Pastural Training centers/PTC.

Integrated Pest Management (IPM) will help to create safe, environmentally friendly, cost effective and socially acceptable plant pests' management support service system that contributes to sustainable increase in crop production, productivity, utilization and marketing significantly. IPM technologies promotion on small holder farmers is getting the best value for the money for both agriculture and the environment. It will also help and increase trust between the small holder farmer the domestic and international customer. Because of its holistic nature and the need to integrate a variety of techniques and disciplines IPM requires significant agronomic and pest management knowledge to implement an effective program based on local farming conditions. Thus, the program will support the training and demonstration on IPM on selected farmer's plot.

The leading implementing institution of this activity will be the plant protection lead executives of the MoA in collaboration with different lead executive within the MoA and also the regional BoA. The activities to be performed under this intervention will be conducting of an assessment to identify new IPM technologies and demonstrate already identified Integrated Pest Management Options. The demonstration will be done on wheat crop and on selected FTC and PTC by the DAs assisted by the plant protection, agronomist and extension experts. The program has also prepared, approved and disclosed IPM plan which will be customized to Climate Resilient wheat projects in the intervention area and will be used along with the IPM strategic guideline.

Trainings and experience sharing visits for federal, Regional, woreda experts, development agents and farmers will be organized by federal and regional level implementing Institutions.

- **Farmers Field Day**

The Objective of Farmers field day is to popularize and evaluate the performance of new technologies and improved agricultural practices tried on Farmers field and Farmer Training centers. As well as to exchange experience among regions and stakeholders for Scaling up the successes.

To increase production and productivity of wheat crop, needs the collaboration with different stakeholders or wheat partners / Governmental and Non-governmental organizations and CREW Project beneficiaries communities/women and youth/ in pre scaling up, Scaling up program and in implementing the Agricultural commercialization cluster/ACC/ strategy promising tasks were carried out in the project implemented regions. To popularize and scale up the project/CREW/ achievements and best practices it is very important to use different extension methods.

Information reaches the farming community mainly through indirect (radio & TV) and direct verbal communication channels which include training, meetings, conferences, field visit, farmers field day and social gatherings, followed by learning through direct observation.

A Farmer's Field Day on rain-fed wheat production is an event organized for farmers to gather and learn about the best practices, techniques, and innovations related to cultivating wheat both in rain-fed or dry land and irrigated wheat, technology demonstration at farmer Training and Pastural Training centers and other climate resilience wheat crop technologies It serves as a platform to disseminate knowledge, exchange experiences, and provide practical demonstrations to help farmers enhance their productivity and resilience in CREW project areas. The event aims to address the specific challenges faced by farmers and scale up good wheat production technologies to peer farmers/Semi agro-pastorals. Therefore, one farmer's field day event planned in 2024 by selecting region which performs best demonstration that will be promoted to scale up/ out the good practice / technology to other farmers. To realized this objective the following activity.

- Identification of the site/best practice done on wheat crop.
- Conducting selection relevant participants for the field day identification

Implementing institutions

Farmer's field days at kebele are conducted, on rainfed and irrigation wheat at National level, regional level, Zonal, Woreda and Kebele level.

The respective Kebele level Development Agents as part of their job at the various stages of crop development and various technologies. The field days and experience sharing visits at woreda level will be organized and managed by woreda Agricultural office and similarly the field days and experience sharing visits from woreda to woreda will be organized and

managed by the regional BoA. The field days and/or experience sharing visits outside of the regions will be organized by the federal relevant Extension service lead executives of the MoA in collaboration with the Regional BoA extension department.

- **Best practice identification and complaining**

The identification and completion of best practice on agriculture sector is used for disseminate on similar agro- ecology has significant role on increase the production and productivity of wheat crop. The small holder farmers and commercial farms; use of improved & new crop technologies is necessary. To enable the producers to use & apply these technologies, such technology has to be released either through research or adaptation. The new and improved crop technologies have been released and developed through research. But the outreach & dissemination of these new/improved technologies especially to the small holder farmers was highly resistant/very low. Therefore, to break this resistance demonstration of improved wheat crop technology at FTC/PTC, on small holder farmers plot, medium and large-scale private farm land and scale up under two production seasons (Meher, /rainfed and irrigated wheat farming condition) is very important & have a vital role to disseminate new/improved best practice technologies easily to the similar agro-ecology via Experience sharing Program.

Implementing Institutions: -The Agriculture and Horticulture Extension service lead Executives, the crop development and crop protection lead Executives of the MoA, the respective department Collaboration with RBoA and woreda Agricultural office will take the responsibility of conducting the following specific activities at federal, regional, woreda and Kebele level

- Selection of best practice done on wheat crop.
- Conducting selection relevant participants for the field day identification

O) Capacity Building of DA and Farmers on wheat agronomy, irrigation, marketing, nutrition, etc.

Capacity building of development agents and farmers on climate-resilient wheat production in Ethiopia is crucial for equipping them with the knowledge, skills, and resources necessary to adopt and implement climate-smart practices. CREW considers the following key elements for capacity building include:

- **Training Programs:**

Develop comprehensive training programs on climate-resilient wheat production tailored to the specific needs and challenges of farmers and development agents. These programs should cover various aspects, including agronomic techniques, water management, soil fertility management, pest and disease control climate change impacts, climate-smart practices, improved, and post-harvest management.

Both theoretical and practical training sessions will incorporate to enhance understanding and hands-on experience. This can include field demonstrations, farm visits, and participatory learning approaches to ensure active engagement and knowledge sharing. ToT training will prepared & deliver to the selected experts from region, Zone, and woreda to further trainee the trainers i.e. DA and farmers. The training content will focus on the- wheat packages both rain fed and Irrigation conditions. Other stakeholders and partners may collaborate and involved in the training process. CREW Project will support the ToT to deliver for the DA, and Farmers/semi-agro pastorals every year before the cropping season is started.

- **Demonstration Plots**:

In collaboration with research institutions, ATI and agricultural extension service demonstration plots will be establish in the selected project kebele of model farmers and, FTC/FPC. The demonstration wills fulfil the standard protocol of demonstrations adopted by MoA. These plots can showcase the implementation of climate-smart practices, such as the use of drought-tolerant varieties, conservation agriculture techniques, and efficient irrigation methods. Demonstration plots provide practical examples for farmers to observe and learn from, helping them understand the benefits and feasibility of adopting climate-resilient practices.

- **Farmer Field Schools:**

Facilitate the establishment of Farmer Field Schools (FFS) focused on climate-resilient wheat production. FFS are participatory learning platforms where farmers can exchange knowledge, share experiences, and learn from each other under the guidance of trained facilitators. FFS can cover various topics related to climate-smart agriculture and provide a supportive environment for farmers to experiment with new practices.

- **Information and Extension Services:**

Strengthen agricultural extension services to provide up-to-date information, guidance, and technical support on climate-resilient wheat production. This can include the development of climate information systems, provision of weather forecasts, and extension materials specifically tailored to climate-smart practices.

Train and equip development agents with the necessary skills and resources to effectively disseminate information and provide advisory services to farmers. This can involve capacity building programs for extension workers, including training on climate change adaptation, communication skills, monitoring and evaluation, and farmer engagement techniques.

- **Access to Inputs and Resources**:

Ensure availability and accessibility of climate-resilient inputs and resources required for wheat production. This includes providing farmers with access to improved seed varieties, appropriate fertilizers, irrigation infrastructure, and other necessary inputs. Collaborate with relevant stakeholders to ensure the timely availability and affordability of these resources.

- **Monitoring and Evaluation**:

Establish systems for monitoring and evaluating the impact of capacity building efforts. This includes tracking the adoption of climate-smart practices, assessing changes in productivity and resilience, and capturing feedback from farmers and development agents to inform continuous improvement of capacity building initiatives.

By focusing on comprehensive capacity building programs, knowledge-sharing platforms, and improved access to resources, Ethiopia can enhance the capacity of development agents and farmers to implement climate-resilient wheat production practices, leading to improved yields, enhanced food security, and increased resilience in the face of climate change.

P) Capacity building DA, Farmers and access to climate smart wheat production

Capacity building of development agents and farmers on climate-resilient wheat production in Ethiopia is crucial for equipping them with the knowledge, skills, and resources necessary to adopt and implement climate-smart practices. CREW considers the following key elements for capacity building include:

- **Training Programs:**

Develop comprehensive training programs on climate-resilient wheat production tailored to the specific needs and challenges of farmers and development agents. These programs should cover various aspects, including agronomic techniques, water management, soil fertility management, pest and disease control climate change impacts, climate-smart practices, improved, and post-harvest management.

Both theoretical and practical training sessions will incorporate to enhance understanding and hands-on experience. This can include field demonstrations, farm visits, and participatory learning approaches to ensure active engagement and knowledge sharing. ToT training will prepared & deliver to the selected experts from region, Zone, and woreda to further trainee the trainers i.e. DA and farmers. The training content will focus on the- wheat packages both rain fed and Irrigation conditions. Other stakeholders and partners may collaborate and involved in the training process. CREW Project will support the ToT to deliver for the DA, and Farmers/semi-agro pastorals every year before the cropping season is started.

- **Demonstration Plots**:

In collaboration with research institutions, ATI and agricultural extension service demonstration plots will be establish in the selected project kebele of model farmers and, FTC/FPC. The demonstration wills fulfil the standard protocol of demonstrations adopted by MoA. These plots can showcase the implementation of climate-smart practices, such as the use of drought-tolerant varieties, conservation agriculture techniques, and efficient irrigation methods. Demonstration plots provide practical examples for farmers to observe and learn from, helping them understand the benefits and feasibility of adopting climate-resilient practices.

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Facilitate the establishment of Farmer Field Schools (FFS) focused on climate-resilient wheat production. FFS are participatory learning platforms where farmers can exchange knowledge, share experiences, and learn from each other under the guidance of trained facilitators. FFS can cover various topics related to climate-smart agriculture and provide a supportive environment for farmers to experiment with new practices.

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Strengthen agricultural extension services to provide up-to-date information, guidance, and technical support on climate-resilient wheat production. This can include the development of climate information systems, provision of weather forecasts, and extension materials specifically tailored to climate-smart practices.

Train and equip development agents with the necessary skills and resources to effectively disseminate information and provide advisory services to farmers. This can involve capacity building programs for extension workers, including training on climate change adaptation, communication skills, monitoring and evaluation, and farmer engagement techniques.

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By focusing on comprehensive capacity building programs, knowledge-sharing platforms, and improved access to resources, Ethiopia can enhance the capacity of development agents and farmers to implement climate-resilient wheat production practices, leading to improved yields, enhanced food security, and increased resilience in the face of climate change

3.2. Component 2: Market Infrastructure, Linkages and Agri-finance

3.2.1. Sub-component 2.1: Market Infrastructure

The main objective of this sub-component is to expand postharvest and market infrastructure (i.e., storage/warehouses, moisture and PH measuring equipment, processing equipment, etc.) at appropriate accessible primary aggregation centers and to enhance a digitally enabled Market Information System. The focus of market infrastructure development will be establishing/strengthening key marketing infrastructures to create convenient environment for proper implementation and improvement of wheat produce marketing system. The key intervention areas include construction of market infrastructures such as warehouses with quality control instruments, processing equipment and enhancement of available digital market information system.

Key activities include:

(i) Assessment and mapping of wheat marketing systems and infrastructure needs and investment opportunities. This activity will assess, and map wheat current marketing systems and infrastructure needs in the potential wheat producing areas. The assessment and mapping interventions will identify current available marketing systems and infrastructures and identify gaps to improve the postharvest handling of wheat production in the target area to deliver the required quality and quantity of wheat grain to the market. The assessment and mapping will help to establish a vigorous and vibrant marketing system with adequate supply chain infrastructure to enhance the changing production and growing marketable surplus of wheat commodity.

(ii) Construction of large warehouses (6) in centralized aggregation centers and medium storage infrastructure (100) at cooperative locations. This activity will support interventions related to increasing competitiveness of wheat by increasing the quality and availability of key infrastructures for postharvest handling and processing. Improving postharvest Infrastructure facilities that will lead to reduction in production losses and improve quality which is crucial for increasing the income of farmers and deliver quality products to the market. Strengthening storage and aggregation facilities for wheat produce such as drying, grading, and packaging will improve linkages among farmers, aggregators and major buyers of wheat.

(iii) Enhancing and expanding a digitally enabled Market Information System/Platform that was developed by the ATI.

Pertinent and timely information is important for wheat market development. Therefore, one of the important activities to provide is to create an appropriate market information system (MIS) that helps to facilitate the communication and the marketing process of wheat in the country. The information includes list of potential areas of production of different types of wheat, details of the actors, daily price, transportation means and fees per unit, among others. An agricultural market information system involves activities including collecting, analyzing, packaging, storing and disseminating prices and other bits of information relevant to farmers, traders, processors and other actors in agricultural commodities. This activity will support to enhance the national market information system (NMIS) currently available and functioning in the country for agricultural commodities. The main purpose of this activity intervention is to increase the capacity of the current NMIS server developed by ATI to accommodate all market information to be collected in the project target Woredas.

(iv) Collection of data on wheat to feed into the market platform. This activity mainly focuses on collection of wheat marketing information in the project target area and feed into the NMIS server and make accessible to key stakeholders engaged in wheat marketing value chain producers and buyers (traders, processors and consumers).

Summary Table 2. Market Infrastructure, Linkages and Agri-finance

Table 8 Outputs and institutional Responsibilities

2.1. Market Infrastructure
Lead Organizations/ Institution: - MoA/Crop and Horticulture Development Sector, **Main Implementing Institutions:** - ATI ,Relevant lead executives/executives of the MoA, Federal Cooperative Commission, Regional Cooperative Authority/Agencies, Farmers Cooperative Unions, **Supporting/collaborating institutions**: Ethiotelecom
• Assessment and mapping of wheat marketing systems and infrastructure needs, and investment opportunities conducted, and wheat marketing system and infrastructure need identified.

• 6 large warehouses Constructed in centralized aggregation centers at cooperative locations
• 100 medium storage infrastructure constructed in centralized aggregation centers at cooperative locations
• 1 digitally enabled Market Information System/Platform that was developed by the ATI Enhanced and expanded.
• Regularly updated Wheat market information Collected and fed into the market platform

Table 9 main activities and time table for Market infrastructure development

No.	Activity	Budget (USD'000)	Implémentation Schedule				
			Year1	Year2	Year3	Year4	Year5
1	Assessment and mapping of wheat marketing systems and infrastructure needs and investment opportunities	---					
2	Construction of large warehouses (3) in centralized aggregation centers at cooperative locations in Amhara and	945					
3	Construction of large warehouses (3) in centralized aggregation centers at cooperative locations in Oromia	945					
4	Construction medium storage infrastructure (30) at cooperative locations in Amhara	900					
5	Construction medium storage infrastructure (30) at cooperative locations in Oromia	900					
6	Construction medium storage infrastructure (20) at cooperative locations in Afar	600					
7	Construction medium storage infrastructure (30) at cooperative locations in Somali	600					
8	Enhancing and expanding a digitally enabled Market Information System/Platform that was developed by the ATI	100					
9	Collection of data on wheat to feed into the market platform	------					

3.2.2. Sub-component 2:2: Market Linkage Interventions

Wheat producers face many constraints that hamper their participation in market and selection of appropriate market outlets. Key challenges include inadequate access to market information; unfair pricing and cheating of farmers; poor market infrastructure and inadequate transportation network. The main objective of this sub-component is to implement the

contract farming policy that was recently adopted, including strengthening linkages (B2B) between farmers-traders/cooperatives-processors and awareness creation. Key activities include:

(i) Conduct of deep-dive detailed wheat value chain and market studies.

Several wheat value chain and market studies have been conducted in Ethiopia for high potential of wheat production areas. However, fragile and changing wheat value chain across all components require interventions such as value chain and market studies to further understand current situations of the project target areas. Therefore, this activity mainly focuses on conducting detailed value chain analysis and market studies through designed surveys.

(ii) Design of a national wheat marketing framework - legalization of wheat trading (developing guidelines and regulations, awareness creation, capacitating regulatory bodies).

Designing and development of wheat marketing and trading is very crucial to benefit both producers, processors, traders and in general wheat value chain actors across all components. Recently a contract farming proclamation has been prepared by Ministry of Agriculture and approved by representatives of the house of council Federal Democratic peoples of Ethiopian Government to enforce producers and buyers' obligations, responsibilities and rights in agricultural commodities produce marketing. To implement and put in practices this proclamation it is mandatory to develop guidelines, regulations, creation of awareness for implementing institutions and building the capacity of regulating bodies. Therefore, this activity supports and mainly focuses on to develop rules and regulations to support the new contract farming guiding policy.

(iii) Dissemination of the new contract farming rules and regulations for farmers, traders and millers/processors.

Following designing and development of new contract farming guidelines, regulations and rules it is very important to disseminate the information to wheat value chain actors (producers/farmers, traders and millers/processors) through developing and provision of quality training materials and trainings. Therefore, this activity supports and mainly focuses on to develop and disseminate quality training materials on standards and grades in collaboration with wheat millers, training on standards and

grades and training on the new contract farming rules and regulations for farmers, traders and millers/processors.

(iv) Establishment of decentralized agro-processing corridors within the wheat production belt to foster direct market linkages with the producers and improve market efficiency through contract farming and warehouse receipt systems (WRS);

(v) Organization of farmers in production clusters at project intervention operational areas.

(vi) Ethiopian agriculture system has long been characterized by poor productivity mainly resulting from following of traditional farming practices. With hopes to rectify these results, the agricultural extension system has introduced various approaches including full package application and agricultural commercialization clusters. Full package focuses on utilizing the optimum amount of inputs and recommended an agronomic protocol that has been developed for each agricultural commodity. Although these practices had demonstrated results by increasing production and productivity among smallholder farmers, there was still limited uptake among the large group of farmers. This was mainly due to poor input supply, lack of capacity building support to extension agents and lack of linkage access to markets. With hopes of promoting the scale up and adoption of full package, the agricultural commercialization approach was introduced to integrate efforts to bring bulk and surplus production by the approach of grouping farmers with adjacent farm plots to consolidate their farm activity and provide targeted support across key points in the value chain more specifically input supply, on-farm production and market linkage. Therefore, this activity mainly focuses and supports to establish and organize farmers in agricultural commercialization clusters in the project target areas.

(vii) Strengthening the wheat supply chain by reinforcing agricultural commercialization clusters and cooperatives for bulk production, aggregation and access markets and

(viii) Promotion of women and youth agribusinesses and start-ups along the wheat value chain with activities including farming innovation programs and Agri-promoter programs (one program per Farmer Hub). The activities under this subcomponent will be implemented by the MoA, ATI and the Ministry of Trade and Regional

Integration, and the Ethiopian Commodity Exchange to roll out the warehouse receipt financing.

Table 10 Outputs and institutional Responsibilities for market linkage interventions

2.2. Market Linkage Interventions
Lead Organizations/ Institution: - MoA/Agricultural Investment and Input Sector **Main Implementing Institutions:** - ATI, Relevant lead executives/executives of the MoA, Ministry of Trade and Regional Integration, and the Ethiopian Commodity Exchange **Supporting/collaborating institutions**: Federal Cooperative Commission
• Deep-dive detailed wheat value chain and market studies conducted and published
• Rules and regulations of the new contract farming policy developed
• 200 farmers, traders and millers/processors Trained on the new contract farming rules and regulations
• National wheat marketing framework designed and legalization of wheat trading in practice
• Quality training materials developed on wheat standards and grades disseminated to stakeholders
• Decentralized agro-processing corridors established within the wheat production belt in Amhara, Oromia, Afar and Somali regions
• 500,000 farmers Organized in production clusters at project intervention operational areas

Table 10 : Main activities of market linkage interventions and their time table

No.	Activity	Budget (USD'000)	Implementation Schedule				
			Year1	Year2	Year3	Year4	Year5
1	Conduct of deep-dive detailed wheat value chain and market studies.	---					
2	Develop rules and regulations to support the new contract farming policy	50					
3	Training on the new contract farming rules and regulations for farmers, traders and millers/processors (200 people)	50					
4	Design a national wheat marketing framework - legalization of wheat trading	----					
5	Develop and disseminate quality training materials on standards and grades in collaboration with wheat millers	50					

6	Training on standards and grades (200 People)	50					
7	Establishment of decentralized agro-processing corridors within the wheat production belt to foster direct market linkages with the producers and improve market efficiency through contract farming and warehouse receipt systems (WRS)	-----					
8	Organization of farmers in production clusters at project intervention operational areas	-----					
9	Strengthening the wheat supply chain by reinforcing agricultural commercialization clusters and cooperatives for bulk production, aggregation and access markets	------					
10	Promotion of women and youth agribusinesses and start-ups along the wheat value chain with activities including farming innovation programs and Agri-promoter programs (one program per Farmer Hub).	-----					

3.2.3. Sub-component 2.3: Access to Agri-Finance

The main objective of this sub-component is to link producers (cooperatives and farmers' unions) to agri-finance opportunities through training and development of financial risk-sharing facilities such as the warehouse receipt system.

Key activities include:

(i) Conduct of a demand-supply study for the wheat sector and in particular storage and finance needs; Farmer Cooperative Unions have limited capacity in infrastructure and working capital to provide the required services to member farmers. This limited capacity hinders the efforts of all value actors to move forward the wheat sector development to attain wheat production self-sufficiency for consumption and import substitution. Therefore, conducting demand-supply study for Cooperative Unions in infrastructure (storage facilities) and financial access opportunities will empower them by identifying gaps and bridging demand -supply needs.

(ii) Scale up ongoing initiatives such as the Commodity Collateralized Financing by establishing warehouse systems (WRS) in the project areas.

(iii) Small-scale farmers have always struggled to pay their debts and they often sell off their produce when harvest season begins so they can hold onto their crops until the lean season, when the price and potential for profits are at their highest. However, improper preservation or drying techniques, coupled with inadequate storage facilities, can force small farmers to let traders reap the rewards of seasonal price changes. With the use of warehouse receipt financing system, small farmers gain an advantage on the playing field. Experience from around the world illustrates that warehouse receipts can make a difference to farmers. By storing their goods in a reliable warehouse until the price increases while using the goods as loan collateral, farmers may access funds before they sell their goods. Warehouse receipts are often administered to producer groups, instead of individuals, which helps the flow of market information. Warehouse receipts also can create price transparency. This empowers farmers to make informed sales decisions rather than waiting for "farmgate" buyers who often offer below-market prices. Therefore, this activity supports the scaling up commodity collateralized financing by establishing warehouse system through training and development of financial risk sharing facilities.

(iv) Strengthening the legal and regulatory framework for the WRS; to implement and make fruitful the warehouse system legal and regulatory framework should be strengthened and guiding principles must be in place.

(v) Development of capacity of financial institutions to support the WRS; and

(vi) Support of the development of financial risk-sharing mechanisms for the financial institutions and the Millers Associations to have better access to financial services – especially working capital. This subcomponent will be linked to the Agri-MSMEs Development for Jobs (AMD4J) project currently under design, to increase access to and use of financial services and products for Agri-MSMEs. Implementation of these activities will be undertaken by the MoA in collaboration with the Ministry of Trade and Industry.

Table 11 Outputs and institutional Responsibilities for access to agri finance

2.3. Access to Agri -Finance

Lead Organizations/ Institution: - MoA/Agricultural Investment and Input Sector, **Main Implementing Institutions:** - ATI, Relevant lead executives/executives of the MoA, Ministry of Trade and Industry **Supporting/collaborating institutions**: Federal Cooperative Commission, Regional Cooperative Authority/Agencies, Farmers Cooperative Unions, ATI
• Commodity Collateralized financing system (CCF) enhanced and scaleup in project target areas
• A demand-supply study for the wheat sector and in particular storage and finance needs Conducted for target project areas
• The legal and regulatory framework for the WRS strengthened in target project areas
• 10 financial institutions capacity developed to support the WRS
• Millers Associations supported in project target areas to have better access to financial services – especially working capital
• Financial risk-sharing mechanisms for the financial institutions developed

Table 122 Main activities and time table for Access to agri finance

No.	Activity	Budget (USD'000)	Implémentation Schedule				
			Year1	Year2	Year3	Year4	Year5
1	Enhance and scaleup commodity collateralized financing system (CCF)	2000					
2	Conduct of a demand-supply study for the wheat sector and in particular storage and finance needs	50					
3	Strengthening the legal and regulatory framework for the WRS	200					
4	Capacity development of financial institutions to support the WRS	1600					
5	Support the Millers Associations to have better access to financial services – especially working capital	500					
6	Support of the development of financial risk-sharing mechanisms for the financial institutions	500					

4. Environmental and Social Safeguards Management

The goal of this section is to describe how the safeguard instruments and tools prepared for CREW project will ensure that proposed interventions comply with the existing environmental protection and social policies/laws, regulations and standards of the Government of Ethiopia as well as with the African Development Bank's Integrated Safeguard system (ISS) and policies so that project activities will not have adverse impact on the environment and the community at large.

As indicated in this PIM, the project development objective is to improve wheat productivity and production thereby increasing the incomes of small-scale wheat farmers in the project areas. It will be implemented in 76 woredas in four regions, namely oromia (36 woredas), Amhara (24 Woredas), Somali (11 Woredas) and Afar (3 Woredas). This Project is aligned to the sustainable development goals (SDGs), and particularly SDG1 (no poverty), SDG2 (zero hunger), SDG 5 (gender equality) and SDG 13 (climate action). It is also aligned to the African Union's (AU) Agenda 2063 for a Prosperous Africa, based on Inclusive Growth and Sustainable Development.

The Project consists of three interrelated components: (i) Climate Smart Wheat Productivity and Production; (ii) Market Infrastructure, Linkages, and Agri-Finance; and (iii) Project Coordination and Management. Sub components with in each component leads to specific project interventions which thereby implies beneficial and adverse environmental and social impacts to be managed during project implementation.

4.1. Legal Frameworks Relevant to Crew Project

Towards ensuring environmental and social sustainability of the CREW, policy and legal frameworks of the Government of Ethiopia (GoE), the African Development Bank (AfDB) and Global Development Agendas and multilateral agreements to which Ethiopia is signatory are integrated and aligned across all the stages of the project interventions.

Legal frameworks of the government of Ethiopia including, but not limited to, the Constitution of FDRE, the Environmental policy of Ethiopia, the Climate Resilient Green Economy (CRGE, 2011) strategy, the Ten Years Development Plan (2020/21-2029/30) of the Agriculture Sector, the Water resource policy, the National biodiversity policy will be considered. Moreover, the Environmental Impact Assessment Proclamation (Proclamation No. 299/2002), the Environmental Impact Assessment Guideline, the Environmental Pollution Control Proclamation (Proclamation No. 300/2002), the Solid waste Proclamation (Proclamation 513/2007), the Pesticide Registration and Control Proclamation No.674/2010 , the Proclamations no 1156/2019 (on The Labour Law), etc. and many other regional context proclamations and directives will also be applied.

The African Development Bank's (the Bank) environmental and social safeguards are a cornerstone of its support to inclusive economic growth and environmental sustainability in Africa. The objectives of the Integrated safeguard system (ISS) of AfDB is to avoid, or when avoidance is not possible, to minimize and mitigate/compensate negative impacts of projects on the environment and the community. In doing so it promotes the sustainability of project outcomes by protecting the environment and people from the potentially adverse effects of projects. It also seeks to help borrowers/clients to strengthen their safeguard systems and develop the capacity to manage environmental and social risks.

To this end, the Bank has adopted five Operational Safeguards (OS), guided by the need to limit their number to just what is required to achieve the goals and optimal functioning of the ISS:

- ***Operational Safeguard 1: Environmental and social assessment.*** This overarching safeguard governs the process of determining a project's environmental and social category

and the resulting Environmental and Social Assessment requirements. OS1 governs the process of determining a project's environmental and social category and the resulting environmental and social assessment requirements. Categorization follows the principle of using the appropriate type and level of environmental and social assessment for the type of operation. At the project identification phase, a screening exercise would be carried with a focus on the environmental and social dimensions of a project to categorize it in one out of following four categories:

Category 1: Projects likely to cause significant environmental and social impacts:

These projects are likely to induce significant and/or irreversible adverse environmental and/or social impacts. Some program-based operations or other regional and sector program loans have significant adverse environmental or social risks and are deemed to be Category 1. In some cases, projects are included in Category 1 because of their potential cumulative impacts or the potential impacts of associated facilities. Any project requiring a Full Resettlement Action Plan (FRAP) under the provisions of the Bank's policy on involuntary Resettlement is also deemed to be a Category 1.

Category 2: Projects that likely cause less adverse environmental & social impacts than Category 1: Such projects are likely to have detrimental site-specific environmental and/or social impacts that are less adverse than those of Category 1 projects. Likely impacts are few in number, site-specific, largely reversible, and readily minimized by applying appropriate management and mitigation measures or incorporating internationally recognized design criteria and standards. An operation that involves resettlement activity for which Resettlement Action Plan (RAP) is required under the ESAPs (Environmental and Social Assessment Procedures) is classified as Category 2. Most program-based operations and regional or sector program loans designed to finance a set of subprojects approved and implemented by the borrower or client are included in this category. *Category 2* projects require an appropriate level of environmental and social impact assessment (ESIA) so that the borrower will prepare and implement an adequate ESMP (for an investment project) or ESMF (for a program operation), to manage the environmental and social risks of subprojects in compliance with the Bank's operational safeguards.

Category 3: Bank operations with negligible adverse environmental and social risks: Category 3 projects do not directly or indirectly affect the environment and are unlikely to induce adverse social impacts. They do not require an environmental and social assessment and beyond categorization, no action is required.

Category 4: Bank operations involving lending to financial intermediaries*:* Category 4 projects involve Bank lending to financial intermediaries (FIs) that on-lend or invest in Subprojects that may produce adverse environmental and social impacts. Financial intermediaries include banks, insurance, reinsurance, and leasing companies, microfinance providers, private equity funds, and investment funds that use the Bank's funds to lend or provide equity finance to their clients.

- ***Operational Safeguard 2: Involuntary resettlement:*** land acquisition, population displacement and compensation. This safeguard consolidates the policy commitments and

requirements set out in the Bank's policy on involuntary resettlement, and incorporates a number of refinements designed to improve their operational effectiveness.

- ***Operational Safeguard 3: Biodiversity and ecosystem services.*** This safeguard aims to preserve biological diversity and promote the sustainable use of natural resources. It also translates the policy commitments in the Bank's policy on integrated water resources management into operational requirements.
- ***Operational Safeguard 4: Pollution prevention and control, greenhouse gases, hazardous materials and resource efficiency.*** This safeguard covers the range of key impacts of pollution, waste, and hazardous materials for which there are agreed international conventions, as well as

comprehensive industry-specific and regional standards that are followed by other multilateral development banks, including greenhouse gas accounting.

- ***Operational Safeguard 5: Labour conditions, health and safety.*** This safeguard establishes the Bank's requirements for its borrowers or clients concerning workers' conditions, rights and protection from abuse or exploitation. It also ensures greater harmonization with most other multilateral development banks.

Operational Safeguards (OSs) Triggered by CREW Project and categorization

All the operational safeguards of AfDB, except Operational Safeguard 2 (OS 2) on Involuntary resettlement, will be triggered based on the nature of the specific activities to be implemented. For explanation the below table can be referred.

Table 1: Operational Safeguards the AfDB to be Triggered by the CREW Project

Operational Safeguards(OSs) of AfDB	**Triggered by CREW**		**Explanation**
	Yes	**No**	
OS 1: Environmental and social assessment.	✓		The CREW project activities across the four regions will involve providing support for increased access to fertilizer and agrochemicals which will intensify its use, conducting maintenance and rehabilitation of irrigation infrastructures, supply of energy efficient pumps, support to strengthen agricultural mechanization service centers and expanding market infrastructures all of which will interact with the physical and social environments. The irrigated farming activities to increase wheat production & productivity will lead to intensive farming requiring the undertaking of land preparation, planting, irrigation water use, application of agrochemicals including fertilizers and pesticides, harvesting and post-harvest management activities. The undertaking of these activities is likely to cause adverse impacts which will need to be properly assessed and mitigated. Hence OS-1 will be triggered.
OS 2: Involuntary		✓	The main activities of the CREW which could potentially lead to land acquisition and involuntary resettlement involve construction activities to

resettlement:			establish the proposed new small scale irrigation schemes, rehabilitation of the existing small and medium irrigation schemes and expansion of the existing ones across the four regions (Mainly in Oromiya & Amhara). Generally, the rehabilitation and maintenance of the small and medium irrigation schemes are going to be carried on existing irrigation infrastructures making the occurrences of new land acquisition and resettlement risks unlikely. The expansion of irrigation infrastructures are going to be carried by extending the canal systems to adjoining farming areas. Primary, secondary & tertiary canals are narrow linear structures with small width (i.e. 0.5 to 1 m). Field assessments and consultations carried with communities revealed that farmers are keen to have access to the irrigation systems. The peripheral lands of farm fields used for canal system extensions are usually felt to be offset by obtaining the access to the irrigation schemes, making land acquisition impacts negligible. Most of the new irrigation schemes to be established are pump irrigation only some are ground water based. As in the existing schemes, usually communal land is used to build the head works and primary canals. For similar reasons stated above, the small peripheral land of farmers to be used for building the secondary & tertiary canal infrastructures are felt to be off-setted by the access to be created to the irrigation schemes. Owing to the common nature of the small & medium scale irrigation schemes which do not require large scale land acquisition for headwork (usually built on communal lands along the river banks) and the linear peripheral farm field lands to be used for canal extension, no major permanent and temporary land acquisition is anticipated to occur. Therefore, OS 2 will not be triggered by the CREW project activities.
OS 3: **Biodiversity and ecosystem services**	✓		The operational activities of the small and medium scale irrigation schemes involve intensive use of water and agrochemicals. Inefficient and over use of water resources from rivers, streams, and ground waters may lead to occurrence of impacts on ecosystem services. The intensive use of agrochemicals may also affect biodiversity and ecosystem services. Thus, these and other related irrigation rehabilitation activities of the CREW project in the region may cause limited impacts to occur on biodiversity resources and ecosystem services. Thus OS 3: Biodiversity, renewable resources and ecosystem services will be triggered by the project
OS 4: **Pollution prevention and control, greenhouse gases, hazardous materials and resource efficiency**	✓		The operational activities of CREW project are expected to release different pollutants to the environment through increased application of agrochemical inputs including fertilizers and pesticides, wastes released from agricultural mechanization service centres, from operation of pumps supplied and construction wastes from rehabilitation and maintenance of irrigation schemes, the residual impacts of the project activities may have the potential to affect the environment and human health. Thus OS 4: Pollution prevention and control, hazardous materials will be triggered by the project.
OS 5: **Labour conditions, health and safety.**	✓		project implementation of CREW will involve the recruitment of both temporary and permanent project management staff members at Federal and regional levels. Besides, the actual wheat production activities are likely to engage large number of laborers working in the farm sites who are usually members of a farmer's household. More skilled and

			semiskilled laborers will also be engaged while strengthening the agricultural service centres. While supporting these and other activities of the CREW project, it is important that labor conditions, working environment, as well as health and safety conditions are taken into consideration. Thus observance of labor recruitment and health and safety rules will become necessary to comply with National laws and OS 5 requirements. Hence OS 5 (Labor conditions, health and safety) will be triggered by the project, across all the regions

4.2. Project Impacts and Categorization of Crew Project:

Project activities CREW across the project intervention areas will involve provision of support for increased access to fertilizer and agrochemicals which will intensify its application, conducting maintenance and rehabilitation of irrigation infrastructures, supplying energy efficient pumps, support to strengthen agricultural mechanization service centres and expanding market infrastructures the implementation of which will interact with the physical and social environments.

The irrigated farming activities to increase wheat production will also lead to intensive farming requiring the undertaking of land preparation, planting, irrigation water use, application of agrochemicals including fertilizers and pesticides, harvesting and post-harvest management activities. Moreover, the project activities are also expected to release different pollutants to the environment through increased application of agrochemical inputs including fertilizers and pesticides, wastes released from agricultural mechanization service centers, from operation of pumps and construction wastes from rehabilitation and maintenance of irrigation schemes. Inefficient and over use of water resources from rivers, streams, and ground waters for irrigation may also lead to occurrence of impacts on ecosystem services.

These and other residual impacts of the project activities may have the potential to affect the environment and human health. However, these impacts are predicted to be site specific, reversible and could be avoided or minimized by applying appropriate management and mitigation measures through the proper implementation of ESMPs (Environmental and social management Plans). These potential detrimental impacts resulting from the project implementation activities are also predicted not to cause any significant land acquisition, involuntary resettlement and population displacement.

Therefore, based on the AfDB ESAP guidelines the project is categorized under category 2 requiring an appropriate level of environmental and social impact assessment (ESIA) to manage the environmental and social risks of subprojects in compliance with the Bank's operational safeguards. On the other hand, according to the EIA procedural guidelines of the Federal Environment Protection Authority (2003), surface and ground water fed irrigation projects covering more than 100 hectares are categorized under schedule-1 activities which require full ESIA report preparation. Generally, As the size of the command areas of the irrigation schemes proposed in the four regions ranges from 15ha to more than 465ha, such schemes falls under schedule-1 which require the preparation of an ESIA report. Accordingly,

the categorization of both guidelines (the AfDB and the national EIA guidelines) appears to agree on the need for preparing an ESIA/ESMP for the Crew project interventions.

4.3. Environmental and social management plan and implementation of safeguard instruments:

Environmental and social safeguard instruments are conducted with an aim to identify and assess the likely adverse environmental and social impacts of the CREW Project and to determine their magnitude and significance and to ultimately define management or mitigation measures. Mainly the safeguard instruments are critical means designed to avoid and minimize where possible, or if not, to compensate for adverse impacts and risks consistent with the relevant environmental requirements of the Government of Ethiopia and the African Development Bank (AfDB) Integrated Safeguard Systems (ISSs).

Accordingly, four ESIAs (Environmental and social Impact assessment studies) are conducted for CREW Project, one for each region. In addition to this a PVMP (Pest and vector Management plan) also conducted to implement Integrated peat management practices across the value chain of the CREW Project. The main purpose of the Integrated pest management plan (IPMP), among many others, is to minimize /control Pest infestation which result reduction of productivity of wheat and to protect the environment and human health during storage and spraying of agro chemicals and pesticides.

Environmental and Social Management Plan (ESMP)

Environmental and Social Management Plan (ESMP) embedded in the ESIA studies narrate a procedure or measures to be taken during the implementation and operation of a project that reduces, eliminate or offset adverse environmental and social impacts and actions needed to implement these measures. It also indicates the expected impact, action to mitigate it, time frame, responsible body and the estimated cost. The objectives of the ESMP are to:

- outline mitigation/enhancing, monitoring, consultative and institutional measures required
  to address project impacts and
- bring the project into compliance with applicable national environmental and social legal
  requirements and ISS and OS policies of the AfDB;

The Project Management Unit at Federal and Regional levels should assign an adequately qualified environment expert to follow up and ensure that the recommended mitigation actions are implemented. The environment expert will also be responsible for conducting trainings, environmental monitoring, inspection and will prepare environmental and social performance reports regularly. The PMU will report every six months on the environmental monitoring to the MoA and the project development partner with a summary of environmental monitoring and the implementation of mitigation measures for all sections of the CREW Project.

The major responsibility for environmental management will be split between several organizations at Federal, regional and woreda levels. However regional Agriculture Bureaus

as implementing agencies of the CREW project have the overall responsibility for the implementation of the Environmental and Social Management Plan(ESMP). Apart from this, most of the project environmental and social management activities will be carried out during the construction phase as they are expected to arise during this phase. The management of this plan will very largely be concerned with managing impacts that may result from the actions of the Contractor, through enforcement of the construction contract clauses related to the protection of the environment, social, health and safety risk management as a whole, and of the components within it. In this respect, it is important to recognize that successful mitigation of construction impacts can only be achieved if the environmental and social management measures, as set out in the construction contract, are properly enforced.

THE CONTRACTOR ENVIRONMENTAL AND SOCIAL MANAGEMENT PLAN (C-ESMP)

In order for the contractor to carry out environmental and social management activities during construction, the contractor should draw up an environmental and social management plan of his/her own which is the contractor environmental and social management plan (C-ESMP). This plan (aligned and derived from the ESIA and ESMP) aims to demonstrate how she/he will address the mitigation measures during the construction period. The contractor is also expected to prepare Management Strategy Implementation Plans (MSIP) or site-specific management plan. These two instruments (C-ESMP and MSIP) will be prepared by the Contractor to be used as a tool to manage anticipated project environmental, social, health, and safety (ESHS) risks and impacts that arise during the construction period and for the proponent to review and supervise the efficiency of the contractor on the management of ESHS risks. Doing so will ensure that all relevant construction impacts of the contractor's operations are mitigated and managed at higher level of concern. Any accident or incident that may happen during the construction work has to be reported to the FPCU through the respective regional safeguard focal persons

The Key Management Strategy and Implementation Plans (MSIPs) to be prepared by the contractor and operationalized during project implementation include, but not limited to, the followings:

- Code of Conduct,
- Gender Based Violence, Sexual exploitation and abuse, Sexual Harassment and Violence against Children (GBV, SEA/SH/ VAC),
- Water resources management plans.
- Occupational health and safety management plans, etc.

4.4. Capacity Building for Implementation of the ESMP

The implementation of project interventions will be carried out in a decentralized fashion and the environmental and social requirements will be applied at Regional and woreda levels. The Regional and Woreda level Agriculture offices staff may not have the necessary capacity to apply the ESMPs effectively. Therefore Staff from Regional and Woreda Counterpart Bureaus/offices (i.e. Bureau of Agriculture and Bureau of Irrigation Development) will need training to strengthen ESMP and PVMP implementation for each project as well as to ensure

adequate monitoring. Additionally, specific training and capacity building of Development Agents (DAs) on ESMP implementations will also need to be provided.

Hence, capacity building training plan should be developed by the safeguard unit of the FPMU to enhance the knowledge and skills of all implementing entities at the Regional, Woreda ,kebele and grass root community levels thorough the government administrative channels.

Given the limited capacity at woreda and kebele levels, effective implementation of the ESMPs will also require strong support from FPMUs and RPMU. The Social and Environmental Risk Management Specialists at FPMU and the focal person at regional level will be responsible to assist the woreda experts and key stakeholders in terms of facilitating awareness and trainings on Safeguards management issues relevant to CREW Project by allocating the necessary budget.

As far as technical capacity building is concerned, Technical training (ToT) will be given by the FPCU and related sectors with in the MoA for concerned regional and woreda actors on Implementation of ESMP and PVMP as appropriate. This training will be cascaded to kebele DAs and Community based institutions by the respective Regional/woreda focal persons.

As the project implementation progress, monitoring , follow up and support on the implementation of the ESIA/ ESMP will be periodically conducted in each region so as to sustainably implement the CREW.

The following themes will be major focus of attention while Capacity buildings trainings are provided for all project implementers at Federal, regional, Woreda and Kebele levels: -

- Legal and policy frameworks of the GoE and AfDB, relevant to environmental and social safeguards issues.
- Training on Environmental and Social Assessment Tools (Environmental and Social Impact Studies, Environmental and Social Management Plan, pest and vector management Plan (PVMP)
- Design and operationalization of Grievances Redress Mechanism.
- Reporting, monitoring and follow-up the implementation of ESMP/ESIA etc;

The safeguard unit at FPMU should plan to provide at least one capacity building trainings every other year for the regions pertinent to environmental and social safeguards management. Following this training, the respective RPMU should support the regional focal person to cascade this training to grass root level to woreda level experts. As an indicative budget, A total of 1,335,000.00 USD is allocated for the implementation of ESMP/PVMP as a whole , as stated in the following table.

Table 2 : Required budget for the implementation of ESMP of the CREW Project

List of activities	Estimated Budget in (US$ '000)					
	1st Year	2nd Year	3rd Year	4th Year	5th Year	Total

ES Screening of sub projects and preparation OF ESMPs. as appropriate		62	56	48		166
Providing ToT to regional focal persons and relevant experts on the implementation of the ESMP and PVMP.	24					24
Cascading the ToT by the regional focal persons down to woreda experts & DAs and support SHFs.	309					309
Refreshment training on ESS and climate resilience for woreda focal persons and DAs , in the 4 regions			12			12
Establish Functional GRM (Grievance Redress Mechanism/ system) for the CREW Project across all the four regions		45				45
Review meeting with key stakeholders including the regulatory (Regional EPAs).		90				90
Monitoring, follow up and support on the implementation of the ESMP/PVMP.		42	76	106	86	310
Annual Environmental and Social Safeguard Compliance Auditing of the CREW interventions, across all regions , by FPCU		62	76	92	148	378
Total	333	301	220	246	234	1335

N.B :The budget source for the implementation of the above activities will be the AfDB.

4.5. Institutional Arrangements

The organizational management of the CREW will facilitate smooth implementation and operation of the project. Besides this, it is also important to manage environmental and social impact of the CREW project. To this end, the key partners for the implementation of the ESMP are indicated as follows:

- During the implementation of the CREW interventions, the FPMU along with the plant protection directorate and the soil health and fertility directorates of the MoA, and the RPMUs will have the primary overall responsibility for the implementation of the ESMP and the PVMP and for ensuring compliance with national legislation and safeguard policies of the AfDB by actively engaging key stakeholders including the community.

- Consultant- The MoA and the PMU can delegate certain tasks to external consultants during the implementation phases of the CREW. Consultants will be needed for some of the specialized monitoring and evaluation activities too.
- Contractors - The contractor (especially the Environmental, Social Health and Safety (ESHS) Unit) will be responsible for avoiding or minimizing, managing and monitoring environmental impacts caused by construction activities. To this end, the requirements associated with the obligation and contribution of the contractors will be integrated with the bid documents.
- The Environmental Protection Authority (EPA) and the line offices in each region from regions, zones and woreda levels are responsible for *enforcing implementation of the environmental policies and legislations through monitoring and compliance auditing of CREW Activities.*
- Financiers/lenders- The CREW Project is likely to be backed by grants and loans from international financial institutions including the AfDB. The proponent and contractor are obliged to comply with the requirement of AfDB policies and operational safeguards throughout the project implementation period. The AfDB will carry out periodic supervision to ensure sound implementation of ESMP and the status of the proponent to compliance with required Bank and standards.

4.6. Grievance Redress Mechanism

Sometimes grievance may happen during project implementation due to construction, compensation, payment modality, pollution, resource use conflict, etc. Good practice in environmental and social performance of projects can be verified by the presence of feedback mechanisms to enable stakeholders to provide input and to make the community being heard for any grievance developed during the project. To materialize this, the project needs to have participatory and culture friendly grievance mechanism. For CREW project a Grievance Redress Mechanism will be established in all regions to ensure that all complaints from local communities are dealt with appropriately, with corrective actions being implemented, and the complainant is informed of the outcome. Concerns will be addressed promptly using an understandable and transparent process that is culturally appropriate and readily acceptable to all segments of affected communities, at no cost and without retribution.

The GRM should be culturally appropriate and, as far as possible, dependable with existing mechanisms both at community level and in the administrative system/contractors. The GRM across all the regions will have the following major objectives:

- provide project-affected peoples (PAPs) with easily accessible procedures for settlement of complaints.
- provide a reliable way for the project to address and resolve disputes.
- contribute to building trust between the project implementers, beneficiaries, PAPs and other stakeholders.
- ensure transparency and accountability throughout the implementation of project
- resolve any emerging environmental and social grievances in project areas.

Grievance Resolution Committee (GRC)

In view of the above, a Grievance Resolution Committee (GRC) shall be set up by the project regions /regional bureau of agriculture at woreda and Kebele levels to inform and coordinate

the relevant stakeholders and to provide resources for resolution of grievances. The Woreda Agriculture bureau or relevant office will be responsible to facilitate and coordinate that grievance related cases are efficiently resolved. The Committee will maintain all records from complaint to final decision for future reference. It will also ensure that public participation and consultation is always central part of the process to promote understanding and prevent unnecessary complaints and disputes. The GRC will be mandated to deal with all types of grievances arising at the community level due to the proposed project activities.

The GRC members shall comprise qualified, experienced, and competent personnel who will be able to interact and gain the trust of the complainants in the local communities. The GRC should consist of both male and female representatives.

The GRC at woreda level will have 5 members, as stated below:

(i) persom assigned by Woreda administrator / Deputy, as Chairperson,
(ii) CREW Woreda Level Focal person (Secretary);
(iii) Woreda expert on social and women affairs (preferably Female); and
(iv) PAPs representatives will be selected by the Woreda the community
(v) Representative from woreda agriculture office (Preferably Female)

The GRC should be able to accept complaints, provide relevant information on the process, discuss the complainants' situations with the concerned person, and explore possible approaches for resolution.

There is a need for clear standard procedure and plan of how to register (how, where, and when) grievance and this GRM shall be disseminated within the project implementation level with focuses to communities at Kebele (Kebele development committee) and Woredas where for concerns/grievances of the program are expected to be presented due its size of the proposed intervention and site specificity of the activities. When affected or concerned persons present their grievance, they expect to be heard and taken seriously so woreda and Kebele levels project stakeholders, particularly kebele Development agent is required to inform the project GRM system and provide adequate information. To facilitate and manage GRM, a very clear and summarized GRM manual will be prepared by the Safegaur officer at the FPCU.

The GRM Process

The grievance resolution process includes four main stages. They are the stage to receive the grievance, to investigate/ review the case, to respond and resolve it and finally to close out the case and report it. The output for the grievance redress mechanism has to be reported to the regional safeguards focal person periodically.

At community level any person who has complaints regarding the CREW Project activities can raise these issues with the Kebele-level contact point (Focal Person (FP), normally the Kebele manager). All received complaints and responses will be documented and copies sent to the Kebele administration and Woreda FP. If the issue cannot be resolved by informal discussion between the project and the complainant, the complaint can be brought to the

Kebele Development Committee (KDC). The complaint and decisions made will be documented and copies sent to the Kebele administration and Woreda FP. Complaints unresolved at Kebele level will be brought to the Woreda Office of agriculture for review, before being sent to the Woreda steering committee for investigation and final decision. At any point, an applicant who is not satisfied by the decision of the committee has the right to apply to the court system.

5. Gender, youth & social inclusion

In the Ethiopia Climate Resilient Wheat Value Chain Development Project, a strong commitment to gender equality, youth empowerment, and social inclusion is evident across all components. Component 1 focuses on enhancing the productivity and production of climate-smart wheat while ensuring the inclusion of women, youths, and vulnerable groups.

Under Sub-component 1.1, efforts will be made to facilitate access to quality seeds for small-scale farmers. Specifically, sub-activity 1.1.1 involves the distribution of small packs of seeds to women and vulnerable groups, such as the elderly and persons with disabilities (PWDs), to encourage their adoption of improved varieties. Additionally, awareness creation sessions will be conducted to educate women farmers on health and safety measures, as well as the proper management of pest control chemicals.

In Sub-component 1.3, the rehabilitation and expansion of irrigation infrastructure are undertaken with a focus on inclusivity. Activities such as training young women on water pump maintenance and establishing youth-led cooperatives for irrigation system operation ensure the involvement of women, youths, and vulnerable community groups.

Similarly, Sub-component 1.4 promotes sustainable farm mechanization while prioritizing the inclusion of marginalized groups. Youth groups will be organized for agricultural mechanization services, and accessible mechanization service centers will be established to reach women, youths, and vulnerable community members, including PWDs and the elderly. Furthermore, flour mills or bakery equipment will be provided to organized women and youth groups for value addition activities.

Under Sub-component 1.5, extension and climate advisory services will be tailored to the needs of women farmers, with activities such as providing technical support and organizing field demonstrations of new wheat varieties with a focus on women's participation.

Moving on to Component 2, which focuses on market infrastructure, linkages, and agri-finance development, efforts will be made to ensure the inclusion of women, youths, and vulnerable community groups. Activities under Sub-component 2.1 include providing access to agribusiness facilities and flour mills for women, youths, and vulnerable groups engaged in wheat processing.

In Sub-component 2.2, market linkage interventions will be designed to address the specific needs and challenges faced by women, youths, and people with disabilities in the wheat value chain. Training modules on value addition techniques and business management will be developed, and a needs assessment will be conducted to inform targeted interventions.

Finally, Component 3 emphasizes project coordination and management with a focus on gender, youth, and social issues inclusion. Activities under Sub-component 3.1 include capacity-building training on mainstreaming gender and social inclusion, conducting community conversations as gender transformative methodologies, and developing guidelines for gender mainstreaming and social inclusion at federal and regional levels.

Overall, the Ethiopia Climate Resilient Wheat Value Chain Development Project prioritizes the inclusion of women, youths, and vulnerable groups throughout its implementation, ensuring that no one is left behind in the journey towards a resilient and sustainable wheat value chain.

Table 13 Women youth & vulnerable group inclusive interventions

Activities	Unit	Program Life	Unit Cost (UA)'000	Physical Plan (quantity) in Years of Implementation					
				2024	2025	2026	2027	2028	Total quantity
Component 1: Women, youths & vulnerable group inclusive Climate Smart Wheat Productivity and Production									

Activities	Unit	Program Life	Unit Cost (UA)'000	Physical Plan (quantity) in Years of Implementation					
				2024	2025	2026	2027	2028	Total quantity
Sub-component 1.1: Facilitate Access to Seeds for women, youths & vulnerable group Small-scale Farmers									
sub activity 1.1.1: Distribute small packs of seeds to women & vulnerable /Elderly, PWD/ farmers and seed producers to encourage adoption.	kilogram	7800 HH		800.00	1,000.00	2,000.00	2,000.00	2,000.00	
sub-Activity 1.1.1: conduct awareness creation sessions for women farmers on health & safety measures and management of pest control chemicals	78 woredas			1.00	1.00	1.00	1.00	1.00	
Sub-component 1.3: Women, youth & vulnerable group inclusive Rehabilitation and Expanding Irrigation Infrastructure development									
Activity 1.3.1 : Training young women on maintenance and repairing water pumps and maintenance skills	7800/	100 per each woreda		800.00	1,000.00	2,000.00	2,000.00	2,000.00	
Activity 1.3.2 : Establish & support youth-led cooperatives or enterprises on the maintenance and operation of irrigation systems	78 cooperatives	1 each per woreda		8.00	10.00	20.00	20.00	20.00	

Activities	Unit	Program Life	Unit Cost (UA)'000	Physical Plan (quantity) in Years of Implementation					
				2024	2025	2026	2027	2028	Total quantity
Sub-component 1.4: Women, youth & vulnerable community group inclusive Sustainable Farm Mechanization									
Activity 1.4.1 . Organizing youth groups for agricultural mechanization service provision and enhancing quality farm machinery maintenance training in agricultural technical and vocational education and training (ATVET) colleges	4 regions	78 WOREDA		8.00	10.00	20.00	20.00	20.00	
activity 1.4.2 establishing accessible/location, hours of operation, and outreach strategies to reach marginalized groups/ mechanization service centers for women, youths & vulnerable community group / PWD, Elderly/...	1.0	1		1.00	1.00	1.00	1.00	1.00	
Activty 1: Providing flour mills or bakery equipment /technology for organized women & youths group /cooperative necessary for value addition activities, .	78 groups	1 per each woreda		8.00	10.00	20.00	20.00	20.00	
Sub-component 1.5: Extension and Climate Advisory Services									

Activities	Unit	Program Life	Unit Cost (UA)'000	Physical Plan (quantity) in Years of Implementation					
				2024	2025	2026	2027	2028	Total quantity
Activity 1. 5.1: Provide technical support tailored to the needs of women farmers in rainfed and irrigated wheat production.	1.0			1.00	1.00	1.00	1.00	1.00	
Activity 1.5. 2 Organize field demonstrations of newly released wheat varieties, with a focus on involving women farmers	1.0			1.00	1.00	1.00	1.00	1.00	
Component 2: Women, youths & vulnerable community group inclusive Market Infrastructure, Linkages and Agri-finance development									
Sub-component 2.1: Market Infrastructure Component									
Activity1: Provide access to agribusiness facilities such as processing units, storage facilities, and marketing support for women, youths & vulnerable group (person with disabilities, elderly, people living with HIV/AIDS….	50 % women & 30 % of youths share from infrastructure			50 % women & 30 % of youths	50 % women & 30 % of youths	50 % women & 30 % of youths	50 % women & 30 % of youths	50 % women & 30 % of youths	
Activity 2: provide flour mills or bakery equipment for women & youths common interst group/cooperatives	78 woredas	1 per each woreda		8.00	10.00	20.00	20.00	20.00	
Sub-component 2:2: Market									

Activities	Unit	Program Life	Unit Cost (UA)'000	Physical Plan (quantity) in Years of Implementation					
				2024	2025	2026	2027	2028	Total quantity
Linkage Interventions									
Activity1: Develop training modules on value addition techniques and business management for women, youth, and people with disabilities engaged in wheat processing.	Doc				1.00				
Activity 2: Conduct a needs assessment and market analysis to identify the specific requirements and challenges faced by women, youth, and people with disabilities in the wheat value chain.	Doc	4 regions			1.00	1.00	1.00	1.00	
Sub-component 2.3: women, youth & social inclusive Access to Agri-Finance									
Activity1: Organize training sessions on financial literacy and entrepreneurship skills for women, youth, and people with disabilities to equip them with necessary knowledge for engaging in commercial wheat production	7,800.0	100 per each woreda		800.00	1,000.00	2,000.00	2,000.00	2,000.00	
Component 3: Project Coordination and Management									

Activities	Unit	Program Life	Unit Cost (UA)'000	Physical Plan (quantity) in Years of Implementation					
				2024	2025	2026	2027	2028	Total quantity
Sub-component 3.1: women, youth & social issues inclusive Monitoring, Evaluation and Learning									
Activity3.1: Supervision and follow up									
Sub Activities 3.1.1: Conduct supportive supervision on the participation and benefits of women, youth & vulnerable community group / Elderly, person with disability etc…) in the project implementation regions	per each quarter			4.00	4.00	4.00	4.00		
Activity 2: Capacity Building training on Mainstreaming Gender, youths & vulnerable group Inclusion in CREW project			250.00	50.00	50.00	50.00	100.00		
Activity 2: Providing Capacity Building Training for Regional and Woreda Gender Focal Persons of the Project Implementation area		50 particiapnt		50 /30 M & 20 F					
Activity 4. Conduct community conversation (CC) as Gender transformative methodologies in the Project Implementation area	78 woreda	7800 participants		800.00	1,000.00	2,000.00	2,000.00		

Activities	Unit	Program Life	Unit Cost (UA)'000	Physical Plan (quantity) in Years of Implementation					
				2024	2025	2026	2027	2028	Total quantity
Sub-component 3.2: Project Management									
Activity 3.1: Preparing CREW Gender mainstreaming & social inclusion Guideline and familiarizing and cascading the guideline at all levels	Doc								
3.1 at federal level	50 / 30 male & 20 female			50.00					
3.2 at regional level									
sub activity 4.1: Develop community conversation (CC) manuals.	doc			1.00					
sub activity 4.2 Conduct ToT for facilitators at regional levels	50 participant per region				200.00				
sub activity 4.3 Conduct community conversation (CC) in selected kebeles (community levels) in project implementation areas	7800	100 participants per each woreda		800	1000	2000	2000	2000	

6. Financial Management

Ethiopian Climate Resilient Wheat Value Chain development project is financed by African Development Bank, Government of Ethiopia to contribute to enhance the production and productivity thereby increasing the incomes of small-scale wheat produces farmers in Ethiopia

to wards Enhance the production and productivity of wheat crop across agro ecology enhance soil productivity across the intervention areas through applying integrated soil fertility

management ensure the women living standard and Job creation for youth all across the intervention woredas of the country To this effect, this short and concise financial management manual is prepared to highlight how financial transitions of the project are recorded and reported by each implementing entity (FPCU, RPCU,EIAR and ATI) from the initial identification of transactions to the final financial reporting. Detailed procedures of fund flow, disbursement, accounts recording, preparing financial report and auditing arrangements are described on the Financial guide line attached an as annex to this PIM.

The objectives of this manual are:

1. To familiarize the accounting policies and standards to be applied.
2. To familiarize the financial function and recording procedures.
3. To enable how to maintain and use financial records and documents.
4. To describe the major duties and responsibilities of the federal and Regional PCU/Project Accountants in the operation of the project.

6.1. Accounting policies and standards

Accounting policies and standards specify how financial transactions are recognized, recorded and summarized. The project uses Interim financial reports (IFRs) as a means of justification of expenditures. In addition to these the project uses modified cash basis of accounting throughout the fiscal period starting from July 8 to July 7 of the subsequent year.

6.1.1. Duties and Responsibilities of Federal Project Accountants

The Project Accounts Division function all financial matters of the project with the direction of the CREW Project Coordination Unit performs the following major duties and responsibilities.

- Opens special USD and local currency bank accounts and notify in writing to the Bank (ADF) the name and account No. of the bank accounts, name and address of the Bank and the name of the authorized signatories with their specimen signatures as soon as the accounts are opened.
- Prepares and maintains accounting records of all project financial flows in conformity to government and Bank's rules and procedures.
- Certifies that disbursements are made in accordance with the government and Bank financial regulations, rules, practices and procedures.
- Effects payments for goods and services purchased for the project and transfers cash to regional CREW bank accounts based on their work plan and request.
- Ensures that payments are for eligible expenditures related with the project.
- Ensures that government taxes are not included for goods, services and works provided in the ADF ,TSF expenditure.

- Prepares disbursement application for direct payments to the third parties for goods, services and works and insure that government taxes are not included in the payments requested.
- Prepares disbursement application for replenishment to special USD bank account and compile SOE (statement of expenditure).
- Receives and verifies expenditure documents from petty cashiers, purchasers
- Prepare Quarterly financial report and ensure the report include in progress report Prepare annual financial statements to the appointed external auditors and the bank
- Makes available all financial documents and records and give adequate explanation to external auditors.
- Provides periodical reports to financer's and concerned authorities.
- Provides training for the regional CREW accountants on the government and Bank financial regulations, rules, practices and procedures.
- Prepared Audit finding action plan
- Assists the regional CREW accountants on financial matters.
- Prepare monthly cash flow statement for approved government matching fund.

6.1.2. Duties and Responsibilities of Regional CREW Project Accounts

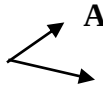
- Opens local currency bank accounts in the nearest Commercial Bank of Ethiopia and notify in writing to federal CREW /PCU the name, account No. of the bank account, bank branch and the authorized signatories as soon as the bank accounts are opened.
- Involve in the planning and budgeting of the project activities.
- Utilize the project fund for planned activities.
- Preparing receipt voucher of funds transferred to their bank accounts from federal Project Fund .
- Prepared and Submits consolidating Statement of expenditure (SOE),RIFR and monthly bank Reconciliation to federal project unit on quarterly basis after 15 days of the month end.
- Acknowledges in writing the receipt of goods and materials or copy of GRN (Goods Receiving Note) issued by federal property Administration Division.
- Give advance (record as receivable of the woreda) for woredas BOFED as per the budget allocated for each component and category and settled the advance within 30 days.
- Effects payments for eligible expenditures and insure that government taxes are not included for the payment of goods, service operating and works in the ADF,TSF expenditures.
- Prepare annual financial statements to the appointed external auditors and FPUC
- Take corrective action based on audit finding action plan and acknowledges in written form based on given time.
- Involve in the planning and budgeting of the project activities.
- Utilize the project fund for planned activities.

- Prepares and maintains separately accounting records and Budget utilization report for all doners based of all project financial flows in conformity to government and Bank's rules and procedures.

6.2. Budget

The Budget of the project is analyzed by category and components as per the grant agreement. No reallocation of budgets allowed without prior approval of the financer in writing, but well-justified proposals may be put forward at the annual work plan and budget preparation stage.

6.2.1. Fund and Financial Report Flow



CREW Resources flow from ADB ⟶ FP CU ⟶ RPCU through the existing arrangement of bank agreement. Under this arrangement each unit will control the release of fund and report the utilization of fund in the form of Statement of expenditure.

When the project implemented institution justify 50% of the first disbursement they submit a request for funds to FPCU by official letter supported by using the appropriate format. After insuring that the funds request is in line with the approved budget, the FPCU authorized the transfer.

6.2.2. Disbursements Operation

All disbursement of funds for Operating, goods, services and works should be made in accordance with the agreement entered with the Bank for eligible expenditures. Disbursements are used from grant account by the following methods.

1. Special Account method
2. Direct payment method

The minimum level for each disbursement request to be submitted to AfDB for direct payment is UA20000.

6.2.3. Initial Advance

Upon effectiveness of the grant fund disbursement application is submitted to the Bank for initial advance. (See the detail on the Financial Management Guidelines)

6.2.4. Replenishment to Special Account

Replenishments are requested when the previous advance is justified up to 50% of the funds advanced to special account (See the detail on the Financial Management Guidelines).

6.2.5. Direct Payments

Direct Payments are payments to be made by the bank on behalf of the project from the grant fund to third parties for goods, services, Operating and works expenditure attaching all the necessary document along the payment request depending on nature of category and degree of task performed (See the detail on the attached Financial Management Guidelines as an annex).

6.2.6. Bank and Petty Cash Operation

Depending upon the type, nature & magnitude of the payment, disbursements made by cash, check and bank transfer. Regarding petty cash when 1/4th of the established petty cash remain it will be replenished and the payment from petty cash float is up to birr 5000 . The establish petty cash amount decide by FPUC.

6.2.7. Bank Account Operation and Check Payment Control

The project maintains a separate special USD and Local Currency bank account for operating the grant funds at the National Bank of Ethiopia. Drawing Cash from these banks under take either by check or transfer supported by the source document that justifies the eligibility of payment and all other routines payment related to goods, service, operating and work are discussed in detail on the financial management guideline. For data entry it is recommended to use Peachtree accounting software and IFMIS . The conversion rates related to grant are used to convert non- Birr Currencies. These rates are the cash buying rates prevailing at the time of bank transfer and will be adjusted at the end of the fiscal period by the buying rates of the bank and gain or loss on holding realized.

6.2.8. Chart of Accounts

The chart of account of CREW is designed according to the Grant agreement entered with the bank and the government of Ethiopia. The accounts include Components, Category, and Location in line to the main account. The main accounts included in the chart of accounts are Cash at Bank and on Hand, Receivables, and advance to regions, Staff Debtors, Sundry, Creditors, Accumulated Expenditure/Fund Balance, Revenue and Expenditure. In connection to these the project should maintain summary ledger, category ledger and contract ledger. Matching Fund from government as Subsidiary accounts can be opened (maintained) by the source of revenue (Grant).

6.2.9. Fixed Asset Register

Fixed Assets have a useful life of longer than one year and should be charged to expense up on purchase and registered separately on Fixed Asset Register. These assets are physically checked periodically and physically verified against records annually. When goods are procured for the project, Goods receiving note (GRN) should be issued. On the other hand, when goods are dispatched Goods issue voucher must be Prepared and original of the GRN vouchers should be submitted to Project finance section and Copy of issue voucher should be submitted to store keeper (recorder)

6.2.10. Financial report and Annual Audit

FPUC, RPUC ,EIAR and ATI Project Management are participating in order to control financial resources. ADF require monitoring based on IFRs. Periodic report shall be submitted on time to the project management and ADf and co –financers.

That includes.

- Funds received from government and utilization during the month and Quarter .
- Funds received from the Bank including direct payments
- Funds utilization report
- Sources and Expenditure report

Every fiscal year the federal project coordinating unit, regional project coordinating units, EIAR and ATT should close their accounts and prepared financial statements so as to make ready for external audit. the project's book of accounts should be audited for each fiscal period by a recognized audit firm appointed by the project and acceptable by the bank.

The draft audit findings should be discussed with all concerned officials. After making the necessary corrections giving explanations the final audit report is produced by the auditors. The final audit report is to reach to the Bank no later than six months after the close of the financial year.

6.3. Capacity building

Climate Resilient Wheat Value Chain Development Project is managing huge resource of fund Grant from ADF for the 4 regions (Oromiya, Amara, Afar and Somale) , EIAR and ATI, The ADF, TSF and others donors as financiers give fund so as to enhance the production and productivity there by increasing the incomes of small-scale wheat producers farmers in Ethiopia. To implement a uniform financial management system across the units at Federal and Regional level, the FPUC should have training to this effect the following capacity building should be undertaken.

1. Accounting Software such as IFMIS ,Peachtree.
2. Project accounting recording and reporting
3. Preparing of Planning and bugting
4. Chart of account

NB: Refer financial management guideline for the detail of the PIM.

7. Procurement arrangements, Federal and regional responsibilities

This Procurement Manual describes the agreed procurement rules and procedures for the effective and efficient implementation of the AfDB Grant to MOA for the ETHIOPIA – Climate Resilient Wheat Value Chain Development Project (CREW), at federal, regional, Woreda and community levels. It provides step-by-step guidance to the procuring entity for the procurement of goods, works and non- consulting services, and presents procedures for selecting consulting firms, contracting and monitoring consultants required for the project.

The Procurement Manual captures the latest development and changes in the procurement environment, and addresses practical procurement issues at all levels. It is a living document, and hence, it is subject to any necessary changes in the future; however, the overall structure of the manual shall be maintained as far as possible. The need to procure the items identified in the program and the use of the funds allocated for the project in an efficient, transparent and fair manner consistent with the basic public procurement principles which agreed with African Development procurement procedures so as to achieve the Project Development Objectives.

7.1. Procurement Institutional Arrangement

At Federal level, the procurement case team which falls under Procurement and Property Administration Directorate within the MOA is the unit which is responsible for the execution of procurement activities of projects financed by the African Development Bank,

CREW procurement at the regional level shall be carried out by each of the procurement unit of the following implementing institutions:

1. Bureau of Agriculture & Natural Resource (BoANR) in Afar region.
2. Bureau of Agriculture (BoA) in Amhara region
3. Bureau of Agriculture (BoA) in Oromia Region
4. Bureau of Livestock and Pastoralist Development (BoLPD) in Somali region

Procurement under CREW at the woreda level shall be carried out through a pooled procurement system by WOFEDs with the necessary support from each respective Region.

Proper organization of the procurement function is critical to achieve procurement objectives. In general, the procurement function should at least involve the Procurement Unit /Procurement Officer, the Bid Evaluation Committee, procurement endorsing committee, the Head of the agency, and an ad hoc technical evaluation committee. Procurement Unit is at the forefront of the procurement process in each implementing agency and its proper organization and functioning is critical for the achievement of procurement and project objectives. Hence, it is imperative to have a commensurate and well-functioning procurement unit with sufficient number of skilled staff/expert, working facilities, communication and other office facilities.

The Heads of the Project Coordination Units at all levels of the CREW are responsible for ensuring that all procurement staffs are trained and are aware of the procurement requirements and procedures applicable in the project

7.2. Actors, Roles and Responsibilities

7.2.1. The National Project Coordination Unit at MoA

NPCU at MoA is responsible for overall CREW coordination and management, and will provide technical support to each actor under the project. It also liaises with other CREW directorates and sections responsible for the project pillars and sub-components to ensure efficient planning and implementation.

The followings are major responsibilities related to procurement under the project.

- Coordinate and oversee the program to ensure that it meets the objective of CREW;
- Coordinate the procurement activities and ensure that all goods and services are procured according to the procurement plan and in accordance with the provisions in the FA;
- Ensure that the project implementing institutions have the required capacity in procurement to fulfil their mandate and provide the necessary procurement guidance and assistance;
- Establish internal control system to check the quality and legality of each procurement process at each stage

7.2.2. Procurement and Property Directorate at MoA

The Procurement and property administration Executive at MOA will play a central role in the coordination of procurement and property administration case teams to conduct procurement activities, management and distribution of goods to the beneficiaries.

The procurement team at MOA has the following major responsibilities

- Check and approve SBD, RFP, ITB, BER and contract agreements.
- Preparation and consolidation of the Project Procurement Plan, and its updating when required.
- Consolidating of periodic procurement and progress reports of the executing agency.

- Publication of General Procurement Notice (GPN) and Specific Procurement Notices (SPN) on the Bank's Web site and in national newspapers of wider circulation;
- Follow-up on the status of all procurement processes and performance of procurement officers and appropriate action, as necessary;
- Evaluation of proposals through Evaluation Committee, negotiations and award of contracts;
- Notification of results of tenders to all successful and unsuccessful bidders;
- Assurance for sound contract administration of international contracts, including termination, contractual claims and arbitration;
- Monitoring of all procurement activities by implementing agencies to assure transparency and efficiency in implementation;
- Handling of procurement-related correspondence with pertinent authorities;
- Clearing of all imported goods of destination; arranging for transportation and dispatch of items to their final destinations;
- Inspection and ensuring of receipt of imported goods to store and follow up on claim settlements;
- Handle all international competitive bidding or international shopping starting from the preparation of bid document up to receiving the goods or services

7.2.3. The Federal Project Procurement Officer

The Federal Project Procurement Officer carries out procurement process on the basis of competitive bidding and ensures that the items meet the basic procurement principles. Shopping procurement method would be used for the procurement goods and works. The procurement officer ensures that the procurement process is carried out in accordance with the African Development Bank Procedures.

- Prepare and consolidate the PP, and update the PP as required and approval to AfDB.
- Prepare General Procurement Notice and forward it to the AfDB for publication in United Nations Development Business (UNDB).
- Prepare bid documents using AfDB's/Public Procurement Agency standard bidding documents and SRFPs providing correct, complete and unambiguous specifications for common user items and preparation of TORs for procurement of Goods and consultancy Services financed by AfDB.

- Submit the bid documents and RFPs to the AfDB for obtaining No Objection and make such modifications in the said documents as the AfDB may recommend.
- In case of International Competitive Bidding (ICB) procurements, prepare "Invitation for Bid (IFB)" and forward to the African Development Bank for publication on the Bank's internet web site to publish and also on the wider circulated newspaper as required.
- Prepare Bidding Documents, Expressions of Interest, RFPs and RFQ, to be sold / issued to prospective bidders.
- If required, provide clarification to queries regarding bid documents raised by prospective bidders
- Attend the opening of bids at stipulated time and place. Record bids as read out. Prepare minutes of Bid Opening and evaluation and Agreed Minutes of Contract Negotiations and Negotiated Contract
- Whenever requested, assist the evaluation committee in the evaluation of bids and provide clarifications.
- Confirm the evaluation report is done in accordance with the African Development Bank's guide lines and submit the evaluation report to the AfDB for "No Objection" in case the contract is undertaken by bank prior review.
- Prepare awards and contracts in conformity with the requirements of the AfDB.
- Maintenance of records of procurement documentation, deliveries and payments in a systematic manner to make them available to bank post reviews;
- Supports and periodically supervises the work of the Regional Procurement Officer.
- Ensures that all procurements are done on a competitive bidding basis to ensure value for money.
- Follows-up on the collections, recording, safeguards and issuance of Bid and Performance Securities/ Bonds submitted by Bidders/ awarded supplier/ contractor including the validity period for extension.
- Closely follow up Bank and Insurance cases and Custom clearance processes for imports.
- Maintains database of suppliers, prices, technical specifications, catalogues, drawings and other information.

- Assumes responsibility for being alert to any indication of unethical behaviour and reporting to immediate supervisor.
- Timely addresses comments provided by External Auditors on procurement related matters.
- Coordinates the provision of procurement training workshops at regional and woreda levels.
- Prepares reports on the status and progress of procurement implementations and consolidates regional procurement performance reports.
- Follows up closely procurement documents and cases submitted to AfDB and ensure timely responses for comments provided and final no-objections.

7.2.4. Procurement Endorsing Committee

The Procurement endorsing Committee (PEC) is one of the most important procurement setups in that it plays a key role in the procurement function by acting as an independent reviewer and to check on the quality of the procurement exercise.

Taking maximum care in selecting members of the PEC based on merit is imperative to ensure that procurement implementation benefits from a well-functioning internal control system. In addition to skill and experience, members of the PEC should have the highest level of integrity.

While it is important to note that every procurement decision should be made based on sufficient technical and market information, the PEC should be kept independent and no interference in its decision should be sought or allowed unless it is justified to do so.

The project coordination unit project coordinator should also ensure that all members of the PEC are aware of, and trained in the agreed procurement rules and procedure of the programme.

The main tasks of the PEC are the following:

- Ensure that proper PP is prepared and approved at different levels
- Review and check the completeness and comprehensiveness of the SPNs, BDs, and RFPs.

- Check and ensure that the procurement request is included in the PP,
- Check for the timely advertisement of the bid,
- Ensure that bids are opened promptly after closing,
- During bid opening, ensure that the required information (name of the bidder, bid amount and currency, bid security amount and discount if any) are read out in public; and those records of bid opening are prepared and signed;
- Ensure that quality BER/TER/CTFER is prepared and the least evaluated bidder is selected in accordance with the requirements stipulated in the BDs/RFPs,
- Endorse the BERs/SLRs/TERs/CTFERs and submit them to the project coordination unit project coordinator for further action.

7.2.5. Federal Project Coordination Unit (FPCU)

Federal Project Coordination Unit (PCU) plays vital role in facilitating proper execution of procurement activities. The head of the project coordination unit has to keep aware of the process, procedures, agreed rules of procurement and should actively lead procurement implementations and ensure the following major responsibilities:

7.2.6. Heads of Project Implementing Institutions/Head of Bureau

- The Head of the project implementing Coordinate the procurement activities and ensure that all goods and services are procured according to the procurement plan and in accordance with the provisions in the Financial Agreement (FA);
- Prepare and consolidate the PP, and update the PP as required in parallel to periodic progress/performance reports in collaboration with the procurement case team;
- Ensure that the project implementing institutions have the required capacity to manage procurement activities in adherence to AfDB' agreed procurement rules and procedures.
- Review and provide feedback on procurement plans, draft SBD, draft contracts and bid evaluation reports submitted by regional project coordination units.
- Provide technical support to regional coordination units on items to be procured under different components and sub-components of the CREW and even to procure items when requested specifically or when purchases are not practical at their level.

- Consolidate periodic CREW procurement progress reports of the implementing agencies.
- Support, monitor and supervise procurement activities at all levels compliance to the AFDB procurement guidelines.
- Facilitate getting No Objection from AfDB for the bid evaluation reports and other documents which are undertaken by AfDB's prior review.
- All procurement staff is trained and is aware of the agreed procurement requirements and procedures applicable in the project.
- All staff should receive the copy of the Project Implementation Manual (PIM), procurement manual and other legal documents of the project.
- The necessary system is established to effectively track procurement implementation
- Continuously track and address procurement capacity problems in the agency and avail training to fill the identified gaps.
- An adequate control and safe keeping of procurement documents and a well-functioning system has been established to track the access and movement of the documents.
- Actively follow and oversee procurement implementation and should also organize the procurement function with clear segregation of duties and responsibilities among different players.

institution has a vital role in setting up a properly functioning procurement system including the Procurement staff and Bid Endorsing Committee. The Head of the project implementing institution has to keep him/herself aware of the process, procedures and agreed rules of procurement and should actively lead procurement implementation

The Heads of project implementing institutions at all levels ultimately have the major responsibility of ensuring that:

- The procurement is handled based on the agreed procurement rules and procedures
- All procurement staffs are trained and are aware of the agreed procurement requirements and procedures applicable in the project
- All staff should receive the copy of the Project Implementation Manual (PIM), procurement manual and other legal documents of the project including the AfDB and GoE system procurement guidelines and SBDs

- The necessary system is established to effectively track procurement implementation
- Proper reporting mechanism is set up so that any pitfall can be traced and corrected in time
- Continuously track and address procurement capacity problems in the agency;
- An adequate control and safe keeping of procurement documents and a well-functioning system has been established to track the access and movement of the documents.
- An internal control system is established to check the quality and legality of each procurement process at each stage.

A reliable internal control and quality assurance mechanisms are very critical to achieve the intended procurement and project objectives. In this respect, Head of the project implementing institutions are responsible to ensure the establishment and proper functioning of internal control and quality assurance system within the procurement environment.

The Heads of project implementing institutions at all levels are particularly responsible to actively follow and oversight procurement implementation and should also organize the procurement function with clear segregation of duties and responsibilities among different players. In addition to this, procurement implementation is over sighted by:

The AfDB through prior review of contracts above defined thresholds and post procurement reviews:

- Internal approval process –approval of evaluation reports, who is PIU accountable to signing of contracts etc. and internal audit, and.
- External oversight – FPPAA's role

7.2.7. Regional Project Coordination Units.

Regional Project Coordination Units (PCUs) play vital role in facilitating proper execution of procurement activities. The RPCU is responsible for preparing a procurement plan based on the approved work plan, procure works and goods through national competitive bidding, local shopping, and direct purchase and force account and submit progress reports to the FPCU. In addition, they will support and supervise procurement activities that would be executed by Woredas and Communities. Procurement for all contracts at Regional levels procured through

NCB and Shopping procedures would be carried out in accordance with federal/regional procurement proclamations and using national SBDs acceptable to the Bank. Thus, the head of the project coordination unit has to keep aware of the process, procedures, agreed rules of procurement and should actively lead procurement implementations and ensure the following major responsibilities:

- Prepare the yearly procurement plan and related budget and submit for review and approval at CREW;
- Consolidate the yearly procurement plan and quarterly, semi-annual and annual procurement progress/performance reports of the respective Woredas and Communities and submit them to CREW for review and consolidation;
- Evaluate bid and service proposals through bid evaluation committees and subsequent approval by the head of the Implementing Agency and negotiation and awarding of contracts in line with the procurement procedures set in this Procurement Manual and approved in the procurement plan;
- Notify the result of the bid to all successful and unsuccessful bidders;
- Follow-up the status of all procurement processes and performance of procurement officers and take appropriate action;
- Provide the required information and follow-up the status of all procurement processes for goods and services;
- Develop and implement procurement monitoring procedures;
- Assume responsibility for being alert to any indication of unethical behaviour and report such behaviour to the appropriate body;
- Provide training and support to procurement staff of the respective Woredas as required;
- Submit purchase requests including the technical specification or TOR to the CREW where goods and services are not available at the regional level; and

7.2.8. Regional Procurement Officers

The Regional Procurement Officer carries out procurement process on the basis of competitive bidding and ensures that the items meet the basic procurement principles. NCB and Shopping procurement methods would be used for the procurement goods and works. The procurement officer ensures that the procurement process is carried out in accordance with regional

procurement proclamations and using national SBDs acceptable to the Bank. She/he shall assist Woredas in procurement and liaise with the Woreda Procurement Officers to ensure that goods and services are procured at a competitive price and in a timely manner. Moreover, s/he shall ensure that procurement, with the proceeds of the project, shall be handled separately from other procurement made with the regular government budget and that reporting of the program-financed procurement shall also be prepared separately. The following are major responsibilities of the Regional Procurement Officer.

- Prepare the yearly procurement plan and related budget and submit it for review and approval to BOA Procurement and Property Administration Directorate.
- Collects pro-forma invoices from at least three bidders for procurement using the Shopping method;
- Prepare bid documents for goods using the Bank/federal SBDs;
- Procure goods, works and services in accordance with the AfDB Procurement Guidelines and procedures stated in this Procurement Manual and deliver to the concerned beneficiaries on time.
- Ensure that goods and services received match the cost and specifications in the pro-forma invoices.
- Maintain lists of suppliers and procurement files for goods, works and services.
- Work with the pertinent authority/person to follow up claim settlements and ensure that appropriate payments are effected on time;
- Assume responsibility for being alert to any indication of unethical behaviour and report such behaviour to his/her immediate supervisor;
- Prepare periodic procurement reports as needed;
- Provide training and support to Woreda procurement staff; and;
- Maintain documents, records, files and registers in a systematic manner and keep in safe custody;

7.2.9. Procurement Officers at Woreda FEDO

The procurement officers at Woreda Finance and Economic Development Office will work in collaboration with their respective Regional Bureau of implementing Agencies Procurement Officers to prepare and consolidate annual procurement plans and periodic updates and reports. The Procurement Officer facilitates procurement requests on the basis of competitive

bidding procedures and ensures that the items meet the cost, quality and specifications requirements of the requesting office using Shopping procedures. The procurement officers shall ensure that procurement with the proceeds of CREW shall be handled as per the agreed procedures and that reporting of program financed procurement shall be made separately.

Major duties and responsibilities of the Procurement Officers at the Woreda levels include:

- Prepare the annual procurement plan in line with their allocated budget in collaboration with the Woreda implementing agencies/offices and submit it for review and approval to the pertinent Regional Bureau of implementing partners.
- Collect pro-forma invoices on the basis of purchase requests forwarded form the Woreda implementing Agencies/Offices from at least three bidders for procurement of goods using Shopping method of procurement;
- Procure goods and other related services on time and deliver them to the concerned beneficiary in an effective manner using the shopping procedures outlined in this Procurement Manual;
- Ensure that goods and other related services received match the cost and specifications in the pro-forma invoices;
- Work with the pertinent authority/person to follows up claim settlements and ensure that appropriate payments are affected on time;
- Assume responsibility for being alert to any indication of unethical behaviour and report such behaviour to his/her immediate supervisor;
- Prepare periodic procurement reports as needed;
- Work closely and harmoniously with the Woreda implementing partners/Agencies/Offices; and

7.2.10. Technical Evaluation Committee (TEC)

Technical and financial proposals, bids and RFPs (that require technical input), should be evaluated by a team of experts with sufficient background and technical knowledge of the items to be procured and the consultancy services required. Hence, TEC members should be formed on an ad hoc basis for each of the procurement packages prior to the bid/proposal opening.

The project coordination unit project coordinator shall select the committee members in consultation with the user agency or department and notify the selected members through official letter ahead of time.

All members of the TEC /Technical person should have sufficient technical background on the procurement items in question, and possess a high level of personal integrity. At least one member should be from the user agency or department.

In case where the project implementation institution does not have staff with the required level of expertise for the equipment or the consultancy service to be procured, it may seek the assistance of other government agencies or NGOs for their technical input.

It is very important to establish a good relationship between the procurement function and the finance and store services. Inspection of the procured items to ensure the quantity and quality of the procured equipment is required and an inspection report is also mandatory for the procured good when they are delivered to the stores. The project coordination unit project coordinator should ensure that there is efficient processing of activities, proper integration and smooth flow of information among procurement, finance and store as well as the existence of effective check and balances.

7.2.11. Internal Controls and Quality Assurance

Reliable internal control and quality assurance mechanisms are critical to achieve the intended procurement and project objectives. The responsibility for the establishment, timely audit and definition of duties of these systems lies with the project coordination unit project coordinator.

Such systems can be achieved, inter alia, through the enhancement and active involvement of all the stakeholders including user agencies/departments, committee members, internal auditors, and technical group or experts at the appropriate stage in the procurement process.

The role of user departments or agencies should be enhanced through the clear identification of their requirements, the preparation of clear technical specification (goods and works) and through the preparation of TORs for the selection of consultants. It should be clearly specified when the goods/services are needed to initiate the procurement stage, and user departments or agencies should be involved in the technical evaluation process, the administering of signed contracts and the inspection of the delivered goods and services to confirm the quality of procured goods.

Well-functioning Bid Evaluation Committee and Technical Evaluation Committees should act as independent reviewers of procurement outcomes, and as the main arm of internal control within the procurement function. It is important that committees can rely on the requisite skills and qualifications to ensure consistent engagement in the internal audit and procurement process.

In conclusion, the project at all levels of CREW shall have core procurement quality assurance responsibilities, and is ultimately accountable to ensure the quality of the procurement process as a whole, including documentation of the services and products delivered.

7.3. Items to procured Under CREW

7.3.1. Procurement of Works

Works procured under CREW includes construction of Rehabilitation of existing SHI Schemes (based on baseline assessment), Construction of Pump house, construction of access road to construction site, Construction of live-stock watering trough, construction of fishery-aquaculture, construction of mechanization service center Construction of Seed stores Permanent.

The procurement process for all constructions will be conducted at regional level. How ever, Contracts of works estimated to the cost greater than or equal to 500,000.00 Unit of Accounts should be procured through International Competitive Bidding (ICB) procedure and it will be procured at the FPMU level. Contracts estimated to cost less than 500,000.00 Unit of Accounts per group of contracts should be procured using national competitive bidding (NCB) procedure and for contracts of works whose estimated cost is less than 50,000.00 UA should be procured through shopping and the project procurement officer shall use AfDB's SBD for ICB, the FPPAA's SBDs for NCB and the FPPAA's SRFQ for shopping with the necessary modifications at RPMU. Direct contracting (single source) may be used in exceptional cases where this method is deemed advantageous in meeting certain specific objective and is justifiable over the other competitive methods.

Works which have Small in values and will be constructed in various remote locations, which are difficult for packaging, with their estimated cost of less than UA 30,000 per contract and

not more than UA 480,000 in aggregate, will be procured through other procedures as shown in detail in the annexed procurement guideline.

7.3.2.**Procurement of Goods**

Goods to be procured under CREW will include Tractors, Disc harrows, Moldboard plows, Row planters, Seed drill, Combine Harvester, Seed thresher, Seed Stores at the research stations, Pumps for pumping from Permanent Rivers, Procurement of maintenance tool kits for Pumps, Hand GPs,Procurement of Water Scheme facilities, soil moisture sensors and Spare Parts, Office Equipment and Furniture- Mobile Extension Services Delivery, Office Equipment and Furniture- Regional PCU, Office Equipment and Furniture- Woreda Office, Activities, Vehicles- pich up d/cab and Motorcycles. The procurement will be done using the AfDB's SBD for all ICB and the FPPAA's SBDs agreed with or satisfactory to the AfDB (with consideration of the NCB exceptions). To the extent practicable, contracts shall be grouped into bid packages estimated more than or equal to 200,000UA shall be procured through ICB procedures. Goods and equipment contracts estimated to cost less than 200,000 UA per contract shall be procured using National Competitive Bidding (NCB), using national SBD agreed with or satisfactory to the Bank. Direct contracting (single source) may be used in exceptional cases where this method is deemed advantageous in meeting certain specific objective and is justifiable over the other competitive methods. Shopping may be used for contracts with estimated values of less than 50,000 UA. The list of goods to be procured under the above procurement modalities is described in detail in the ANNEXED procurement guideline.

7.4. Selection of consultants.

The consultancy services to be undertaken for CREW are Baseline survey, Mid Term and Impact Assessment Evaluations and recruiting external auditor, market infrasture, and procurement audit.The project procurement officer shall use AfDB's SRFP document for selection of consulting firms and may use relevant selection methods depending on the nature of the assignment and threshold.

Selection of consultants (firms) for all consultacy service procurement process will be prior reviewed by the bank.The detail procedures for each selection method are described in the Annexed procurement guideline.

7.5. Capacity Building

To enhance the CREW Procurement activities, the project procurement officers at federal and regional levels should have trainings and certified on African Development Bank and other Development partners' procurement procedures on procurement of goods, procurement of works, selection of consultants and contract administration. The selection of institutions for such specialized trainings will be done on the basis of quality and therefore will use the Qualifications Based Selection method. Venues and materials for workshops will be on the basis of at least three quotations.

7.6. Applicable Documents for the implementation of procurement

The following documents are governing procurement documents which are applicable under the project:

- Financing Agreement of CREW.
- The Bank's Procurement and Consultant Guidelines of dated May 2008 Edition and revised July 2012.
- Project Appraisal Document dated October 10, 2012.
- The Procurement Manual.
- The Procurement Plan of CREW approved by the African Development Bank.
- AfDB Standard Bidding Documents/Request for Proposals for goods procured under the procurement modality of ICB and Consultancy services.
- CREW Project Implementation Manual

7.7. List of SBD, SRFPs and Different Templates

The following lists of Bank's and the Federal Governments Standard procurement documents are applicable to the project.

For goods, Works and non-consulting services:

- SBD: Procurement of Goods: September 2010, revised July 2012.
- Bid Evaluation Form: Procurement of Goods or Works.

- Pre-qualification Document: September 2010, revised July 2012
- Procurement of Small Works – September 2010, revised July 2012
- Request for Quotation (RFQ) Format
- Standard Bidding Document for the Procurement of Goods and related services (NCB) issued by the FPPA (Version 1, April 2011)

For consultancy Services:

- Standard Request for Proposals: Selection of Consultants – May 2013
- Sample Form of Evaluation Report for Selection of Consultants
- REOI Templates
- Standard Procurement Planning Format. All these documents are ANNEXED in the procurement guideline.

7.8. General Provisions

The specific procurement rules and procedures to be followed in the implementation of the project depend on the circumstances of the particular case; however, five considerations generally guide the project's implementation requirements:

- The need for economy and efficiency in the implementation of the project,
- The need for high quality service (consulting service);
- The project's interest in giving all eligible bidders/consultants from developed and developing countries the same information and equal opportunity to compete in providing goods, works, and non-consulting services and consultancy service;
- the project's interest in encouraging the development of domestic contracting and manufacturing industries in Ethiopia and use of national consultants; and
- The importance of transparency in the procurement and consultant selection process.

7.9. Applicable Procurement Threshold and Methods

7.9.1. Procurement Methods for Goods, Works and Non-consulting Services

The method of procurement that the project will adopt depends on the nature and size of procurement. The agreed methods of procurement for goods and non-consulting services include:

a. International Competitive Bidding (ICB),
b. National Competitive Bidding (NCB),
c. Direct Contracting,
d. Shopping
e. Community Participation

Table 14 Procurement methods for goods, works and none consulting services as per the protocol of agreement and technical document the project

Expenditure Category	Contract Value (Threshold) in UA	Procurement Method	Contracts Subject to Prior Review	Contracts Subject to Post Review
1. Works	≥500,000.00	ICB	All Contracts	---
	≥200,000	NCB	All contracts	---
	<200,000.00	NCB	--	all contracts shall be post reviewed except the first three contracts as indicated on PP.
2. Goods	≥200,000	ICB	All Contracts	all contracts shall be post reviewed except the first three contracts as indicated on PP.
	<200,000 and ≥50,000	NCB	The first three contracts as indicated on the procurement plan	all contracts shall be post reviewed except the first three contracts as indicated on PP.
	< 50,000	Shopping	None	all contracts

7.9.2. Selection Methods for Consultancy Service

The following are the agreed selection methods, to be used for the selection and employment of consultants' services. The Procurement Plan shall specify the circumstances under which such methods may be used:

(a) Quality and Cost based Selection (QCBS).
(b) Quality Based Selection (QBS)
(c) Least Cost Selection (LCS)
(d) Single sources Selection (SSS)
(e) Individual Consultant (IC)
(f) Selection Based on Consultants' Qualifications (CQS)

Table 15 The agreed threshold for the selection and employment of consultancy service

Expenditure Category	Prior Review Threshold in UA	National Short-List Max Value UA
Consultants (Firms) - QCBS	All contracts	N/A
Consultants (Individuals)	All contracts	N/A

7.10. Procurement Processes and Procedures

7.10.1. Procurement Process for Goods, Works and Non-Consulting Services

The Procurement process itself starts with the preparation of the bidding documents. Among other information, which must appear in the documents, it is particularly important to ensure that information on technical and functional specifications are complete and sufficiently detailed, and that it accurately reflects the needs of the end-users. Technical specifications must be drawn by a competent technical authority or person, with advice and external technical assistance if required.

The Bank's Procurement Guideline specifies the essential elements of the tendering process. The major steps in the procurement of goods, works and non-consulting services for different procurement methods are as follows.

7.10.2. International Competitive Bidding (ICB)

In as much as possible ICB is the best procurement method (preferred method) compared to other procurement methods. However, when ICB is not efficient or economical or if other methods are otherwise deemed more appropriate, then methods other than ICB can be used. The ICB method will be used for works valued at ≥ UA five hundred(500,000.00), goods valued greaterthan UA two hundred(200,000.00) and consultancy services greater UA 200,000.00

The following points are noteworthy in respect to ICB application:

- All ICB procurements shall be done at Ministryof Agreeculture whether it may goods or construction of any works.
- Required to advertise the invitation to bid in the form of the Specific Procurement Notice (SPN) for any particular contracts under ICB, in at least one newspaper of general circulation in the Borrower's country and in the "Development Business" published by the United Nations.
- Standard Bidding Documents (SBDs) of the African Development Bank shall be used for procurements
- The goods, works or services must be organized and/or grouped in such a way that they attract the interest of foreign suppliers or contractors. However, domestic preference should apply to goods manufactured and works carried out in Ethiopia.
- AfDB's Standard Bidding Documents will be used for all ICBs.
- Bidding documents shall be issued which will clearly state the type of contract, size, scope, technical specifications, payment method and the general and specific conditions governing the procurement. The bidding documents should also describe the criteria and methodology for evaluation of bids and selection of successful bidder.
- ICB contracts will be subject to prior review by the Bank

7.10.3. National Competitive Bidding (NCB)

National Competitive Bidding (NCB) is a procedure normally used for public procurement and may be the most appropriate method of procurement of goods, works and non-consulting services which, by their nature or scope, are unlikely to attract foreign competition.

NCB may be the most appropriate method of procurement where foreign bidders are not expected to be interested because:

- Of the size and value of the contract for goods and non-consulting service
- Works are scattered geographically or spread over time
- Works are labour intensive; or Goods, works and non-consulting services are available locally at prices below the international market.

NCB procedures may also be used where the advantages of ICB are clearly outweighed by the administrative or financial burden involved.

For National Competitive Bidding (NCB) the SBD developed by Federal Public Procurement and Property Administration Agency (FPPAA) may be used subject to the exceptions agreed to AfDB

7.10.4. Preparation of Bidding Document for NCB

The BD shall furnish all information necessary for a potential bidder to prepare a bid for the goods or works to be procured. The details and complexity of the information required may vary with the nature of the proposed bid package.

The project procurement officer shall comply with the following instructions when preparing BDs. BDs shall be worded to permit and encourage open competition and shall set out clearly and precisely:

- The work to be carried out and the location of the work.
- The goods to be supplied and the place of delivery or installation
- The schedule for delivery and completion
- The minimum performance requirements
- The warranty and maintenance requirements, and
- Any other relevant terms and conditions.

All prospective bidders shall be provided the same information and be assured of equal opportunities to obtain additional information promptly upon request.
The advertisement, bid closing and opening, evaluation, award and signing of contracts are the same with the provisions and steps described under ICB procedure.

7.10.5. Shopping (Local and International)

Shopping is the least competitive procurement method. Shopping is an appropriate method only for procuring readily available off-the-shelf goods or standard specification commodities in quantities of small value, services and in some cases, very small simple works.

Shopping occurs in the immediate market area and may extend geographically to available suppliers. Shopping does not require formal BDs, and is carried out by requesting written quotations using the Request for Quotation (RFQ) from several local suppliers' contractors – at least three - to ensure competition. Telephone or verbal quotations are not acceptable.
The following steps should be followed during procurement of Goods, Works and Non-Consulting Services using Shopping method:

Step 1: Preparation of Technical Specifications

Step 2: Preparation of Request for Quotations

Step 3: Invite Suppliers/Contractors to Quote

Step 4: Receiving Quotations (minimum three)

Step 5: Quotations Opening and Comparison

Step 7: Contract Award and Issuance of Purchase Order

Step 8: Contract Management and Implementation

Step 9: Delivery and Acceptance

For the procurement of labor contractor, under this method the advertisement, evaluation and award shall follow the above mentioned provisions.

The request for quotations shall spell out the scope of work, place, start and completion date, the criteria for selection and the deadline for submission of quotations. All quotations will be opened at the same time and in the presence of a bid opening committee. The Bid Evaluation committee will evaluate the quotations based on qualifications, experience, skills and price.

After the initial evaluation, the committee may invite the qualified and experienced contractor who submitted the lowest evaluated bid to ascertain his experience and skills. When satisfied with his experience and qualifications, the committee will proceed to negotiate a contract with the contractor. The construction/labor contract will be signed by the project coordination unit and the awarded contractor. It is recommended if payments terms is made in stages e.g. at foundation, window, roof and finishing levels.

7.10.6. Direct Contracting (DC)

DC is contracting without competition (single-source) and may be an appropriate method under the following circumstances:

- An existing contract for goods, works, and non-consulting services, awarded in accordance with procedures acceptable to the African Development Bank (AfDB), may be extended to, a maximum of 15% of the original amount, for additional goods, works, and non-consulting services of a similar nature. The AfDB shall be satisfied in such cases that no advantage could be obtained by further competition and that the prices on the extended contract are reasonable. Provisions for such an extension, if considered likely in advance, shall be included in the original contract.

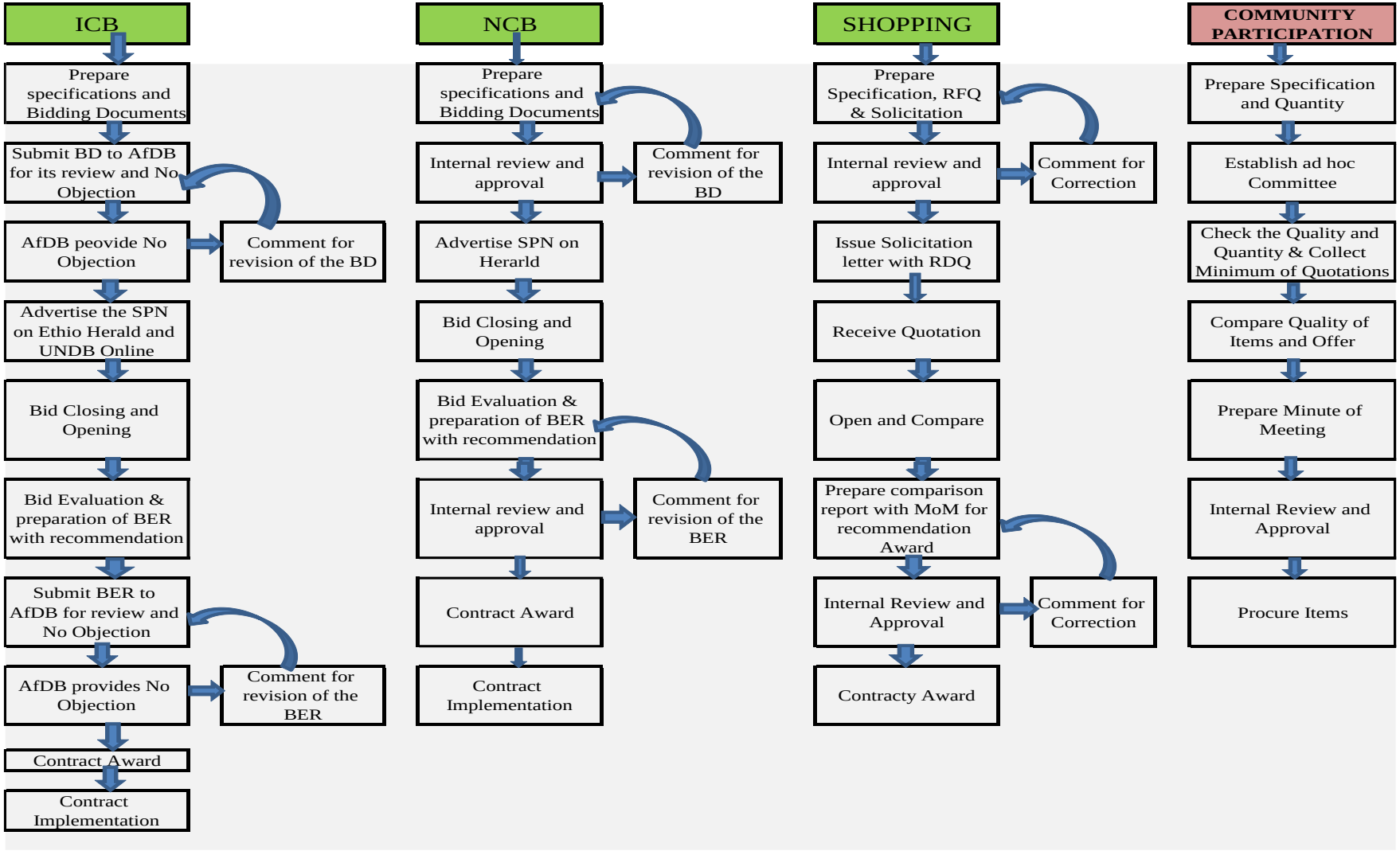
- Standardization of equipment or spare parts, to be compatible with existing equipment, may justify additional purchases from the original supplier.
- The required equipment is proprietary and obtainable only from one source.
- The procurement of certain goods from a particular supplier is essential to achieve the required performance or functional guarantee of an equipment or plant or facility.
- In exceptional cases, such as, but not limited to, in response to natural disasters and emergency situations declared by the project and recognized by the AfDB. For detail description procurement guideline is ANNEXED

7.11. Procurement under Community participation.

As part of the community participation, the project requires the procurement of locally available items from not well established shop and from suppliers/service providers that do not have formal invoices. Those items may be procured from individuals, small enterprises and small local associations within the community. Examples of these items are local stone, sand, donkey back transport, soil, grass, various seedlings, local skilled laborers, etc. The recommended steps for the procurement of these items are as follows

- the user agency or department will prepare specification/requirement of the items
- the Head of the project implementing institutions will establish an ad hoc team/committee having at least three members; one from the end user agency or department
- the committee shall request for prices from at least three service providers / suppliers of those items and collect the price and visually observe the quality of the item/service
- the committee prepares minutes of meeting with detail explanation of the quality, price and recommendation to award the contract to the supplier/service provider with reasonable quality and least-cost
- the Head of the project implementing institutions shall examine the minutes and shall approve the committee's recommendation
- The project implementing institutions will procure the items from the recommended supplier/service provider and keep all documentation safely.
- Notification of contract award Selection Process for Consultancy Services

Figure 7-1 Procurement workflow for Goods & Works



7.12. Procurement Process for Selection of Consulting Services

7.12.1. Quality and Cost based Selection (QCBS)

QCBS uses a competitive process among short-listed firms that takes into account the quality of the proposal and the cost of the services in the selection of the successful firm. Cost as a factor of selection shall be used judiciously. The relative weight to be given to the quality and cost shall be determined for each case depending on the nature of the assignment.

The selection process shall include the following steps:

(a) Preparation of the TOR;

(b) Preparation of cost estimate and the budget, and short-listing criteria.

(c) Advertising the Request for Expression of Interest (REOI);

(d) Preparation of the short list of consultants;

(e) Preparation and issuance of the RFP (which should include: the Letter of Invitation (LOI), Instructions to Consultants (ITC), the TOR, and the proposed draft contract);

(f) Receipt of proposals;

(g) Evaluation of technical proposals: consideration of quality;

(h) Public opening of financial proposals;

(i) Evaluation of financial proposal;

(j) Combined technical and financial evaluation; and;

(k) Negotiations and award of contract to the selected firm. Refer the detail selection processes from the ANNEXED procurement guideline.

7.12.2. Quality-Based Selection (QBS)

All provisions in QCBS shall apply for other methods of selection under all others selection methods unless a different provision has been specifically identified in the respective selection methods, in which case, the latter shall apply. Project coordination unit shall use the applicable standard RFP issued by the AfDB with minimal changes,

acceptable to the AfDB, as necessary to address project-specific conditions, except as otherwise provided in this manual.

QBS is appropriate for the following types of assignments:

a) complex or highly specialized assignments for which it is difficult to define precise TOR and the required input from the consultants, and for which the client expects the consultants to demonstrate innovation in their proposals (for example, country economic or sector studies, multi-sector feasibility studies, design of a hazardous waste remediation plant or of an urban master plan, financial sector reforms);
b) assignments that have a high downstream impact and in which the objective is to have the best experts (for example, feasibility and structural engineering design of such major infrastructure as large dams, policy studies of national significance, management studies of large government agencies); and
c) Assignments that can be carried out in substantially different ways, such that proposals will not be comparable (for example, management advice, and sector and policy studies in which the value of the services depends on the quality of the analysis).

In QBS, the RFP may request submission of a technical proposal only (without the financial proposal), or request submission of both technical and financial proposals at the same time, but in separate envelopes (two-envelope system). The RFP shall provide either the estimated budget or the estimated time of key experts, specifying that this information is given as an indication only and that consultants shall be free to propose their own estimates.

If technical proposals alone were invited, after evaluating the technical proposals using the same methodology as in QCBS, the Project coordination unit shall ask the consultant with the highest ranked technical proposal to submit a detailed financial proposal. The Project coordination unit and the consultant shall then negotiate the financial proposal and the contract. All other aspects of the selection process shall be identical to those of QCBS, including the publication of the award of contract as described above, except that only the contract price of the winning firm is published. If consultants were requested to provide financial proposals initially together with the technical proposals, safeguards

shall be built in as in QCBS to ensure that the financial proposal of only the selected firm is opened and the rest returned unopened, after the negotiations are successfully concluded.

7.12.3.Selection under Fixed Budget (FBS)

Similar with that of QCBS and QBS methods, the steps to be flowed under FBS are the same. However, the consultant who has obtained the highest rank on the technical proposal within the stated budget will be called for negotiation. Proposals that exceed the indicated budget are discarded.

7.12.4.Least-Cost Selection (LCS)

This method is generally appropriate for selecting consultants for assignments of a standard or routine nature (audits, engineering design of noncomplex works, and so forth) where well-established practices and standards exist. Under this method, a "minimum" qualifying mark for the "quality" is established. Proposals, to be submitted in two envelopes, are invited from a short list. Technical proposals are opened first and evaluated. Those securing less than the minimum qualifying mark are rejected, and the financial proposals of the rest shall be opened as stipulated in QCBS method. The firm with the lowest price shall then be selected. Under this method, the minimum qualifying mark shall be established, understanding that all proposals above the minimum compete only on "cost". The minimum qualifying mark shall be stated in the RFP. The award of contract shall be published using the standard template.

7.12.5.Selection Based on Consultant's Qualifications (CQS)

This method may be used for small assignments or emergency situations declared by the Project coordination unit and recognized by the AfDB for which the need for issuing an RFP, and preparing and evaluating competitive proposals is not justified. In such cases, the Project coordination unit shall prepare the TOR and obtain expressions of interest that include information on their experience and qualifications, eventually through an REOI as may be needed, from as many firms as possible, and at least three qualified firms with relevant experience. Firms having the required experience and competence relevant to the assignment shall be assessed and compared, and the best qualified and experienced firm shall be selected. Only the selected firm shall be asked to submit a combined technical and financial proposal and, if such proposal is responsive and acceptable, be invited to

negotiate a contract. Both technical and financial aspects of the proposal may be negotiated. If the negotiations fail with the selected firm, the provisions outlined in QCBS will apply. The minutes of negotiations shall be prepared and signed by both parties. Awards of contract shall be published.

The steps to be followed under CQS, in accordance: with Section 3.7 of the Bank Guidelines, are as follows:

Step 1: Detailed TORs are prepared by the relevant technical experts;

Step 2: The Procurement Unit send the draft TORs for all contracts subject to prior review to the AfDBfor its comments and no objection.

Step 3: After receipt of AfDBcomments, the Procurement Unit requests expressions of interest and qualified information relating to the experience and competence of the consultants relevant to the assignment in the local and/or international media. It is highly desirable to advertise the advert in UNDB on line. It is also possible to identify candidates from those who have sent CVs or information in response to other procurement notices such as the GPN or from other lists for as long as the information is up-to-date.

Step 4: The Procurement Unit can select six or less candidates with the best qualification experience and competence relevant to the assignment from the list of possible candidates.

Step 5: The implementing agency requests the selected consultant by sending TOR to submit a combined technical and financial proposal

Step 6: The implementing agency invites the consultant to negotiate the contract if the technical proposal proves acceptable.

7.12.6.Single Source Selection (SSS)

Single-source selection of consultants does not provide the benefits of competition in regard to quality and cost, lacks transparency in selection, and could encourage unacceptable practices. Therefore, single-source selection shall be used only in exceptional cases. The justification for single-source selection shall be examined in the context of the overall interests of the client and the project, and the Bank’s responsibility

to ensure economy and efficiency and provide equal opportunity to all qualified consultants.

Single-source selection may be appropriate in the following cases, and only if it presents a clear advantage over competition: (a) for tasks that represent a natural continuation of previous work carried out by the firm (see next paragraph); (b) in exceptional cases, such as, but not limited to, in response to natural disasters and for emergency situations both declared by the Project coordination unit and recognized by the Bank; (c) for very small assignments or (d) when only one firm is qualified or has experience of exceptional worth for the assignment. In all such cases, the Project coordination unit is not required to issue an RFP and shall submit to the AfDB for its review and no objection the TOR of the assignment, a sufficiently detailed justification including the rationale for single-source selection instead of a competitive selection process, and the basis for recommending a particular firm, except for contracts below a threshold defined on the basis of risks and the scope of the project, and set forth in the Procurement Plan.

When continuity for downstream work is essential, the initial RFP shall outline this prospect, and, if practical, the factors used for the selection of the consultant shall take the likelihood of continuation into account. Continuity in the technical approach, experience acquired, and continued professional liability of the same consultant may make continuation with the initial consultant preferable to a new competition subject to satisfactory performance in the initial assignment. For such downstream assignments, the Project coordination unit shall ask the initially selected consultant to prepare technical and financial proposals on the basis of the TOR furnished by the Project coordination unit, which shall then be negotiated.

If the initial assignment was not awarded on a competitive basis or was awarded under tied financing or if the downstream assignment is substantially larger in value, a competitive process acceptable to the AfDB shall normally be followed in which the consultant carrying out the initial work is not excluded from consideration if it expresses interest. The AfDB will consider exceptions to this rule only under special circumstances and only when a new competitive process is not practicable. The award of contract shall be published.

The evaluation reporting and publication of contract award standard format for all other methods is similar to QCBS.

7.12.7. Individual Consultant

- Teams of personnel are not required.
- no additional outside (home office) professional support is required, and
- teams of personnel are not required
- no additional outside (home office) professional support is required, and
- The experience and qualifications of the individual are the paramount requirement

Selection Steps:

Step 1: Prepare TOR.

Step 2: Send the TOR to the AfDB for its clearance and no-objection if the estimated service cost **is > UA 100,000** or the individual consultant is to be recruited to work on procurement or legal, regardless of the contract amount.

Step 3: Prepare REOI and publishing the local media or identify candidates from consultants' database. Locally advertise in widely circulated, national newspapers. The advert shall be floated for a period of at least for two weeks for the potential candidates to submit their CVs.

Step 4: Assess the CVs and rank the interested applicants on the basis of their qualifications and experience for the assignment. Capability is judged on the basis of academic and professional background, relevant experience and, as appropriate, knowledge of the local conditions, such as local language, culture, administrative system, and government organization.

Step 5: Request the top ranking consultant to submit a financial proposal and negotiate and sign contract if the negotiation is concluded in agreement.

Step 6: If the negotiation with the first ranked consultant failed, invite the second ranked consultant for negotiation and follow the above steps.

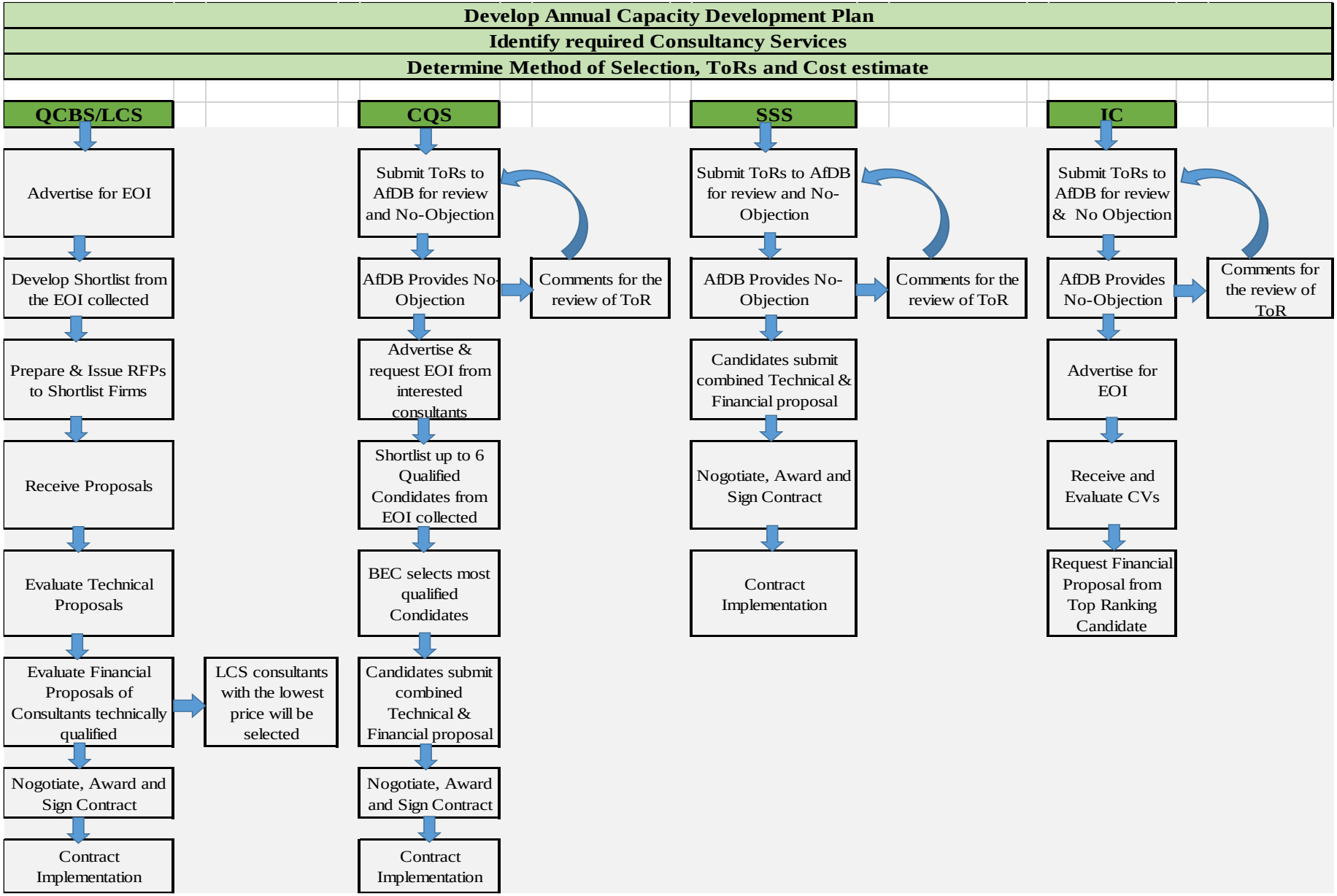
Selection process for the individual consultant with estimated budget of UA 100,000 and above is subject to prior review of the AfDB.

Government officials and civil servants may only be hired under consulting contracts, either as individuals or as members of a team of a consulting firm, if they are on leave of absence without pay

- Are not being hired by the agency they were working for immediately before going on leave, for at least six months.
- Their employment would not create a conflict of interest.

  University professors or scientists from research institutes can be contracted individually provided that they have a full time employment contract with their institution and have regularly exercised their function for a year or more before they are contracted under the project.

Figure 7-2 procurement workflow for selection of consultancy



7.13. Procurement Planning

The first and most important step in the procurement cycle is the identification of the required items and services to be procured and determine when those goods/services are required for the project. This requirement shall be reflected in the procurement plan and the project coordination unit shall perform proper planning for all procurement activities in order to ensure that they meet their needs in the most effective, economical and timely manner.

A procurement plan is a legal document required in the procurement process. Annual PPs shall coincide with budgets and targets to be achieved. Plans shall address the technical, operational, management, financial and other significant considerations. This should be carried out by the user section/entity and should result in clear identification of the items and corresponding specific functional and service requirements. Officers at all levels that are engaged in planning and managing procurement must be aware of all governing legislation and the agreed procurement rules and procedures in order to inform decision-making processes.

Once procurement needs have been identified and specified, the concerned project coordination unit should also ensure that procurement is carried out in time. For this a PP should be prepared giving sufficient lead time for the process. Project coordination unit should ensure that appropriate approvals at each level are in place. Separate Procurement Plan should be prepared for Consulting service, Goods, Works and non- consulting services.

The procurement plan should be prepared and approved by the authorized institution or body. Procurement should be carried out in accordance with the provisions in the approved procurement plan. Procurement activities carried out without an approved procurement plan are considered as ineligible expenditures and are not financed through project fund.

The duration of time required to properly carry out each stage of procurement activities should be estimated carefully to develop a realistic plan.

7.13.1. Procurement planning steps

- For the preparation of a PP, the following steps are essential
- Listing of the various goods /works to be constructed and services to be obtained for the project in that period; each of them constitutes an “item” to be procured.
- Identifying key dates when each project component made of several items should be ready for use or available on site;

- Package the items on the list in logical contract groups (for example, if the list includes a laptop computer and a desktop computer, these together would make a "computers contract package" for procurement purposes).
- deciding the contents of each procurement "package" which will constitute a specific procurement operation;
- Obtain updated unit prices of items included in each package from suppliers, shopkeepers, catalogues, previous contracts, internet, etc.; using this information, cost each package.
- Identifying the interrelationships between dependent items;
- Determining the lead time for each package and scheduling the target date for each package in the procurement process; and
- Developing a monitoring system to ensure timely procurement decisions and follow-up actions;
- Project Management Unit will use the Bank's standard template for PP to prepare and submit the PP.

7.13.2. Procurement planning packaging

The project coordination unit should decide contract packages by answering the following key questions:

- What to procure: (like items, large, small, values slice and packages)
- How to procure; What method to apply ((ICB, NCB, shopping, DC, other agreed procurement methods,)
- When to do each step: (each step)
- Who: (who is the implementing institution, who reviews the request, who approves)

Packaging:

- Each product/goods, work or service can be considered as an "item" to be procured.
- Each item/package could be procured separately under a specific procurement process.
- Suppliers' interest is likely to increase with the contract value. For the benefit of the project coordination unit and the supplier, it is desirable to group items which are technically similar and would interest the same type of firms.
- All items similar by nature or origin of supply should then be grouped together in a "lot" of goods, or "slice" of works
- Bidders are invited to submit a bid and quote price for entire lots and slices. This grouping of items in lots or slices will reduce the number of procurement operations.

Bidders could nevertheless be invited to indicate unit prices for each item in a lot or slice. This is useful in the event of quantity variation

- In order to obtain best prices group by lots or slice, or package, items which are reasonably homogenous and can be offered from a single source.
- The size of lot, slice or package should be appropriate to foster maximum competition to obtain the most economic contracts
- If similar items are to be procured for several sectors/sub-components in a project it may be convenient to combine them in one package in order to achieve economies of scale
- If large quantities of a single commodity are being procured, packaging should take into consideration
- Market practices for the item.
- Transportation costs,
- Unloading capabilities,
- Storage capacities and, etc.
- Goods and works supplied from different sources and to be delivered and erected at different dates during the project implementation should be packaged separately. This is because of potential problems with contract interfacing and assigning responsibility for any subsequent defect.
- A large number of small lots or slices, encourages small suppliers and domestic contractors, but discourages larger and more efficient firms from bidding on a package of small contracts. It is recommended against an excessive number of lots or slices in a package.

7.13.3. Procurement Method

The list of goods and works required for the project has to be analyzed to determine which methods (ICB, NCB, Shopping, Direct Contracting,) are the permitted methods for the project and will result in economic and efficient procurement.

7.13.4.Scheduling

Once the bid package and the procurement method are determined, the next step is to fix the dates when procurement action should take place to ensure the availability of goods, works and none consulting services.

- This requires the determination of the lead-time for each package and this consists of both administrative time and the contractor's or supplier's delivery time.

- Scheduling may involve some or all steps of the procurement cycle like:
- Preparation of bidding document
- Advertisement for Invitation for bid
- Preparation and submission of bids
- Preparation and submission of evaluation report
- Contract award
- Signing of contract and contract performance
- Arrival of goods
- Each of the activities must be examined and the time to complete the activity will be determined
- Some of the activities may overlap; e.g. an advertisement for IFB could be sent to the daily press while the bidding documents are being finalized and printed;
- An implementation chart should then be prepared indicating the critical path and dates for the completion of each activity
- An implementation chart should then be prepared indicating the critical path and dates for the completion of each activity

7.13.5. Updating the Procurement Plan

- Once the project starts, the procurement plan needs to be updated. This means a revision of the "plan" duration for each contract in accordance with the actual starting date of the project.
- The updated procurement plan should be submitted to the AfDB for review and clearance.

"Plan" versus "Actual"

- Once the procurement process for the contract starts, the actual dates for each procurement step are inserted in the row designated as "Actual".
- The procurement officer at project level should keep record of the reasons for any delays, for supervision purposes and the same will done for the actual implementation of contracts.

7.14. Disclosure Policy

As a requirement of meeting the basic procurement principles, the opportunity to bid and the awarded of contract shall be disclosed to public. The opportunity to bid shall be done according to the explanation made in the previous sections and summarized the table below. The award result shall be done as per the following guidelines:

7.14.1. Goods and works

Within two weeks of receiving the Bank's "no objection" to the recommendation of contract award for ICB, Modified ICB and Direct Contracting (DC), CREW PCU shall publish in UNDB online the results identifying the bid and lot numbers and the following information:

- Name of each bidder who submitted a bid;
- Bid prices as read out at bid opening;
- Name and evaluated prices of each bid that was evaluated;
- Name of bidders whose bids were rejected and the reasons for their rejection; and;
- Name of the winning bidder, and the price it offered, as well as the duration and summary scope of the contract.

With regard to NCB and Shopping, publication of contract awards could be done by posting on local newspapers and local notice boards respectively

Table 15 Summary of Disclosure of procurement information at different stages (Goods, works and non- consulting services)

Table 16 Summary of disclosure of procurement information at different stages

What	Where	By whom	When
General Procurement Notice (GPN)	The bank's internet website	CREW PCU/MOA	Effectiveness date
Procurement Plan (PP) and updates	The bank's internet website	CREW PCU/MOA	Effectiveness date and after updates are cleared by AfDB
Specific Procurement Notice (SPN)	The bank's internet website and wide circulated national news gazette or streaming media	CREW PCU/MOA	As soon as the Bid Document is ready
Expression of Interest (EOI)	The bank's internet website and wide circulated national news gazette or streaming media	CREW PCU/MOA	As soon as the TOR and RFP is ready
Contract Award	The bank's internet website if necessary	CREW PCU/MOA CREW PCU/BoFED CREW PCU/WoFED/ BoFED	After contract signature
Direct Contracting	The bank's internet website	CREW PCU/MOA	After contract signature

(DC) award information	if necessary		

7.14.2. Consultants

After the award of contract for consultancy services, the client shall publish on the bank's internet website the following information:

The names of all consultants who submitted proposals;

- The technical points assigned to each consultant;
- The evaluated prices of each consultant;
- The final point ranking of the consultants;
- The name of the winning consultant and the price, duration, and summary scope of the contract.

The same information shall be sent to all consultants who have submitted proposals.

Table 17 Summary of disclosure of procurement information

What	Where	By whom	When
Expression of Interest (EOI)	UNDB/DG market Image bank (contract value of >UA200,000National news paper	CREW PCU/MOA	After preparation of ToR is finalized and approved
Contract Award	UNDB online (Contract value of >UA200,000 National news Paper	CREW PCU/MOA	After contract signature

7.15. Procurement Process Trail and Accountability

7.15.1.Review requirements

- Prior and Post Review

The AfDB has a fiduciary responsibility to ensure the procurement process is carried out as per the FA. The AfDB's review process ensures that the project funds are used for the purposes intended and that procurement procedures outlined in the FA are followed. AfDB staff's fiduciary responsibilities involve prior review and post review of procurement documentation as the procurement process is carried out.

The prior review process protects the project coordination unit at each stage of the procurement transaction and also provides advice and suggestions to the project coordination unit for enhancing the quality of the documents and procedures. In the project, prior review

involves review of the procurement plan (PP) and review of proposed procurement documentation and decisions by the project coordination unit by AfDB staff before implementation. Post review is similar in scope but is related generally to smaller value contracts. Post review is carried out by AfDB staff or by a consultant designated by the AfDB on a sample basis after action has been taken by the project coordination unit.

- Scope of Prior Review

The AfDB shall review Procurement Plans and their updates that are prepared by the Project coordination unit. In addition, all ICB contracts, the first three NCB contracts for each goods &works, all consultancies services whose threshold is greater than or equal to 200,000 UA and the first three consultancy service contracts whose threshold is less than 200,000 UA are prior reviewed by AfDB.

With respect to all contracts which are subject to the AfDB's prior review:

- In cases where prequalification is used, the CREW PCUshall, before prequalification submissions are invited, furnish the AfDB with the draft documents to be used, including the text of the invitation to prequalify, the prequalification questionnaire, and the evaluation methodology, together with a description of the advertising procedures to be followed, and shall introduce such modifications in said procedure and documents as the AfDB shall reasonably request. The report evaluating the applications received by the CREW PCU,s, the list of proposed prequalified bidders, together with a statement of their qualifications and of the reasons for the exclusion of any applicant for prequalification, shall be furnished by the CREW PCUto the AfDB for its comments before the applicants are notified of the CREW PCU's decision, and the CREW PCUshall make such additions to, deletions from, or modifications in the said list as the AfDB shall reasonably request.
- Before bids are invited, CREW PCUshall furnish to the AfDB for its comments, draft bidding documents, including the invitation to bid, instructions to bidders, including the basis of bid evaluation and contract award, and the conditions of contract and specifications for the civil works, supply of goods, or installation of equipment, etc., as the case may be, together with a description of the advertising procedures to be followed for the bidding (if prequalification has not been used), and shall make such modifications in the said documents as the AfDB shall reasonably request. Any further modification shall require the AfDB's no objection before it is issued to the prospective bidders.

- After bids have been received and evaluated, CREW PCUshall, before a final decision on the award is made, furnish to the AfDB, in sufficient time for its review, a detailed report on the evaluation and comparison of bids received together with the recommendation for award and such other information as the AfDB shall reasonably request. The AfDB shall, if it determines that the intended award would be inconsistent with the Financing Agreement and/or the Procurement Plan, promptly inform CREW PCUs and state the reasons for such determination. Otherwise, the AfDB shall provide its no objection to the recommendation for the award of contract. CREW PCU shall award the contract only after receiving the no objection from the AfDB.
- If after publication of the award the CREW PCU receives protests or complaints from bidders, a copy of the complaint, the CREW PCU,s' comments on each issue raised in the complaint, and a copy of the CREW PCUs response shall be sent to the AfDB for its review and comments.
- If as result of analysis of a protest CREW PCU changes its contract award recommendation, the reasons for such decision and a revised evaluation report shall be submitted to the AfDB for no objection. CREW PCUs shall provide a republication of the contract award.
- The terms and conditions of a contract shall not, without the AfDB's prior no objection, materially differ from those on which bids were asked or prequalification of suppliers, if any, was invited.
- One conformed copy of the contract, and of the advance payment security and the performance security if they were requested, shall be furnished to the AfDB promptly after its signing and prior to the submission to the AfDB of the first application for withdrawal of funds from the Loan Account in respect of such contract. When payments for the contract are to be made out of a Special Account (SA), copies of the contract, and the advance payment security and the performance security if they were requested, shall be furnished to the AfDB prior to the making of the first payment out of the SA in respect of such contract.
- All evaluation reports shall be accompanied by a summary of the procurement on a form provided by the AfDB. The description and amount of the contract, together with the name and address of the successful bidder, shall be subject to public disclosure by the AfDB upon receipt of the signed copy of the contract from the CREW PCU.

- CREW PCU shall retain all documentation with respect to each contract during project implementation until two years after the closing date of the Loan Agreement. This documentation would include the list of documents outlined under section Record keeping.

Modifications of the signed contract: In the case of contracts subject to prior review, before agreeing to (a) a material extension of the stipulated time for performance of a contract; or (b) any substantial modification of the scope of services or other significant changes to the terms and conditions of the contract; or (c) any variation order or amendment (except in cases of extreme urgency) which, singly or combined with all variation orders or amendments previously issued, increase the original contract amount by more than 15% (fifteen percent); or (d) the proposed termination of the contract, CREW PCU shall seek the AfDB's no objection. If the AfDB determines that the proposal would be inconsistent with the provisions of the Financing Agreement and/or Procurement Plan, it shall promptly inform the CREW PCUand state the reasons for its determination. A copy of all amendments to the contract shall be furnished to the AfDB for its record.

- Post Review

Post Procurement Reviews (PPRs) are normally carried out by the AfDB or by the consultant designated by the AfDB. The Project coordination unit at all levels shall retain all documentation with respect to each contract during project implementation and up to two years after the closing date of the Financing Agreement. This documentation would include, but not be limited to, Copies of Invitation for Bids/Quotations/EOI;

- Invitations for Bids/Quotations/REOIs
- Bidding Documents/RFQ/RFP;
- Clarifications requested and issued;
- Addenda to bidding documents RFP/RFP issued;
- Minutes of Bid/quotation/proposal opening;
- Bids/quotations/proposals received;
- Clarifications requested and received;
- Bid/quotation/proposal Evaluation Report;
- Complaints received and responses;
- Notification of Award/Letter of Acceptance/LPO;
- Signed Contract;

- Securities (Mobilization/Performance);
- Payment certificates/Invoices;
- Goods received note/Delivery notes;
- Inspection Certificates/Acceptance Certificates/Certification of Completion;
- Progress Reports; and
- Publication of contract award (for NCB and consultants) if applicable.

Implementing agencies are required to facilitate the post procurement review of procurement documents and shall furnish such documentation to the AfDB or its delegate upon request. In order to facilitate the review implementing agencies at all levels should maintain approved procurement plans, procurement contract registers and the above mentioned procurement related documents;

7.16. Complaints Handling

It is likely that the project coordination unit can receive complaints from bidders/consultants at any stage in the procurement process. The complaint may come direct from the complainant or it may come through other agencies including the AfDB. It is very important to give due attention and handle complaint properly and transparently.

Any complaint received should be directed to the Head of the CREW PCU whose responsibility is to handle the complaint after a thorough investigation and analysis of the facts.

The Head of Procuring entity shall ensure that the following steps are taken in handling complaints.

7.16.1.For Prior Reviewed Contracts

In addition to handling the complaint, the project coordination unit is required to inform the TTL of the project in the AfDB through Head Project coordination unit about the complaint and the response given thereof at the appropriate stage in the procurement process.

The following table shows the action Head Project coordination unit and other implementing agencies should take when receiving complaints at different procurement stages.

Table 18 Action to be taken by procuring entity when complain is received

SN	Stage when complaint is received	Actions to be taken by PROCURING ENTITY

SN	Stage when complaint is received	Actions to be taken by PROCURING ENTITY
1	After short listing	Provide response to the complainant and copy to the AfDB. Hold debriefing session with any unsuccessful applicant or consultant who requests for it. Copy of the debriefing report shall also be forwarded to the AfDB.
2	Prior to bids/proposal submission	Provide response to the complainant and copy to the AfDB
3	During evaluation of bids/proposal/ prequalification	Provide response to the complainant (address the complaint the response given in the evaluation report)
4	Before receiving "no objection" to bid/proposal evaluation report	Provide response to the complainant and copy to the AfDB
5	After receiving "no objection" to bid/proposal evaluation report	Provide response to the complainant and copy to the AfDB and ensure that the AfDB is satisfied with the response given thereof before awarding the contract.
6	After contract award	Provide response to the complainant and copy to the AfDB. Hold debriefing session with any unsuccessful bidder who request for it. Copy of the debriefing report shall also be forwarded to the AfDB

7.16.2. For Post Reviewed Contracts

The procedure to be followed in handling procurement complaints for post reviewed contracts is also similar with the prior reviewed contracts as shown above except the federal PCU, regional and woreda project implementing institutions are not required on their own initiation to notify the AfDB the nature of the complaint and the response given during the procurement process. Federal, regional and woreda project implementing institutions/procuring entities shall keep complete documents of complaints in the procurement files and provide all the information associated with the complaint if requested by the AfDB during project supervision.

7.17. Record Management

Absence of procurement documentation in all contracts financed by the project may affect the effectiveness of the contract management process, the quality of procurement audit and the ability of the management to make timely and appropriate decision.

Proper recordkeeping of procurement processes is essential for a successful outcome. The experience is that lack of an efficient filing system causes long and costly delays in the search for documents at critical moments when procurement issues arise or when procurement post reviews need to be done.

Procurement filing has specific characteristics. Files must be based on the three procurement categories (works, goods and consultant services/training) and span the entire procurement process per contract in each category in an efficient and easily accessible manner (from advertising to contract signature and management.). This system should best be maintained professionally in order to remain useful.

Procurement files should be managed by the procurement team at all levels; procurement files should separate and distinct from financial records. While the finance department may need to keep the original receipts to support payments, however, the procurement files must contain photocopies of the receipts and the original procurement documents (bids, quotations, etc.).

Filing structure

- Create File Boxes or drawers according to the procurement categories Works/labour contract, Goods and Consulting Services/Training.
- a general box or large solid folder, labelled a general box or large solid folder , labelled as Works, Goods and Consulting Services/Training, as well the list of reference numbers for bulky bid documents and contracts for the three categories. These bulky documents should be kept in a separate box;
- Computer Printed labels should be used on all filing boxes/drawers for clarity
- Each contract is filed in the relevant box for the project (a contract for works/labour contract goes in a “Works” box, and a consultant contract goes in a “Consultants” box), and each box carries the contract name.
- Each procurement category box per contract contains individual folders for (i) each procurement step, (ii) a folder for complaints and (iii) a folder for procurement reviews or technical audits

- For each contract, separate larger procurement document boxes are created to keep bulky bidding and contract documents, which are given reference numbers. The reference numbers of the prequalification, bidding, consultant proposal and contract documents are listed on a "Bidding Document Sheet" or "Proposal Document Sheet", which is placed in each Procurement General box for the project. This is to ensure that such documents will always be rapidly traceable.

Among others the important records to be kept for the procurement of goods, works and non-consulting services are the following:

Table 19 Important records to be kept for procurement of goods, works and non consulting services

S.n	Description	ICB	NCB	Shopping
1	Copy of the GPN with publication date	√	√	
2	Copy of the SPN with publication date	√	√	
3	No objection from AfDB to draft BD	√		
4	Modifications/addenda to BDs (if any)	√	√	√
5	Final BDs/RFQ issued to bidder	√	√	√
6	Bid opening report	√	√	
7	Head of project coordination unit notice on composition and start of BEC	√	√	
8	Clarifications required from bidders and written replies	√	√	√
9	BER/quotation comparison report	√	√	√
10	Analysis of BER and recommendation	√	√	
11	Comments by the AfDB and/or no objection to the BER	√		
12	Contract award/Letter of Acceptance	√	√	√
13	Copy of signed contract	√	√	√
14	Copy of letter to the AfDB transmitting signed contract for disbursement	√		
15	Publication of Award in UNDB online	√	√	
16	Contract amendments	√	√	√
17	Inspection report	√	√	√
18	Payment authorization	√	√	√

S.n	Description	ICB	NCB	Shopping
19	Completion certificates	√	√	√
20	Copies of guarantees	√	√	
21	Claims and disputes	√	√	√

The important records to be kept for selection of **consultants** (firms and IC) are the following:

Table 20 Important records to be kept for selection of consultants

S.n	Description	Selection of consulting firm	Selection of IC
1	Terms of Reference	√	√
2	Advert for request for EOI and publication date	√	√
3	List of Consulting firms having expressed their interest	√	√
4	Documentation provided by consultants (EOI)	√	√
5	report on the evaluation of EOIs (short listing report)	√	
6	Comments/no-objection from the AfDB to proposed short list	√	
7	Approved RFP as sent to the consultants	√	
8	No-objection/Comments by the AfDB to draft RFP	√	
9	copy of the proposals received and minutes for opening of technical proposals	√	
10	letter for the appointment of BEC	√	
11	Technical Evaluation Report	√	
12	No objection/comments by the AfDB on the TER	√	
13	Minutes of Opening of Financial Proposals.	√	
14	Combined Technical and Financial Evaluation Report	√	
15	No-objection by the AfDB to negotiated initialed draft contract and minutes of negotiation	√	
16	Copy of signed contract	√	√
17	Publication of Award	√	√
18	Contract Amendments	√	√
19	Consultant/technical assistance Reports	√	√

S.n	Description	Selection of consulting firm	Selection of IC
20	Claims and Disputes	√	√
21	Folders for Contracts with ICs for technical assistance and training: Name on a folder for each ICs, TOR, request for EOI, short list, CVs and evaluation report		√

7.17.1. Contracts (procurement) register

All project coordination and implementation units are required to maintain a "Contract Register" which provides basic contract information to internal and external users.

Implementing Institutions should ensure that the procurement unit and the assigned person for record management are provided with the facility and responsibility to register contracts and update the list regularly.

The following includes some of the basic information to be provided in the contract register.

- Name and address of supplier or consultant
- Description of the Item or consulting service procured
- Contract amount, reference number and date
- Procurement/consultant selection method
- Delivery or completion date, etc.

7.18. Reporting

The reporting of procurement activity is essential for procurement planning and budgeting, and for the monitoring and tracking of procurement status at all levels. Regular monthly or quarterly reporting of procurement as part of the financial performance report shall be submitted to the Head of the IAs as follows:

- The quarterly report of procurement shall be submitted by the Woreda implementing institution in collaboration with the Woreda Finance and Economic Development office (WoFED) to the Regional CREW Coordination Office within 15 days of the end of each calendar month. The report should comprise the procurement process, contracts signed or procurements cancelled and contracts completed in the reporting month.
- In like manner, CREW Regional Coordination Office will consolidate the reports, and communicate to the Federal CREW Project Coordination Office three weeks after the end of the quarter.

- CREW Federal Project Coordination Office will consolidate all the procurement activity reports submitted from different regions and shall communicate to the Bank every 30 days after the end of the quarter of the fiscal year.

Table 21 Summary of typical time required to process procurement activities

S.n	Description of the job to be done	Types of procurement				
		ICB	**NCB**	**Direct contracting**	**National Shopping**	**International Shopping**
1	Prepare SBD & submit to the AfDB for its no objection	10 days	10 days	10 days	10 days	10 days
2	Get Bank No-objection for the submitted SBD	14 days	-	14 days		14 days
3	Publish the specific procurement notice on wider circulate local Gazette and bank website	7 days	7 days	solicit officially the request (3days)	solicit RFQ officially (3days)	solicit RFQ officially (3days)
4	Bid closing/opening	42 days	28 days	28 days	10 days	28 days
5	Bid evaluation	28 days	28 days	10 days	7 days	14 days
6	Submit BER to AfDB for its no objection	1 day	-	-	-	1 day
7	Get Bank No-objection for the submitted BER	14 days	-	14 days	-	14 days
8	Get internal approval for the PER from Endorsing Committee.	3 day	3 day	3 day	3 day	3 day
9	Award	1 day	1 day	2 day	5 day	5 day
10	contract signature	15 days	15 days	5 days		15 days
11	Delivery of goods	120 days	90 days	60 days		45 days
Total required days to accomplish the contract		**255 days**	**182 days.**	**149 days**	**38 days**	**141 days**

Table 22 Summary of typical time required to process procurement for short listing consulting firms

S.n	Description of Activities	Prior review threshold	Post review thresholds

		consultancy services	consultancy services
1	Prepare and submit EOI to AfDB	10 days	10 days
2	Get No-objection for the submitted EOI from AfDB	14 days	-
3	Publish the specific procurement notice for EOI on wider circulate local Gazette and bank website	7 days	7 days
4	Bid closing/opening	14 days	14 days
5	Short listing consulting firms	28 days	28 days
	Submit short listing report and RFP to AfDB for its no objection	1 day	-
6	Get No Objection for short listing report and RFP from AfDB	14 days	-
7	Get internal approval for the short listing report and RFP from PEC.	3 day	3 day
8	Invite the RFP for the shortlisted firms	3 day	3 day
9	Bid closing/opening	42 days	28 days
10	Technical evaluation report	28 days	28 days
11	Submission of TER to AfDB	1 day	-
12	Get Bank No objection for TER	14 days	-
13	Prepare and Submit minute of negotiation report and draft contract agreement to AfDB	7 days	7 days
14	Get Bank No objection for the minute of negotiation and draft contract agreement.	14 days	-
15	Contract Award for the winner	5 days	5 days
16	Signature of contract	15 days	15 days
	The total required time for procurement process	**220 days**	**148 days**

8. Monitoring, Evaluation and Learning (MEL) system

Monitoring, evaluation, and learning (MEL) are essential components of assessing and improving the climate-resilient wheat value chain project (CREW). MEL processes provide a systematic approach to gather data, measure progress, evaluate impact, and facilitate learning to enhance the effectiveness and sustainability of the project interventions.

The Monitoring and evaluation system for CREW Project is based on the result in which a visual representation of the logical sequence of outcomes or results that a project aims to achieve have been defined. It focuses on the cause-and-effect relationship between project inputs, activities, outputs, outcomes, and impacts. The key features of a result framework are:

i. **Outcome orientation**: Emphasizes the desired results or outcomes that the project intends to achieve, such as changes in policies, conditions, or behavior.

ii. **Linear logic**: Represents the logical flow of results from inputs to impacts in a linear, step-by-step manner.

iii. **Hierarchical structure**: Organizes results in a hierarchical structure, typically with multiple levels of outcomes, starting from immediate outputs to broader impacts.

iv. **Indicators:** Identifies specific indicators or measures that can be used to monitor and assess the achievement of each result level.

v. **Emphasis on results measurement**: A result framework helps in setting targets, tracking progress, and evaluating the effectiveness of project interventions.

- The M&E system will also help to restructure monitoring, evaluation and reporting procedures, responsibilities and processes essential to track performance and progress of the various components and activities of the project in comparison with the set targets in the Annual Work Plans and Budgets. The system will work as a useful tool to guide management during implementation of the Project.
- A detailed M&E framework will be developed to guide M&E processes within the project. The M&E framework developed at the beginning of implementation will detail the scope, organization and contents of the M&E system; roles and responsibilities; a plan for data (sex- and age-disaggregated where appropriate) collection, analysis, reporting, use and management; timeline for M&E-related activities; staffing and capacity building plan; budget; etc.; The

Framework will also include an M & E plan that will set out guidelines for progress monitoring of implementation as well as evaluation of the performance of the Program in relation to the performance indicators in the Program Log frame.

- The monitoring and evaluation (M&E) system of CREW will be based on the Results Framework and will be used to track and monitor progress and impact. The M&E system is also used to measure the performance progress of the project activities. CREW M&E system has the general goal of providing relevant information to all key stakeholders on the implementation progress with specific purpose of:
- Supporting the project key stakeholders in effective decision-making by collecting, analyzing and communicating reliable and timely M&E information on all aspects and subcomponents of the project;
- Undertaking routine and periodic measurement and observation of the inputs, activities and outputs of CREW implementation;
- Keeping track of the daily activities of the project on a continuous basis and identifying any shortcomings with regard to delivery of inputs, execution of activities and achievement of outputs as early as possible.;
- Establishing rapid problem identification as well as an internal and external reporting system among the stakeholders;
- Promoting importance, the need for systematic data collection and utilization of M&E results by the stakeholders;
- Ensuring systematic provision of basic management information essential for coordination and control of implementation decisions, to permit effective use of financial and other project resources;
- Enabling project management and other project stakeholders undertake timely corrective action and/or make needed adjustments to implementation processes, techniques and approaches;
- Empowering and building the staff capacities of implementing organizations to guide, manage, implement, and monitor development activities effectively and efficiently;
- Ensuring accountability of implementing agencies in project delivery, in terms of budgets, expenditures, actions, outputs and results; and
- providing objective evidence on project results, in terms of outputs and outcomes to permit lessons to be learned towards planning future investments and development initiatives.

Monitoring and Evaluation Functions

Monitoring and Evaluation of CREW shall have the following functions:

- Translate PDO into operational plans and coordinate the preparation of annual work plans with quarterly breakdowns for implementation of the project;
- Conduct regular monitoring and evaluation of the project activities at subcomponents level;
- Collect and analyze the project data from implementers;
- Carry out trends' assessment, identify weaknesses in implementation of plans and guide the project team to take corrective action in order to bring operational effectiveness to the project; and
- prepare interim, quarter and annual reports to the management, government and the African Development Fund (AfDB) and the Netherlands Government on the progress of the project.

Roles and Responsibilities of Implementers in Monitoring and Evaluation

A Federal Project Steering Committee (PSC) will be established to ensure inter-ministerial coordination and provide overall coordination, policy guidance and oversight for the implementation of the project. The Federal PSC, chaired by the State Minister of Agriculture (MOA) will include membership from the relevant sector ministries and institutions including Ministry of Irrigation and Low Lands, the Ministry of Water and Energy; Ministry of Finance; Ministry of Industry (MOI); Ethiopian Institute of Agricultural Research (EIAR); Federal Environment Protection Authority (EPA); Agriculture Transformation Institute (ATI), Ethiopian Cooperative Commission (ECC); representatives from the Regional Bureau of Agriculture in Afar, Amhara, Oromia and Somali Regions, a representative from the Millers' Associations and a representative from farmers' cooperatives.

a) PMU and PIU at Federal Level

The PMU & Executives' in MoA and the above mentioned organizations at Federal level will be responsible for implementing, monitoring and evaluating, and reporting the project implementation under their respective subcomponents by monitoring for effective and efficient project implementation. Their major responsibilities in monitoring CREW activities under their respective subcomponents include:

- preparing and adopting standardized data collection and reporting formats to be used for quantitative and qualitative part of the project.
- developing an outcome-oriented approach that would allow corrections during implementation to achieve the objectives and efficient incorporation of lessons learned;
- Preparing all periodic reporting requirements including progress of different activities in the annual work plan, key performance indicators, financial reports and annual audits, procurement and any administrative matter;

- Coordinating timely preparation, processing and approval of annual work plan and budget (AWPB), and follow up the implementation;
- Organize and conduct stakeholders' meetings, workshops and seminars to monitor and evaluate project progress;
- Prepare self-assessment project Implementation Completion Report (ICR) of CREW prior the Bank's Implementation Completion Report;
- Facilitate project mid-term and terminal review; and actively participate in the ICR.
- In addition, PMU in MoA will be responsible for supporting the PIUs in the regions and Cluster Coordination Units at zones for undertaking the project implementation, monitoring and evaluation as well as performance reporting. PMU is also responsible for tacking the performance of project environmental and social safeguards activities as per Environmental and Social Management Framework (ESMF) in coordination with regions and cluster units, and ATI and EIAR.

The overall M&E activities will be coordinated by a Monitoring and Evaluation Specialist of PMU-MoA in collaboration with M&E officers based at the project implementation units of regions, ATI and EIAR. The coordination of all M&E activities of the program comprises overseeing data collection, analysis, and reporting on the implementation and progress of the project. It also includes managing project evaluations and impact evaluations to be carried out and supporting regions M&E staff with regard to M&E requirements. The Monitoring and Evaluation Specialist of PIU-MoA will coordinate the monitoring and evaluation of the project at federal level.

b) PIUs at Regions and Cluster Coordination Units, ATI and EIAR

- Collecting data in standard report format.
- Preparing all periodic reporting requirements including progress of different activities in the annual work plan, key performance indicators, financial and procurement reports;
- Providing the work plan and budget for timely preparation, processing and approval of annual work plan and budget (AWPB) by PMU at federal level and follow up the implementation as per the approved AWPB;
- Organizing and conducting stakeholders' meetings, workshops and seminars to monitor and evaluate project progress;
- providing input for self-assessment project Implementation Completion Report (ICR) of CREW prior the Bank's Implementation Completion Report;
- participating in project mid-term and terminal review; and in the ICR.

- tacking the performance of project environmental and social Monitoring and Evaluation Officers at regions will coordinate M&E for their respective region ensuring that M&E data are collected regularly. The M&E Officers will compile and cross-check reports submitted by cluster coordination units and relevant regional sector bureaus. They will also provide M&E technical assistance and training. It will also conduct once a year a qualitative analysis of M&E data. The PIUs of regions will submit quarterly report to PMU in MoA.

At cluster level, the project implementation progress will be reported by the project coordinate and the financial information and non-financial information management and reporting will be the responsibility of the project accountants.

The CREW M&E System

The implementation of the CREW M&E system will ensure that timely data is collected to inform project activities, and that all data necessary for the reporting on the results indicators is collected. The responsibility for reporting will mainly rest with the PIU & PMU in MoA and the key staff at the Regional PMUs will include a Regional Project Coordinator, Procurement Officer, Monitoring and Evaluation Officer and an Accountant; and also, Project Focal persons, Monitoring and Evaluation Officer and an Accountant in ATI and EIAR .

a) Conceptual Framework

The CREW M&E system is guided by the following conceptual framework of M&E. Both monitoring and evaluation are interrelated but distinct concepts.

Monitoring is a regular collection and analysis of information to assist timely decision-making, ensure accountability and provide the basis for evaluation and learning. It helps continual self-evaluation. It is the process of continuously collecting information about the progress of the project. Collecting the information is a normal part of day-to-day work for monitoring and evaluation staff. The purpose of doing this is helping to decide whether activities are being implemented as planned. The information is then used to make decisions about improving the management and implementation of the project.

Evaluation is a systematic examination of planned, on-going or completed activities of the project. It aims to answer specific management questions and to judge the worthiness of an endeavor and supply lessons learned to improve future actions, planning and decision-making.

Evaluation in its broadest sense tells us that “to assess or judge the worth or value of the project’s outcome”. Furthermore, in the project context it is a process to measure the outcomes, impacts and

effectiveness of a project in order to use lessons learned. We do this by determining the achievement of goals and objectives. Project evaluations are separately scheduled activities performed at specific intervals. For example, in the middle of the implementation period this is known as mid-term evaluation. The other one is terminal evaluation which is conducted at the end of the project. Both the Mid-term and terminal evaluations are conducted by external bodies or financier of the project. However, the internal evaluations of the project are performed by project staff and the implementing body at any time convenient for the project management. Key elements of monitoring and evaluation are summarized in Table below in order to frame the M&E system of CREW.

Table 23 Comparison of monitoring and Evaluation of CREW

Issues/ questions	Monitoring	Evaluation
What	Monitoring is the process of routinely gathering data/information on all CREW result indicators. Basically, for the use of project management decision.	Evaluation is an assessment to value results achieved due to the CREW intervention, as systematic & objective as possible, to gain a deeper perspective on what results and impacts have been achieved. It informs major decision-making.
Purpose	➢ Analyse the input utilized and outputs delivered at all levels, ➢ Identify problems encountered while the project is ongoing and recommend solutions, ➢ Discover trends and patterns due to the intervention of CREW, ➢ Keep project activities on schedule as planned, ➢ Based on data/information gathered measure progress towards intermediate goals and propose revision action in order to achieve the goals set, ➢ Formulate key questions for monitoring at all CREW implementation levels, ➢ Make decisions on the proper utilization of project's resources.	➢ CREW evaluation enables to determine how effective the project has been, ➢ Determine the extent to which PDO have been achieved, ➢ Learn how efficiently things are being done and relevant, ➢ Draw & share lessons learned so that future projects of a similar nature can be improved.
When	Monitoring of the CREW is continuous (daily, monthly, bi-annual, annual). Monitoring activities should be scheduled as part of the project's work plan and be a routine part of project implementers.	As any AFDB financed projects the CREW evaluation is conducted periodically: ➢ At or near the mid-term of implementation ➢ Terminal - at the end of a project ➢ At a point well after a project has been completed (ex-post) to see the project's impact. ➢ PMU and PIUs will conduct internal evaluation quarterly, bi-annually and annually.

Issues/ questions	Monitoring	Evaluation
How	Monitoring of CREW is carried out by staff of PMU, PIUs, beneficiaries and stakeholders based on formal reports, field visits/observations, review of service delivery, meetings with stakeholders, conduct interviews, etc.	PMU and PIUs will carry out internal evaluations by developing suitable terms of reference. Mid-term, Terminal and ex-post evaluations are conducted by financiers (AfDB) or independent consultants,
Why	Monitoring provides the project management team with information needed to analyse the current project performance/situation, identify problems and find solutions, discover trends and patterns, keep in schedule and measure progress towards expected outcomes.	Evaluation in the context to CREW the formal process of documenting the project achievements: ➢ Progress in the work plan (in line to physical, finance and procurement), ➢ Establishment of project management system (Institutions, financial management, procurement, M&E, and other units/sections) ➢ Implementation of planned activities ➢ Achievement of results ➢ Effectiveness of the project ➢ Impact of the project ➢ Efficiency or cost effectiveness & sustainability
Information type	More emphasis on quantitative data collection, conduct comparisons against plans but some qualitative data are gathered to complement the quantitative analysis.	More emphasis on qualitative data but quantitative data is gathered in indicators. Qualitative analysis often supports the quantitative assessment.
Information Category	Based on the M&E Reporting Formats descriptive, analytical and documentary data should all be used to observe outputs and results/outcomes attained due to the project intervention.	Descriptive, analytical and documentary data should all be used to observe and record the ‘what’ and understand the ‘why’ of the project achievements.
How the result is used?	The result of monitoring of CREW is used to improve quality of implementation and adjust planning. Also serves as an input for evaluation.	To judge the impact on the target population and the spill overs; decide about the future of the Project

A monitoring and evaluation procedure should be introduced to regular review progress of activities to be implemented by the project on the basis of identified/developed indicators in order to meet the objectives of the system and resolve issues raised by evaluators. To this end, indicators have been identified and targets are set to allow monitoring and evaluation at different levels, i.e., at the activity, input, output, outcomes and impacts level (intermediate results and PDO).

Periodic review meetings will be among the procedures to be employed to monitor progress of the activities. Participants of such meetings include, among others, representatives of key partners/ stakeholders including project management key staff possibly representatives of project implementers and CREW clients.

Project's self-assessment evaluation shall also be conducted by the CREW staff before external body starts to conduct the terminal evaluation. The information analysed during the project's management self-assessment could be used by the external evaluators for validation purpose or data used as an input.

b) Major Features of CREW M&E System

CREW M&E system is an operational working guide & management tool for the whole project implementation processes (physical & financial), which recognizes the need to maintain an efficient, regular and user-friendly information flow between all key stakeholders.

- It focuses on regular collection, analysis and communication of a mix of quantitative and qualitative information on all aspects and components of the project;
- It employs a bottom-up system of information collection and analysis with active participation of the target destinations. The system is designed to allow timely verification of data and identification of operational problems and improve performance and facilitate faster learning process;
- It will use a bottom up periodic (monthly, quarterly, and annual) reporting on input, output and result indicators;
- Data/information used to measure performance will be collected regularly by PMU and PIUs staff at different times of the project period. The monitoring schedules will be entered directly into the system and will be adhered to by all. All data collected from regions and cluster coordination units and also from ATI and EIAR will be used as inputs for measuring results using the indicators.

Monitoring of CREW

Monitoring refers to a routine checking of the day-to-day CREW activities to determine whether or not activities are being performed on schedule; resources are utilized effectively; and short-term targets for output are being achieved in accordance with project work plan. It involves routinely gathering of information on all aspects of the project, analysis and making decision to improve performance, ensure accountability and provide the basis for continuous learning and adjustment. Generally, monitoring of CREW is the process of collecting data/information on regular basis, analysing, summarizing, and communicating/reporting of project status using prior set indicators.

The following are key questions for an effective monitoring system:

(a) Are we achieving what we intend? How? Why or Why not?

(b) What are the obstacles to implementation?

(c) Has the management of the project been adequate?

(d) What are the strengths and limitations?

(e) Were activities managed well?

(f) Were resources expended according to plan?

(g) Were intended target beneficiaries reached by the project?

(h) Did key project implementers' coordination occur according to plan?

(i) Was the timing and quality of the technical assistance adequate?

(j) Are PCU and PIUs coordinating effectively? and

(k) What are the implementation gaps? How are we going to address them?

Input Monitoring (Progress Monitoring)

Input monitoring or process (activity) monitoring tracks the use of input and resources, the progress of activities for the delivery of output. It examines how activities are performed – the efficiency in time and resources. Moreover, it is concerned with use of staff, material resources and finance that are necessary to produce the intended output of the project.

Regularly, the input monitoring compares targets against actual performance. Regular reports (monthly, quarterly, bi-annual and annual reports) are main sources of data/ information for the input monitoring. PMU and PIUs staff are responsible for the input monitoring activities report as per the agreed reporting procedure. The M&E specialist is responsible whether the appropriate reporting formats have been issued and applied to all concerned partners and he/she is in charge to organize, collect information gathered from different sources, and organize relevant timely meetings/workshops to discuss on the project status.

Output and Results Monitoring

Results monitoring tracks effects and impacts to determine if the project is on target towards its intended results (output, outcome, impact) and whether there may be any unintended impact (positive or negative). Results monitoring enables assessment of what have been achieved as a result of the delivered output during the implementation of the project in terms of reaching the different stakeholders, partners and beneficiaries. Results monitoring will be carried out on the basis of identified and agreed on intermediate results indicators and the corresponding targets.

Evaluation

Evaluation is a process used to determine the effectiveness, efficiency, relevance, impact and sustainability of the project in the light of its stated objectives. It is an important process in identifying the major achievement or success of the project and bottlenecks that hindered the project not to move as it was planned.

It would be necessary to conduct internal evaluation at the project level periodically while the project is ongoing. See Table below that shows the time for internal and external evaluations while the project is under implementation.

Table 24 Evaluation Schedule for CREW

What to Evaluate	Responsible body	When to evaluate				
		Yr 1	Yr 2	Yr 3	Yr 4	Yr 5
Internal evaluation	Implementing body (PMU and PIUs)					
Mid-term review	External (the Bank or consultant)					
Terminal/end	External (the Bank or consultant)					
Impact evaluation	External (The African Development Fund (AfDB))					

The evaluation should answer to the following key questions: -

(i) What did we do & deliver?

(ii) What did we achieve?

(iii) Did we attain what we initially intended to achieve?

(iv) what lessons drawn from this intervention?

(v) Have we fully addressed the causes and effects of the problems of the target end users? If not, what are the gaps?

(vi) what capacity the target groups develop in exploiting the economic potentials of their area?

(vii) Who benefited from the project?

(viii) What factors explain successes/failure and unintended results due to the project intervention, the impact?

Like monitoring of CREW, all the key stakeholders are involved in evaluation of the project. However, unlike the monitoring and internal evaluations, the mid-term review, terminal and post implementation evaluations will be carried out by external evaluators for the purpose of avoiding bias, conflict of interest, and for ensuring objectivity and transparency.

During the process of evaluation of CREW, the PMU and PIUs together with the evaluator should establish a procedure to keep the project team informed of the progress and findings during the evaluation.

Terminal evaluation

Terminal Evaluation will be conducted by external body when the project ends, which assesses the overall performance of the project in terms of relevance, effectiveness, efficiency, impact and sustainability. Lessons learned and experiences drawn from implementation of the project; information generated from project monitoring and review reports, baseline studies and mid-term evaluation will make important input for terminal evaluation of the project.

In general, the final evaluation reviews the sustainability of the project after AfDB's financing ends and proposes CREW exit strategy to enable mainstream the project long term objectives with the regular government planning & implementation system.

Impact Evaluation

The impact evaluation will be operated and managed by African Development Bank (AfDB) research team and an empirical analysis will typically build on survey data that will be collected for the project. In addition to measuring results of project activities, the impact evaluation will involve testing innovative new pilots and rigorously measuring their impact.

CREW Monitoring and Evaluation Plan

Monitoring and Evaluation Plan is prepared for ensuring real progress towards achievement of project results and its development objectives within the time frame, allocated resources and quality of the output.

CREW M&E Plan identifies indicators for various levels of results, data collection methods, frequency of data collection and reporting, and responsible bodies. It guides the planning process for data collection,

analysis and use for measuring the progress of the project. Table 6.3 below shows data collection, analysis and use built in CREW results framework.

Table Results Framework Monitoring and Evaluation Plan

- The project development objective is to improve wheat productivity and production thereby increasing the incomes of small-scale wheat farmers in Ethiopia.

Table 25 PDO indicators

Indicator Name	Unit OF Measurement	Baseline	End Target
to improve wheat productivity	t/ha	3.04	4.0
to improve national wheat production	million	6.58	8.2
Number of Farmers practicing climate-smart agriculture	Number Men Women	100,000 100,000	250,000 250,000
Increased average household incomes for wheat growers	USD/ year	1,216	1,680
Share of wheat production marketed commercially by households	%	30	60

Table 26 Immediate Result indicators

Indicator Name	Unit OF Measurement	Baseline	End Target
Wheat breeder/pre-basic seeds produced	Mt	0	4,000
New farmer households benefitting from extension and climate advisory services	Number (HHS)	0	600,000 (25% WHHs)
. Soils of wheat growing areas analysed and mapped for acidity and salinity	Ha	0	1,000,000 (25% belonging to WHHs)
Area under irrigation and improved water management	Ha	0	24,092 (25% belonging to WHHs)
. New Farmers registered in Farmer Production Clusters (FPCs)	Number	0	106
Storage and warehouses constructed	Number Men Women	0	300,000 200,000
Financial Institutions receiving capacity development to support the Warehouse Receipt System	Number	0	10
Women trained as wheat researchers and extension agents	Numbers	0	200
Women trained as leaders in wheat cooperatives	Numbers	0	200
Women in Agri Booster program	Number	0	10,000

Activity Planning

The PMU and PIUs will coordinate the planning, implementation and reporting of all project activities. Project activities in the 5-year budget will be programmed into annual work plan and budget. Then quarterly work plan and budget will be prepared. Detailed monthly activity planning will be the basis for implementation monitoring and reporting on monthly basis.

The project team is responsible to prepare the list of sub tasks for detail activity planning for each contract of the project. The action plan for each contract needs to be developed and agreed up on with winning contractor/consultant of each contract so as to employ supervision and monitoring the implementation of the contract.

Detailed timing of tasks, sub-tasks and activities under each subcomponent will be prepared at the beginning of each year and updated regularly.

Activity Implementation

The project activities are implemented by MoA and other Stakeholders respective to their responsibilities of subcomponent implementation and when the technical expertise lack to conduct an activity, expertise and consultants' input will be engaged to assist in activity implementation or produce specific output. The project will actively seek and encourage cooperation, technical assistance and implementation support from the African Development Bank (AfDB).

Management Information System of CREW

This management information system (MIS) involves how the data are to be managed and reported to management to assist in monitoring and controlling the project performance and responsibilities, resources and results/ output which will form the basis for assessment of the outcomes and impacts of the project.

There are two sets of data/ information: (i) Data related to input delivery that is financial/budget, human and material resources required to carry out the intended project activities. This data will be needed to track and assess the project performance; (ii) Data related to development aspects of the project such as output/results and impacts with the corresponding indicators.

CREW Data Source

Project data will be drawn basically from the project activities. The project activity reports from PIUs at regions, PIU of other aligned governmental offices with Integration of MoA (ATI and EIAR). The data from regions will be collected and report will be prepared by PIUs at regions. Similarly, the reports from PIUs at regions, PIU and other aligned governmental offices with Integration of MoA

(ATI and EIAR) will be consolidated by PMU at MoA and submitted to the African Development Bank (AfDB), the Netherlands and MoF.

The information flow of the CREW among project implementers, beneficiaries, and other stakeholders is shown in figure 1.1.

CREW Reports

The CREW monitoring and evaluation system includes reporting of project progress on monthly, quarterly, biannual and annual reports as well as non-routine reports that include mid-term review report, survey/analytical report and project completion report. Reports are prepared on regular basis as well as upon completion of the project activity. The reports of PIUs & PMUs at MoA and PIUs & PMUs at Region, and ATI and EIAR will be consolidated by the M&E Specialist into quarterly and annual reports that will be submitted to Project Steering Committee, MoF and the African Development Bank (AfDB), and the Netherlands.

Interim Financial Reports (IFR) are prepared by PCU and submitted to the African Development Bank (AfDB) on quarterly basis. Annual audit reports are submitted to the African Development Bank (AfDB), the Netherlands, Project Steering Committee and MoF.

i. Quarterly Project Progress Reports

The quarterly report is expected in quantitative and qualitative form, the narrative parts of the progress reports should provide information on:

- Work plan for the period and activities completed by the subcomponents of the project;
- Financial progress - in terms of commitments, payments and disbursements from AfDB financing (ADF and TSF);
- Physical progress in terms of physical performance indicators and targets;
- Work plan and related budgets for the next two quarters; and
- Problems/ constraints experienced and management action exercised.

ii. Annual project progress reports

The quarterly progress reports will be consolidated to form the annual project progress reports; and will have similar contents as the quarterly progress reports. The financial report component of the annual progress report contains Audited Project Financial Statements to be submitted to the AfDB and The Netherlands.

The financial report will include statement of sources and utilization of funds, indicating funds of AfDB (ADF and TSF) and project expenditures with appropriate schedules classifying project expenditures by subcomponent, showing yearly and cumulative balances and Special Account reconciliation statement.

M&E Internal Learning

The purpose of promoting monitoring and evaluation and learning is to ensure that learning through practices, experiences and good practices is improved and to ensure that beneficiaries learn from other beneficiaries' experiences and good practices and to take advantage of the opportunities for learning from each other.

The results from monitoring will potentially feed scaling up of good practices, allow adaptation of the program design as necessary, especially in early stages, also support a productive dialogue on findings and will empower them to take corrective actions. Information collected from M&E will be kept at woreda level and compile once a year by the Regional CREW M&E Officer, ATI and EIAR M&E officers.

Thus CREW will promote ME and L by organizing community learning platforms. Thus, farmers, in communities where CREW is active, will discuss results achieved, progress on intended objectives and implementation problems and/or good practices using simple visual formats. This will include cross community monitoring (farmers monitor other farmers' activities/sub-programs); using report finding, stocktaking, case studies (gender, micro irrigations, market information,), and dissemination of M&E findings and other relevant thematic studies (e.g. irrigation sub-programs, impact on women, etc) undertaken within the program. The ME and L will be conducted continuously throughout the life of the project as important. To enhance this different monitoring and evaluation tools can be used

Integration of Cross Cutting Issues to Project Development initiatives

9. Annexes

Annex 1 Monitoring & Evaluation Plan: Intermediate Results Indicators

Annex 2 The Reports Required from CREW

Annex 3 Project target woredas

Annex 1 Monitoring & Evaluation Plan: Intermediate Results Indicators

Indicator Name	Definition/ Description	Frequency	Data Source	Methodology for data collection	Responsibility
Wheat breeder/pre-basic seeds produced					
New farmer households benefitting from extension and climate advisory services					
. Soils of wheat growing areas analysed and mapped for acidity and salinity					
Area under irrigation and improved water management					
. New Farmers registered in Farmer Production Clusters (FPCs)					
Storage and warehouses constructed					
Financial Institutions receiving capacity development to support the Warehouse Receipt System					
Women trained as wheat researchers and extension agents					
Women trained as leaders in wheat cooperatives					
Women in Agri Booster program					

Annex 2 The Reports Required from CREW

Who prepares the report	To Whom the report is submitted to			
	PCU, Federal	MOA PPE	African Development Bank (AfDB)	MOFED
PCU, Federal		➢ Monthly Project Progress Report for internal management use ➢ Quarterly progress report ➢ Annual progress report; ➢ Annual financial audit	➢ Monthly Loan Report ➢ Quarterly IFR ➢ Quarterly progress report ➢ Annual progress report; ➢ Annual ➢ financial audit	➢ Quarterly progress report ➢ Quarterly IFR ➢ Annual progress report; ➢ Annual ➢ financial audit
Federal CREW implementers	➢ Quarterly progress report ➢ Biannual/ ➢ Annual progress report; ➢ Annual financial audit			
PIUs, regions	➢ Quarterly progress report ➢ Biannual/ ➢ Annual progress report; ➢ Annual financial audit			

Annex 3 **Ethiopia: National Wheat Value Chain Project- list of woredas and production potential**											
No	**Region**	**Zone**	**Woreda**	**Data on most recent wheat production**				**Data on most recent wheat production**			
				Area planted (Ha)	**Total Production (Qt)**	**Yield (Qt/ha)**	**Rainfed/ Irrigated)**	**Area planted (Ha)**	**Total Production (Qt)**	**Yield (Qt/ha)**	**Rainfed/ Irrigated)**
	Oromia	Arsi	Limmu fi Bilbilo	20,782.00	665,024.00	32	Rainfed	1,697	61,092	36	Irrigated
			Digalu Xiijoo	22,980.00	735,360.00	32	,,	1,647	59,292	36	,,
			Muunessa	19,138.00	612,416.00	32	,,	1,098	39,528	36	,,
			Roobee	17,943.00	574,176.00	32	,,	477	17,172	36	,,
			Heexosa	15,953.00	510,496.00	32	,,	50	1,800	36	,,
			Loode Heexosa	9,474.00	303,168.00	32	,,	471	16,956	36	,,
			Siree	7,820.00	250,240.00	32	,,	548	19,728	36	,,
			Shirka	16,336.00	522,752.00	32	,,	1,386	49,896	36	,,
			Xiyoo	9,351.00	299,232.00	32	,,	3,270	117,720	36	,,
			Jajuu	10,245.00	307,350.00	30	,,	4,943	168,062	34	,,
		West Arsi	Gadab Haasaasa	29,187.00	992,358.00	34	,,	1,908	83,952	44	,,
			Dodolaa	18,722.00	636,548.00	34	,,	1,300	57,200	44	,,
			Arsii Nagellee	16,513.00	528,416.00	32	,,	2,852	102,672	36	,,
			Adaabbaa	12,629.00	429,386.00	34	,,	3,319	132,760	40	,,
		Bale	Sinaanaa	46,830.00	1,592,220.00	34	,,	2,195	83,410	38	,,
			Gaasara	30,584.00	1,039,856.00	34	,,	2,310	87,780	38	,,
			Gooroo	18,260.00	620,840.00	34	,,	2,275	86,450	38	,,
			Agaarfaa	13,161.00	447,474.00	34	,,	1,320	50,160	38	,,
			Goobba	8,150.00	277,100.00	34		1,345	51,110	38	
		East Bale	Gindhiir	17,059.00	580,006.00	34	,,				
			Gololcha	18,011.00	612,374.00	34	,,				

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				Area planted (Ha)	**Total Production (Qt)**	**Yield (Qt/ha)**	**Rainfed/ Irrigated)**	**Area planted (Ha)**	**Total Production (Qt)**	**Yield (Qt/ha)**	**Rainfed/ Irrigated)**
		East Shewa	Gimbichu	17,796.00	569,472.00	32	,,	10,165	365,940	36	,,
			Dugdaa	30,718.00	982,976.00	32	,,	6,977	251,172	36	,,
			Luumee	15,075.00	482,400.00	32	,,	8,862	319,014	36	,,
			Booraa	23,128.00	740,096.00	32	,,	5,892	212,112	36	,,
			Adaami tuluu jiddo kombolchaa	17,864.00	500,192.00	28	,,	1,936	61,952	32	,,
			Liiban cuqaalaa	21,872.00	656,160.00	30	,,	6,678	227,052	34	,,
		Oromia special	Sabataa hawaas	24,224.00	775,168.00	32	,,	4,624	166,464	36	,,
			Walmaraa	12,360.00	395,520.00	32	,,	4,738	170,568	36	,,
			Barrak	8,581.00	274,592.00	32	,,	1,550	55,800	36	,,
			Aqaaqii	12,089.00	386,848.00	32	,,	4,634	166,824	36	,,
			Sululta	4,913.00	157,216.00	32	,,	1,654	59,544	36	,,
		North Shewa	Kuyyuu	12,180.00	365,400.00	30	,,	856	29,104	34	,,
			Alaltuu	7,696.00	230,880.00	30	,,	1,215	41,310	34	,,
			Qimbibiit	6,355.00	190,650.00	30	,,	2,109	71,706	34	,,
			Wucaalee	6,369.00	191,070.00	30	,,	3,668	124,712	34	,,
		West Shewa	Amboo	14,653.00	439,590.00	30	,,	1,298	44,132	34	,,
			Jalduu	13,345.00	400,350.00	30	,,	1,990	67,660	34	,,
	Afar	Awsiresu (zone-1)	Dubti					12,250.00	417,500.00	34.08	,,
			Afambo					183.00	486.00	18.00	,,

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				Area planted (Ha)	**Total Production (Qt)**	**Yield (Qt/ha)**	**Rainfed/ Irrigated)**	**Area planted (Ha)**	**Total Production (Qt)**	**Yield (Qt/ha)**	**Rainfed/ Irrigated)**
		Gebiresu (zone-3)	Amibara					7,300.00	244,000.00	33.42	,,
	Amhara	South Wello	Borena	21,842.00	611,576.00	28	Rain fed				,,
			Wereilu					4,150.00	132,800.00	32.00	Irrigated
			Jama	20,578.00	576,184.00	28	Rain fed				
			Legambo	15,290.00	412,830.00	27	Rain fed				
			Delanta	9,140.00	255,920.00	28	Rain fed				
		West Gojam	Wenberema	18,966.00	568,980.00	30	Rain fed				
			North Mecha					10,242.00	327,744.00	32.00	Irrigated
			Bahirdar					12,690.00	406,080.00	32.00	,,
			Demebecha	14,638.00	409,864.00	28	Rain fed				,,
			Jabi					12,460.00	348,880.00	28.00	Irrigated
		North Shewa	Menjar Shenkora	16,330.00	489,900.00	30	Rain fed				
			Mojanaena wedera	14,098.00	451,136.00	32	Rain fed				
			Basoena werana	11,669.00	373,408.00	32	Rain fed				
			Angolelana enatera	8,320.00	266,240.00	32	Rain fed				
			Moertena Jiru	12,150.00	388,800.00	32	Rain fed				
		East Gojam	Baso Liben	18,526.00	574,306.00	31	Rain fed				
			Debere Elias	22,950.00	711,450.00	31	Rain fed				
			Andede	10,180.00	315,580.00	31	Rain fed				
		Awi	Ayehu Gagussa					8,000.00	264,000.00	33.00	Irrigated
			Dangela					6,500.00	214,500.00	33.00	,,

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			Fageta					5,500.00	181,500.00	33.00	,,
			Gagussa					5,500.00	181,500.00	33.00	,,
		Central Gonder	East Denebia					6,075.00	194,400.00	32.00	,,
			Gunder Zuria					7,224.00	231,168.00	32.00	,,
	Somali	Fafen zone	Tuli Guleed	61,689.00	1,517,549.40	24.60	Rainfed				,,
			Awbare	41,250.00	1,014,750.00	24.60	Rainfed				,,
			Harorays	18,750.00	525,000.00	24.50	Rainfed				,,
		Liben	Dekasuft	4,878.00	42,179.00	23.00	Rainfed				,,
			Bokolmayo					140.00	1,800.00	25.00	Irrigated
			Qarsadula					9,240.00	105,950.00	24.00	,,
			Dolo-Ado					200.00	1,080.00	25.00	,,
		Afder	Dolobay					156.00	150.00	25.00	,,
		Shabele	Gode					370.00	50.00	27.00	,,
			Berano					139.00	880.00	26.00	,,
			Adadle					185.00	500.00	25.00	,,
		Total	**78**	**969,590.00**	**29,781,024.40**		**-**	**211,760.50**	**6,976,774.00**		**-**